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SLEEP

15 SLEEP PROTOCOLS

1. View Sunlight as Soon as Possible After Waking
2. View Sunlight in the Late Afternoon
3. Avoid Viewing Bright Lights in the Final Hour Before Bedtime
4. Find Your Ideal To-Sleep and Wake-Up Times and Strive to Adhere to Them
5. Lower Your Core Body Temperature at Night
6. Increase Your Core Body Temperature in the Final One to Two Hours of Sleep Before You Wake Up
7. Nap (Correctly) to Offset Minor Sleep Deficits
8. Use Non-Sleep Deep Rest (NSDR) to Offset Sleep Deficits (and to Get Better at Falling and Staying Asleep at Night)
9. Consider Supplements to Support Sleep
10. Approach Prescription Sleep Drugs with Caution
11. Adjust Your Caffeine Schedule to Improve Your Sleep
12. Avoid Alcohol Within Eight Hours Before Sleep
13. Avoid Cannabis as a Sleep Aid
14. Avoid Screens Within an Hour or More Before Bedtime
15. A Tip Sheet for Avoiding Jet Lag

Sleep is the subject of the first chapter of this book because sleep profoundly impacts every aspect of our physiology and psychology. Put simply, sleep is the foundation of our mental health, our physical health, and our performance.[1] We win by sleeping.

I came to that realization pretty early but only because I struggled with sleep. As a kid I slept like a log—always a solid eight to nine hours, rarely waking up in the middle of the night even once. As a teen I would stay out late and sleep in on the weekends, as most of us did. I never thought about my sleep.

When I reached my junior year in high school, however, everything changed. I would lie down, fall asleep right away, but then wake up three to four hours later, wide awake. Then

[1] Meir H. Kryger et al., Kryger's Principles and Practice of Sleep Medicine, 7th ed. (Elsevier UK, 2022); Alexander J. Scott et al., "Improving Sleep Quality Leads to Better Mental Health: A Meta-Analysis of Randomised Controlled Trials," Sleep Medicine Reviews 60 (December 2021): 101556, https://doi.org/10.1016/j.smrv.2021.101556; Esther Yuet Ying Lau et al., "A Longitudinal Investigation of the Bidirectional Relationship of Sleep Quality with Emotional Stability and Social Cynicism in a Large Community Sample," Sleep Health 7, no. 3 (2021): 384–89, https://doi.org/10.1016/j.sleh.2021.03.007. Luciana Besedovsky et al., "Sleep and Immune Function," Pflügers Archiv: European Journal of Physiology 463 (2012): 121–37, https://doi.org/10.1007/s00424-011-1044-0; Ken A. Paller et al., "Memory and Sleep: How Sleep Cognition Can Change the Waking Mind for the Better," Annual Review of Psychology 72 (January 2021): 123–50, https://doi.org/10.1146/annurev-psych-010419-050815; Bryce A. Mander et al., "Wake Deterioration and Sleep Restoration of Human Learning," Current Biology 21, no. 5 (2011): R183–84, https://doi.org/https://doi.org/10.1016/j.cub.2011.01.019.

I would toss and turn. Eventually I would fall back asleep but then needed to sleep really late into the morning to feel rested. School started at 8:00 a.m., so that wasn't an option most days. That had me falling asleep in class, feeling wide awake in the evening, and then the whole cycle would repeat.

Toward the end of high school when I started getting into fitness, I learned that we recover from exercise during sleep. That was the early 1990s and the public was not yet aware of the importance of sleep. There was some science on it—in fact some of the best work on the role of sleep for health and cognition was coming from the Stanford Sleep Lab just a few miles up the road from where I lived. But it was from a prominent trainer in the world of fitness that I first learned to take sleep seriously. He gave me one mostly good suggestion for sleep, and one terrible one. The mostly good suggestion was to read the work of Allan Hobson—a sleep researcher from Harvard—and I eventually did, although I discovered Allan's focus was mainly on the importance of dreams, which felt like a bit of a tease for someone with sleep issues. The terrible advice he gave me? He told me if I woke up in the middle of the night and couldn't fall back asleep that I should drink a half mug of white wine. I was seventeen! I tried it anyway, and it didn't work for me. Thankfully. We now know alcohol greatly reduces the

quality and restorative properties of sleep, even if it does help some people fall asleep.

Eventually I figured out how to master the art of getting a great night's sleep, but it took time. Years, in fact. The science progressed too, of course, and we now know a lot about both the importance of sleep and ways to remedy pretty much any sleep issue. I wish I had the information you are about to read back in high school, and college, and graduate school, and my postdoc. The reality is, the busier we are, the more important sleep is to our health and performance. But no matter how busy or calm our life is, a great night's sleep is crucial to our mind and body.

If you are like the majority of people, and you struggle to get a great night's sleep, you want to remedy that as soon as possible.[2] But even if you are one of those rare individuals who sleeps deeply every night and wakes up feeling fully refreshed every day, there are still ways to improve your sleep and thus, your quality of life. Regardless of which category you fall into, this chapter will help.

I'd like to mention here that this chapter is a long one. We go deep into the science of sleep, but I promise you it's all made

[2] Healthy People 2030, "Increase the Proportion of Adults Who Get Enough Sleep—SH-03," data from the National Health Interview Survey, accessed September 30, 2024, https://health.gov/healthypeople/objectives-and-data/browse-objectives/sleep/increase-proportion-adults-who-get-enough-sleep-sh-03/data?tab=data-table#data-table.

accessible. I know that understanding this science will help you get a great night's sleep. I wrote this chapter so you can browse by topic or tool. Start with the parts that interest you most, or the protocols that you think will help you right now.

I prefer to focus on the benefits of doing things correctly, but when it comes to sleep, it's worth thinking a bit about the downsides of getting things wrong too, and learning to avoid those. Sleep deprivation—a term that means getting too little sleep or waking up too many times during the night—has been linked to significantly elevated risk of heart disease, stroke, and numerous hormonal imbalances involving—just to name a handful: cortisol, estrogen, testosterone, and insulin, the latter of which can cause type 2 diabetes.[3] Obesity, mental health issues, physical injuries, cancer, car accidents, depression, hypertension, and age-related

[3] Endurance O. Evbayekha et al., "Sleep Deprivation Is Associated with Increased Risk for Hypertensive Heart Disease: A Nationwide Population-Based Cohort Study," Cureus 14, no. 12 (2022): e33005, https://doi.org/10.7759/cureus.33005; Chun Seng Phua et al., "Relationship Between Sleep Duration and Risk Factors for Stroke," Frontiers in Neurology 8 (August 2017): 392, https://doi.org/10.3389/fneur.2017.00392; Eve Van Cauter et al., "The Impact of Sleep Deprivation on Hormones and Metabolism," Medscape Neurology, April 28, 2005, https://www.medscape.org/viewarticle/502825; Ali Darraj, "The Link Between Sleeping and Type 2 Diabetes: A Systematic Review," Cureus 15, no. 11 (2023): e48228, https://doi.org/10.7759/cureus.48228; Yuan Pan et al., "The Association Between Sleep Deprivation and the Risk of Cardiovascular Diseases: A Systematic Meta-Analysis," Biomedical Reports 19, no. 5 (September 2023): 78, https://doi.org10.3892/br.2023.1660; Francesco P. Cappuccio and Michelle A. Miller, "Sleep and Cardio-Metabolic Disease," Current Cardiology Reports 19, no. 11 (September 2017): 110, https://doi.org/0.1007/s11886-017-0916-0; Peter Y. Liu and Radha T. Reddy,

dementia have all also been linked to poor sleep.[4] Sleep deprivation doesn't necessarily directly create these problems; rather it impairs cellular mechanisms that cascade to produce multiple issues within and between different organs and systems.

To be clear, everyone has a bad night's sleep now and again. But here's a good goal for you: Aim to sleep well 90 to 95 percent of the nights of your life, and allow the remaining 5 to 10 percent of those nights to be bad due to uncontrollable negative circumstances, like a work conflict, or joyful experiences, like a great party. Do your best to make sleep a

"Sleep, Testosterone and Cortisol Balance, and Ageing Men," Reviews in Endocrine and Metabolic Disorders 23 (September 2022): 1323–39, https://doi.org/10.1007/s11154-022-09755-4; Peter Y. Liu et al., "Clamping Cortisol and Testosterone Mitigates the Development of Insulin Resistance During Sleep Restriction in Men," Journal of Clinical Endocrinology and Metabolism 106, no. 9 (May 2021): e3436-48, https://doi.org/10.1210/clinem/dgab375; Suling Zhang et al., "Effects of Sleep Deprivation on Heart Rate Variability: A Systematic Review and Meta-Analysis," Frontiers in Neurology—Sleep Disorders 16 (2025): 1556784, https://doi.org/10.1177/25424823251391704; Jaruwan Vierra et al., "Effects of Sleep Deprivation and 4-7-8 Breathing Control on Heart Rate Variability, Blood Pressure, Blood Glucose, and Endothelial Function in Healthy Young Adults," Physiological Reports 10, no. 13 (July 2022): e15389, https://doi.org/10.14814/phy2.15389.

[4] Vijay Kumar Chattu et al., "The Global Problem of Insufficient Sleep and Its Serious Public Health Implications," Healthcare 7, no. 1 (2019): 1, https://doi.org/10.3390/healthcare7010001; Healthy People 2030, "Sleep," accessed September 30, 2024, https://health.gov/healthypeople/objectives-and-data/browse-objectives/sleep; Alberto R. Ramos et al., "Sleep Deprivation, Sleep Disorders, and Chronic Disease," Preventing Chronic Disease 20 (August 2023): 230197, https://dx.doi.org/10.5888/pcd20.230197.

top priority so that the rest of your life can be that much better.

Perhaps you have heard some of this before. Most people know a bit about the benefits and importance of sleep. As I mentioned above, that is a fairly recent development. It was about a decade ago when Matthew Walker, who formerly directed the Center for Human Sleep Science at the University of California, Berkeley, and is currently at the University of Texas in Dallas, wrote the now famous book, Why We Sleep. At that time, the "I'll sleep when I'm dead" or the "I'll catch up on sleep this weekend" mentalities prevailed.[5] Thanks in large part to Matt's book and other public education efforts, people now recognize they must sleep, yet the majority of people still don't sleep well and far too many people sleep poorly, with nearly half of adults reporting their sleep satisfaction as an "F"—meaning the lowest possible grade.[6] It is true that we all have slightly different sleep needs. Some of us consider ourselves morning larks. We prefer to rise early. Others are night owls who prefer to stay up late. These are what sleep researchers call chronotypes: genetically determined features of the chemicals, cells, and circuits that regulate when we prefer to

[5] Matthew Walker, Why We Sleep: Unlocking the Power of Sleep and Dreams (Scribner, 2017).

[6] "2025 Sleep in America Poll," National Sleep Foundation, March 6, 2025, https://www.thensf.org/wp-content/uploads/2025/03/NSF_SIA_2025-Report_final.pdf.

go to sleep and wake up. You can blame your parents for your bedtime (and wake time) preference.

Separate from your chronotype is the total amount of sleep you need. Some of us need six hours to feel fully rested, others need eight, and some people feel great with just five. I should point out, however, that the feel-great-on-five folks are exceedingly rare. Most people need seven to eight.[7] But as mentioned above, according to recent reports from the National Sleep Foundation and the Centers for Disease Control and Prevention (CDC), at least a third of Americans do not get nearly enough sleep—and that statistic is likely to be a vast underestimate given the enormous increase in time spent on social media and online nowadays.[8] In fact, that figure skyrockets to more than 75 percent among high school students. That's right, approximately three-fourths of teens are not getting enough sleep,[9] which is extremely concerning given that our adolescent and teen years are when much of our brain wiring related to mood and behavior regulation, as well as attention and learning, gets solidified. From birth

[7] Nathaniel F. Watson et al., "Joint Consensus Statement of the American Academy of Sleep Medicine and Sleep Research Society on the Recommended Amount of Sleep for a Healthy Adult: Methodology and Discussion," Journal of Clinical Sleep Medicine 11, no. 8 (2015): 931–52, https://doi.org/10.5664/jcsm.4950.

[8] Healthy People 2030, "Increase the Proportion."

[9] Tanner J. Bommersbach et al., "Insufficient Sleep Among US Adolescents Across Behavioral Risk Groups," JAMA 335, no. 13 (March 2026): 1173–76, https://doi.org/10.1001/jama.2026.1417.

until about age twenty-five is an especially sensitive and time-limited phase, or what is called a critical period.[10] And the thing about critical periods is eventually they close: While our brains remain plastic our entire lives, the amount of neuroplasticity possible in those first twenty-five years far exceeds what is possible afterward, so teens can't simply make up for their lost brain wiring later on.[11] We have no idea what the long-term consequences of this kind of chronic sleep deprivation are, but we do know that far too many young people are missing out on sleep at exactly the time when they need it most.

Compulsive overuse of phones and other screen-based technologies at night is a major factor for the poor sleep of adolescents and teens, but also for the poor sleep habits of adults.[12] A central feature of life in the twenty-first century

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[10] J. Miguel Cisneros-Franco et al., "Critical Periods of Brain Development," Handbook of Clinical Neurology 173 (2020): 75–88, https://doi.org/10.1016/B978-0-444-64150-2.00009-5.

[11] Valerie J. Sydnor et al., "Investigating Hierarchical Critical Periods in Human Neurodevelopment," Neuropsychopharmacology 51, no. 1 (January 2026): 67–85, https://doi.org/10.1038/s41386-025-02246-5; Takao K. Hensch, "Critical Period Regulation," Annual Review of Neuroscience 27 (July 2004): 549–79, https://doi.org/10.1146/annurev.neuro.27.070203.144327; Eric I. Knudsen, "Sensitive Periods in the Development of the Brain and Behavior," Journal of Cognitive Neuroscience 16, no. 8 (October 2004): 1412–25, https://doi.org/0.1162/0898929042304796; Bart Larsen et al., "A Critical Period Plasticity Framework for the Sensorimotor-Association Axis of Cortical Neurodevelopment," Trends in Neurosciences 46, no. 10 (October 2023): 847–62, https://doi.org10.1016/j.tins.2023.07.007.

[12] Bommersbach, "Insufficient Sleep"; Maria Fiore et al., "Digital Devices and Sleep Use in Adolescents: An Umbrella Review," International Journal of

is to always be connected online and by text. So much so that by the classic definition of shift work, many—perhaps even most people—worldwide, now meet the definition of a shift worker, that is, awake for two to three hours between 10:00 p.m. and 5:00 a.m. Shift work is not just about staying up late but also involves reduced amounts of exposure to sunlight during the day, lack of sufficient darkness at night, and a nocturally-shifted eating schedule. That has well-established negative health effects, including poor gut health, mental health deficits, and even decreased lifespan.[13] So this is no trivial matter. Smartphone technologies are not to blame per se, but when we use those technologies certainly is. We can scroll 24-7 and unless we resist, the light from our phone, tablet, or computer screen impairs our sleep. Not to mention that the content we absorb on those screens often affects our sleep, too, because of the elevated levels of arousal—aka stress—that content creates. Even if being on our phones or tablets in the hours before sleep doesn't stress us out, it can disrupt our sleep.[14] We rarely hear how texting alters our sleep patterns but I would argue that for many

Environmental Research and Public Health 22, no. 10 (October 2025): 1517, https://doi.org/10.3390/ijerph22101517.

[13] Isabel Silva and Daniela Costa, "Consequences of Shift Work and Night Work: A Literature Review," Healthcare 11, no. 10 (2023): 1410, https://doi.org/10.3390/healthcare11101410.

[14] Cheng Wang et al., "The Association Between Blue Light Exposure and Incidence of Type 2 Diabetes: A Prospective Study of UK Biobank," Environmental Research 246 (April 2024): 118070, https://doi.org/10.1016/j.envres.2023.118070; Izere Salomon et al., "Artificial

people, it is at least as pernicious as social media in this regard. To be clear, texting is a wonderful tool for communication; it allows us to connect quickly. And social media can be a good source for learning, social connection, and entertainment. The downside, however, is that we can have conversations and observe events at any time of day or night. If we wake up in the middle of the night, we know we should roll over and go back to sleep, but many of us instead start reading and sending texts, or scrolling the internet or browsing social media.

So it comes as no surprise that most of us now qualify as shift workers but very few of us are sacrificing our sleep hours by doing actual work.[15] We have all become so habituated to this new normal of consuming content and communicating at all hours that we don't see how out of sync it really is with our sleep and therefore our health.

The good news is, we can fix our sleep duration, quality, and even determine its timing—regardless of our

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Light Exposure at Night: A Hidden Risk Factor for Type 2 Diabetes," Sleep Medicine: X 10 (June 14, 2025): 100146, https://doi.org/10.1016/j.sleepx.2025.100146; Cappuccio, "Sleep and Cardio-Metabolic Disease," 110; Zhang, "Effects of Sleep Deprivation on Heart Rate Variability: A Systematic Review and Meta-Analysis," 1556784.

[15] Andrew Huberman, "Dr. Satchin Panda: Intermittent Fasting to Improve Health, Cognition and Longevity," Huberman Lab (podcast audio), March 13, 2023, https://www.hubermanlab.com/episode/dr-satchin-panda-intermittent-fasting-to-improve-health-cognition-and-longevity.

chronotype—without having to disengage too much from these technologies.

And it's oh-so-worth it because when I say, "We win by sleeping," I am referring to the fact that sleep shapes everything about us—how we look, how we think, and how we feel. But I also mean that if you get your sleep right 90 percent of the time, you'll be absolutely amazed how your appearance, health, and performance will improve. The data say you'll likely live longer as well.[16]

This chapter offers protocols you can easily integrate into your life to get far more restful and restorative sleep. That is, to "sleep better." None take much time, and many of them are downright pleasurable. Let's consider these protocols in order of their importance.



SLEEP PROTOCOL

View Sunlight as Soon as Possible After Waking

WHAT TO DO

- Get morning sunlight in your eyes as soon as possible after waking. Go outside and look in the direction of the

[16] Diego Robles Mazzotti et al., "Human Longevity Is Associated with Regular Sleep Patterns, Maintenance of Slow Wave Sleep, and Favorable Lipid Profile," Frontiers in Aging Neuroscience 6 (June 2014): 134, https://doi.org/10.3389/fnagi.2014.00134.

sun for at least five to ten minutes each morning. Your goal should be to get sunlight in your eyes within the first thirty to sixty minutes of waking up.

- A few things to keep in mind. If you wake up before the sun rises, you'll obviously have to wait. If you can view the sun as it comes up across the horizon, great, but that's not vital for this protocol to work. Morning sunlight, as we will review, differs in quality from sunlight at other times of the day. It is best viewed by facing the sun without any physical barriers, like awnings or buildings, obscuring your view. Don't wear a brimmed hat, as it blocks some of the light that would otherwise reach your eyes from overhead. Wearing eyeglasses or contacts is fine but, alas, this protocol won't work nearly as well through a window or windshield, as that will filter out many of the relevant wavelengths of light. Skip the sunglasses while doing this protocol. What about clouds? No matter how overcast, there is always some sunlight coming through once the sun is up. In fact, the more accurate way to state this protocol would be "get as-early-in-your-day-as-possible daylight in your eyes," but that is a mouthful. The point is that the sun provides daylight even on overcast days. Just compare how bright it is at 8:00 a.m. on an overcast day to midnight the night before. So forget the "there is no sun out in winter where I live" argument. There is daylight, and you want to view

it. Indeed, this protocol is especially important on cloudy days. Caution: Do not stare at the sun. Never force your eyes to stay open. Allow your natural squint and blink reflexes to protect your eyes. If the sun still feels too bright, look a few degrees to the side of the sun.

HOW IT WORKS

Correctly timed sunlight viewing—especially morning sunlight viewing—is immensely beneficial to our overall health. This simple, zero-cost protocol powerfully affects your daytime mood, focus, and alertness, and improves your sleep, which further improves your health and performance in myriad ways.

The power of this protocol arises from its impact on three synergistic elements: your circadian clock, your eyes (or more specifically, the nerve cells in your eyes that detect sunlight and communicate its presence to your circadian clock), and specific qualities of light that come from the sun, especially when it is low, relative to the horizon. Together, these three elements make up a robust network that drives many of the biological processes in your brain and body.

I first learned about the importance of viewing morning sunlight from one of the world's preeminent sleep researchers, Seiji Nishino, MD, PhD. I was working in his lab at Stanford during a summer home from college, and I

noticed he would always stop to face the sun for a few minutes before going into the lab for the day. Even if he arrived at the lab before sunrise (not uncommon for researchers in a sleep lab!), he would make sure to step outside shortly after sunrise and face the sun for a few minutes before returning to work. That was way back in 1996—when our understanding of the mechanisms of human circadian biology was rudimentary compared to now. Seiji explained that viewing morning sunlight felt good, that it gave him more energy for the day, and that it improved his sleep. I started joining him and joked that we resembled the many ground squirrels around us up on their haunches doing the very same thing. Little did I know that the exact same morning sunlight-sensing cells, circuits, and mechanisms were at play in us and those squirrels and, in fact, in every mammal where they have been examined.

I quickly noticed that on days I viewed morning sunlight, I also felt good, had more energy for the day, and slept better at night, so I kept with it. You should experience those benefits, too, because of the way morning sunlight—whether on a clear or cloudy day—kickstarts a powerful set of hormonal and neural mechanisms that positively shift the state of your entire brain and body for many hours afterward. It can work the first time and every time, and once you understand a bit about those mechanisms, it will allow you to easily adjust this protocol to your schedule and

circumstances to derive the maximum benefits with minimal time investment.

Your Circadian Clock

Located a few centimeters above the roof of your mouth is a small cluster of neurons called the suprachiasmatic nucleus, or SCN.[17] The SCN is your master circadian clock. Its job is to coordinate the activities of all the cells and organs in your body into a coherent twenty-four-hour rhythm so that certain cells and organs in your body are more active, and other cells and organs are less active, depending on the phase of that twenty-four-hour rhythm. The SCN keeps your body and brain on schedule.

The first documented discovery of a circadian clock came not from studies of humans or even of rodents, but of plants. In 1729, a French astronomer named Jean-Jacques d'Ortous de Mairan noticed that a mimosa plant kept in a totally dark room continued to rhythmically fold and unfold its leaves every twenty-four hours, just as if the sun was shining through the window.[18] It was a mystery: If the plant was not unfurling to absorb sun or other light, then what was guiding

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[17] Melinda A. Ma and Elizabeth H. Morrison, "Neuroanatomy, Nucleus Suprachiasmatic," StatPearls, last updated July 24, 2023, https://www.ncbi.nlm.nih.gov/books/NBK546664/.

[18] Rong-Chi Huang, "The Discoveries of Molecular Mechanisms for the Circadian Rhythm: The 2017 Nobel Prize in Physiology or Medicine," Biomedical Journal 41, no. 1 (2018): 5–8, https://doi.org/10.1016/j.bj.2018.02.003.

its movements? About two hundred years later, German biologist Erwin Bünning tracked the movements of bean plant leaves and reported that they followed a cycle lasting 24.4 hours. Observations such as these confirmed that plants had internal, or what scientists call “endogenous,” clocks that followed what later became known as… a circadian rhythm, named for the Latin circa diem, or “around a day.” And as decades of research would eventually confirm, the vast majority of organisms on Earth—not just plants but also insects, rodents, reptiles, and all mammals, including humans—have an endogenous timekeeper that keeps a twenty-four-hour circadian rhythm, even in the absence of any external cues about the status of the outside world.[19]

Our core body temperature, the release of certain hormones, and most notably our sleep-wake cycle, all follow a twenty-four-hour circadian rhythm.[20] Even when scientists extracted individual cells from the body, the cells continued

[19] Martha Hotz Vitaterna et al., “Overview of Circadian Rhythms,” Alcohol Research & Health 25, no. 2 (2001): 85–93, https://pmc.ncbi.nlm.nih.gov/articles/PMC6707128/.

[20] Roberto Refinetti, “Circadian Rhythmicity of Body Temperature and Metabolism,” Temperature 7, no. 4 (2020): 321–62, https://doi.org/10.1080/23328940.2020.1743605; Davide Gnocchi and Giovannella Bruscalupi, “Circadian Rhythms and Hormonal Homeostasis: Pathophysiological Implications,” Biology 6, no. 1 (2017): 10, https://doi.org/10.3390/biology6010010; Sujana Reddy et al., “Physiology, Circadian Rhythm,” StatPearls, last updated May 1, 2023, https://www.ncbi.nlm.nih.gov/books/NBK519507/.

to follow this rhythm.[21] And which genes they expressed and when, the proteins downstream of those genes, the cells' enzymatic activities—indeed not just some things but nearly everything about cellular function—appeared to vary, not randomly, but rather systematically every twenty-four hours. That was remarkable.

But what is equally remarkable is that all cells in our body can coordinate their individual rhythms in a manner that is timed perfectly with one another—like a well-practiced symphony, except this symphony involves not a few hundred or even a thousand participants, but trillions. And just like a symphony, it is not as if every cell simply needs to start and end its activity at the same time. Some cells need to be active at the same time, or in-phase with one another; some need to be active at opposite times, or in anti-phase, and still others repeat their cycles of activity faster, then slower, then faster, again with a unique signature and yet, still repeating once every twenty-four hours. And the start and end time of each cell's twenty-four-hour rhythm must be tightly bound to all the other cells of the body, just like the start and end of a symphony. Without this coordination, you end up with noise rather than music, illness rather than wellness. The only difference is the symphony that is your

[21] Hitoshi Okamura, "Clock Genes in Cell Clocks: Roles, Actions, and Mysteries," Journal of Biological Rhythms 19, no. 5 (2004): 339–458, https://doi.org/10.1177/0748730404269169.

physiology never ceases; it repeats every twenty-four hours, from the day you're born until the day you die.

The basis of that persistent and extraordinarily precise biological coordination remained a mystery until somewhat recently. Clues came from multiple sources. First, studies in which people were placed into constant dark environments—such as caves or windowless bunkers—confirmed the existence of a circadian rhythm in humans. Back in 1938, two physiologists spent a month in Mammoth Cave, Kentucky (perhaps figuring this experiment was best done on themselves before asking anyone else to try it).[22] There, they confirmed that human body temperature has a circadian rhythm whereby it increases upon waking, decreases approximately ten to twelve hours later, drops to its nadir during sleep, then rises once again—all once every twenty-four hours, even though the subjects/researchers had no access to sunlight or other cues about time of day. And other studies confirmed that, in the absence of an alternating pattern of ambient light and darkness, the human body still maintains a circadian rhythm for numerous aspects of our physiology.[23]

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[22] "Mammoth Cave," Collections & Exhibits, University of Chicago Library, 2016, https://www.lib.uchicago.edu/collex/exhibits/discovering-beauty/mammoth-cave/.

[23] Kristin D. Hussey, "Timeless Spaces: Field Experiences in the Physiological Study of Circadian Rhythms, 1938–1963," History and Philosophy of the Life Sciences 45, article no. 17 (2023), https://doi.org/10.1007/s40656-023-00571-w.

In 1966, a German physician named Jürgen Aschoff initiated what became known as the bunker experiment, in which subjects (not researchers this time!) spent several weeks in a sealed bunker, living life as usual but without any windows or other ways to determine the passage of time.[24] They found that subjects generally spent about one-third of each twenty-four-hour period sleeping and two-thirds of it awake, just as most humans living aboveground do. However, the wake-up and to-sleep times of the people in the bunker drifted—they woke up slightly later and went to sleep slightly later each day. Thus, the bunker experiment confirmed the existence of a circadian rhythm, but one that requires some external cue to anchor it. What remained mysterious was the biological basis of the circadian clock—was it a bloodborne factor? In the liver? In the brain? In the pineal gland? Theories abounded but none of them held up to experimentation. And another major question remained: In the environment outside the bunker, which external cues were responsible for setting that clock to a specific start and finish time each day?

Subsequent discoveries, first in rodents and later in humans, revealed the existence of a small cluster—or what is called a

[24] Maria-Angeles Bonmati-Carrion et al., “Living Without Temporal Cues: A Case Study,” Frontiers in Physiology 11 (February 2020): 11, https://doi.org/10.3389/fphys.2020.00011; Jürgen Aschoff, “Circadian Rhythms in Man,” Science 148, no. 3676 (1965): 1427–32, https://doi.org/10.1126/science.148.3676.1427.

"nucleus"—of neurons located a few centimeters above our mouths that had a key part to play. It was later named the suprachiasmatic nucleus (SCN) for "above the chiasm"—the chiasm being the location at the base of the brain where the nerve fibers from both eyes partially cross. As I mentioned earlier, the SCN is the master circadian clock: the structure that coordinates the activities of all our cells and organs. In other words, the SCN is the orchestra conductor that makes sure each cell's twenty-four-hour rhythm starts on time.[25] In 1972, Irving Zucker's lab at UC Berkeley and Robert Moore's lab at the University of Chicago independently discovered that lesions to the SCN render organisms' circadian pattern arrhythmic and disorganized.[26] Naturally occurring SCN lesions caused by tumors eventually showed the SCN is necessary for sleep-wake, hunger, and other bodily rhythms to obey circadian cycles in humans as well.[27] Moreover, animal studies showed that transplanting even just a small number of healthy SCN neurons into an

[25] Ma and Morrison, "Neuroanatomy."

[26] F. K. Stephan and I. Zucker, "Circadian Rhythms in Drinking Behavior and Locomotor Activity of Rats Are Eliminated by Hypothalamic Lesions," Proceedings of the National Academy of Sciences U. S. A. 69, no. 6 (1972): 1583–86, https://doi.org/10.1073/pnas.69.6.1583; Robert Y. Moore and Victor B. Eichler, "Loss of a Circadian Adrenal Corticosterone Rhythm Following Suprachiasmatic Lesions in the Rat," Brain Research 42, no. 1 (July 1972): 201–06, https://doi.org/10.1016/0006-8993(72)90054-6.

[27] Robert M. Haugh and William Markesbery, "Hypothalamic Astrocytoma: Syndrome of Hyperphagia, Obesity, and Disturbances of Behavior and Endocrine and Autonomic Function," JAMA Neurology 40, no. 9 (September 1983): 560–63, https://doi.org/10.1001/archneur.1983.04050080060011.

arrhythmic brain restored a coherent twenty-four-hour rhythm, proving the SCN is both necessary and sufficient for endogenous circadian rhythms.[28]

If ever there was a brain structure whose function is beautifully organized to help ensure our survival as a species, it is the SCN. Even the SCN's seeming imperfection turns out to be perfect after all.

Stay with me here. For example, the SCN generates a period—meaning a cycle length—that is close to but not quite twenty-four hours. The SCN's period tends to be a bit longer than that, up to 24.5 hours.[29] By virtue of this, it can shift the start of the circadian rhythm forward or backward according to seasonal changes in day length or other survival-based needs, without making a mess and setting our cells on different schedules.[30] In other words, allowing a little room in its period duration affords us more flexibility

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[28] Ehab Tousson and Hilmar Meissl, "Suprachiasmatic Nuclei Grafts Restore the Circadian Rhythm in the Paraventricular Nucleus of the Hypothalamus," Journal of Neuroscience 24, no. 12 (2004): 2983–88, https://doi.org/10.1523/JNEUROSCI.5044-03.2004.

[29] Charles A. Czeisler et al., "Stability, Precision and Near-24-Hour Period of the Human Circadian Pacemaker," Science 284, no. 5423 (1999): 2177–81, https://doi.org/10.1126/science.284.5423.2177. Kryger et al., Kryger's Principles.

[30] Kryger et al., Kryger's Principles; Anita Jagota et al., "Morning and Evening Circadian Oscillations in the Suprachiasmatic Nucleus in Vitro," Nature Neuroscience 3 (2000): 372–76, https://doi.org/10.1038/73943; Horacio O. De La Iglesia et al., "Antiphase Oscillation of the Left and Right Suprachiasmatic Nuclei," Science 290, no. 5492 (2000): 799–801, https://doi.org/10.1126/science.290.5492.799.

and overall precision, not less, which is great because as we all know life is rarely the same from one day to the next. What's more, it is also what allows for phase shifting,[31] or the ability to change the start time of the daily cycle. If you're wondering why this doesn't have us waking up later and going to sleep later each day, it's because viewing morning light and evening light nudges your clock back and forth perfectly, to keep it regular. Later I'll explain how you can leverage that to overcome jet lag or any new schedule fast, but for now, all you need to know is that the not-exactly-twenty-four-hour rhythm is a feature, not a flaw.

As you probably already realize from the call to action that is Sleep Protocol 1, our SCN cares about whether it is light or dark out. And of course light-dark cycles are set by the spin of the earth. But how is it that our SCN—a tiny cluster of neurons embedded deep in the darkness of our brains—"knows" about the spin of the earth? Let's consider how to answer that question with insights from planetary science and cosmology.

We're all on this big sphere in space, spinning around at about one thousand miles per hour, causing us to face the sun, then away from the sun, then back toward the sun every

[31] Stephanie J. Crowley and Charmane I. Eastman, "Phase Advancing Human Circadian Rhythms with Morning Bright Light, Afternoon Melatonin, and Gradually Shifted Sleep: Can We Reduce Morning Bright Light Duration?," Sleep Medicine 16, no. 2 (February 2015): 288–97, https://doi.org/10.1016/j.sleep.2014.12.004.

twenty-four hours. Because the earth is round, the exact duration of sunlight we can experience each day depends, of course, on where we happen to be on earth. The variability is large, ranging from rather equal light-dark splits at the equator to nearly full days and/or nights of darkness or sunlight exposure at the poles. But no matter where we are on earth, inside each of our brains, our SCN steadily churns out that slightly longer than twenty-four-hour rhythm of coordinated cellular activity across our whole body.

Ultimately it is our SCN that tells us when to sleep and when to wake. We are a diurnal, not a nocturnal, species. We can stay up late and sleep all day if we choose, but even for night owls, all our bodily systems of hormones and neurotransmitters, our immune system, our levels of focus and quality of that focus, all of it, function much better when we live diurnally and wake just before or during the early daylight hours and go to sleep some hours after sundown.[32] (Yes, night owls would be wise to do this too—that is, if mental and physical health are your priority.)

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[32] Renske Lok et al., "Perils of the Nighttime: Impact of Behavioral Timing and Preference on Mental Health in 73,888 Community-Dwelling Adults," Psychiatry Research 337 (July 2024): 115956, https://doi.org/10.1016/j.psychres.2024.115956; Kristen L. Knutson and Malcolm von Schantz, "Associations Between Chronotype, Morbidity and Mortality in the UK Biobank Cohort," Chronobiology International 35, no. 8 (2018): 1045–53, https://doi.org/10.1080/07420528.2018.1454458; Raha West et al., "Sleep Duration, Chronotype, Health and Lifestyle Factors Affect Cognition: A UK Biobank Cross-Sectional Study," BMJ Public Health 2, no. 1 (July 2024): e001000, https://doi.org/10.1136/bmjph-2024-001000.

Keep in mind these are independent phenomena; the earth's spin and travel around the sun is not what causes our SCN to have a twenty-four-hour rhythm, nor does it tell us when to sleep and when to wake, but they do need to be linked. So how do we link—or what is called entrain—our SCN's rhythm with the light-dark cycle generated by the earth's spin and travel around the sun? Remember, our SCN resides deep in our brain. It lives in the darkness of our skulls. To synchronize our circadian clock with the day-night cycle cast on us by our location on earth, we need a way for the presence or absence of sunlight to be conveyed to the SCN. And the best way to do that is to call upon the one part of our brain that has direct access to sunlight: our eyes.

Your Eyes

Wait, did I say that your eyes are part of your brain? Yes, that is correct. Lining the inside-back part of each of your eyeballs is a thin sheet of brain tissue called the neural retina. Your neural retinas were squeezed out and away from the rest of your brain during the first trimester of your life, while you were still in your mother's womb, and then grew wirelike extensions into the brain proper, forming two optic nerves. Your optic nerves convey electrical signals about the presence or absence of sunlight, objects, faces, moving

vehicles, and on and on to different brain areas, including—you guessed it—the SCN.[33]

The connections between your eyes and your brain are truly a spectacular product of biological engineering. First, they solve the problem of how to inform image-forming regions of your brain about "what is out there" in the world, so you can see faces, moving objects, and colors. They also solve the problem of letting your SCN know "when it is out there" in the world, by the presence of sunlight. The neurons that convey the presence of sunlight to the SCN are not of the image-forming kind; they don't respond to the shape of the sun, but instead to its brightness and to the specific wavelengths of sunlight that reach your eyes when the sun is low in the sky. We refer to these specific wavelengths as low solar angle sunlight.[34]

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[33] Mohit Gupta et al., "Neuroanatomy, Visual Pathway," StatPearls, last updated March 23, 2026, https://www.ncbi.nlm.nih.gov/books/NBK553189/; Onkar S. Dhande and Andrew D. Huberman, "Retinal Ganglion Cell Maps in the Brain: Implications for Visual Processing," Current Opinion in Neurobiology 24, no. 1 (February 2014): 113–42, http://doi.org/10.1016/j.conb.2013.08.006; Alyssa M. Rivera and Andrew D. Huberman, "Neuroscience: A Chromatic Retinal Circuit Encodes Sunrise and Sunset for the Brain," Current Biology 30, no. 7 (April 2020): R316–18, http://doi.org/10.1016/j.cub.2020.02.090.

[34] Ignacio Provencio et al., "Melanopsin: An Opsin in Melanophores, Brain, and Eye," Proceedings of the National Academy of Sciences 95, no. 1 (1998): 340–45, https://doi.org/10.1073/pnas.95.1.340; Rivera and Huberman, "Neuroscience"; Sara S. Patterson et al., "S-Cone Circuits in the Primate Retina for Non-Image-Forming Vision," Seminars in Cell & Developmental Biology 126 (June 2022): 66–70, https://doi.org/10.1016/j.semcdb.2021.05.004.

Have you ever noticed that low solar angle sunlight appears more yellow than midday sun? Or even orange? Check out a sunrise or sunset and compare the color of sunlight to the color of sunlight at noon. Of course, sunlight doesn't change according to where it is in the sky, but what portions of sunlight are visible to us does.

Sunlight is always full spectrum: It contains all wavelengths of light, red-orange-yellow-green-blue-indigo-violet, or ROYGBIV, just as they taught us in elementary school. The R(ed), O(range), and Y(ellow) are longer wavelengths of light, the G(reen) is medium wavelength, and the B(lue), I(ndigo), and V(iolet) are shorter wavelengths. And of course, the sun emits UV and far-red, or infrared wavelengths, too, but we can't see those (although many other animals and insects can). When the sun is high in the sky, it appears white, because all the wavelengths of light arrive at our eyes at roughly the same intensities. When the sun is low in the sky, we see the sun as more yellow, orange, or red, because molecules in the atmosphere preferentially redirect the shorter blue, indigo, and violet wavelengths of light—a process called Rayleigh scattering. (We'll discuss the properties of light in depth in chapter 5.)

All mammals, us humans included, have evolved a special system of light detection that responds specifically to the presence and contrast of short and long wavelengths of

bright light, which is exactly what reaches your eyes if you view sunlight early or late in the day, while the sun is fairly low in the sky. In other words, we all have a sunrise and sunset detecting system in our eyes that communicates the time of day to our SCN.[35]

As I mentioned earlier, this timekeeping eye-to-brain circuit is also a fairly recent discovery. And by recent, I mean recent: In the late 1990s, there still wasn't general agreement among circadian biologists whether light—and not some other cue like social interactions—was the dominant force entraining our SCN. Then, in 1998, neuroscientist Ignacio "Iggy" Provencio made a key discovery. Iggy found a molecule in the skin of an African frog species that caused its skin to darken when exposed to sunlight was also present in the eyes of mice.[36] The molecule was an opsin, which is a kind of protein that absorbs light, and when it does, it changes its conformation to set off a signaling cascade.[37] Since the same opsin was known to cause darkening of dermal melanophores in frog skin, Provencio called the

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[35] Ignacio Provencio et al., "A Novel Human Opsin in the Inner Retina," Journal of Neuroscience 20, no. 2 (January 2000): 600–605, https://doi.org/10.1523/JNEUROSCI.20-02-00600.2000; Ignacio Provencio et al., "Photoreceptive Net in the Mammalian Retina," Nature 415, no. 493 (January 2002): 493, https://doi.org/10.1038/415493a; Rivera and Huberman, "Neuroscience."

[36] Provencio et al., "Melanopsin"; Provencio et al., "A Novel Human Opsin."

[37] Akihisa Terakita and Takashi Nagata, "Functional Properties of Opsins and Their Contribution to Light-Sensing Physiology," Zoological Science 31, no. 10 (2014): 653–59, https://doi.org/10.2108/zs140094.

mouse retinal opsin melanopsin. Other labs confirmed melanopsin-containing cells are abundant in the human retina, too, mostly in the lower portion that views the upper visual field, which, you might have guessed, corresponds to the sky.[38]

So we have a frog-skin-like set of cells in our eyes? Not quite. To explain what makes melanopsin-containing retinal cells so special, first a bit about photoreceptors. Photoreceptors are neurons that convert light into electrical signals the rest of the retina and brain can interpret. But the retinal cells that contain melanopsin are not like other photoreceptors. The melanopsin-containing retinal neurons Iggy discovered turned out to be both photoreceptive and retinal ganglion cells—the neurons that connect the eyes to the brain and the very cells whose wirelike extensions, or axons, make up the optic nerves. Soon after, a group of neuroscientists, David Berson and Felice Dunn at Brown University and Samer Hattar and King Wai-Yau at Johns Hopkins University, along with several of their students and postdocs, confirmed that melanopsin RGCs send their axons to the SCN and that they fire electric signals, thereby

[38] Provencio et al., “A Novel Human Opsin”; Ignacio Provencio et al., “Photoreceptive Net in the Mammalian Retina. This Mesh of Cells May Explain How Some Blind Mice Can Still Tell Day from Night,” Nature 415, no. 6871 (January 2002): 493, https://doi.org/10.1038/415493a.

releasing neurotransmitters, when light is shone on them.[39] Remarkably, they did this when they were isolated from all other retinal cells. As long as the light shone was very bright and long lasting, they fired electrical signals. These cells were therefore named intrinsically photosensitive RGCs (ipRGCs) to differentiate them from the other 99 percent of RGCs that can't respond directly to light on their own. Some years later, work from the Neitz Color Vision Lab at the University of Washington in Seattle and by others showed that the optimal stimulus for ipRGCs isn't just bright light.[40] They discovered that the best way to activate the ipRGCs that connect to the SCN is with sustained very bright light, similar to what we see as blue or green light, contrasted with medium wavelengths of light, or what we typically see as yellow or orange—in other words, features prominent in low solar angle sunlight. The color contrasts in sunrises and sunsets tell your brain about what time it is in the twenty-four-hour cycle. That collection of work, from the discovery of melanopsin to ipRGCs, and how they work,

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[39] David M. Berson et al., "Phototransduction by Retinal Ganglion Cells That Set the Circadian Clock," Science 295, no. 5557 (February 2002): 1070–73, https://doi.org/10.1126/science.1067262; S. Hattar et al., "Melanopsin-Containing Retinal Ganglion Cells: Architecture, Projections, and Intrinsic Photosensitivity," Science 295, no. 5557 (February 2002): 1065–70, https://doi.org/10.1126/science.1069609.

[40] Sara S. Patterson et al., "S-Cone Circuits in the Primate Retina for Non-Image-Forming Vision," Seminars in Cell & Developmental Biology 126 (June 2022): 66–70, https://doi.org/10.1016/j.semcdb.2021.05.004; Rivera and Huberman, "Neuroscience."

was revelatory. It meant that there was a specific neural circuit extending from the eye to the brain dedicated to entraining the SCN and our circadian rhythms. That circuit is tuned to—neuroscience lingo for “responds to”—the presence of low solar angle sunlight. The SCN then handles the job of body- and brain-wide synchronization of circadian rhythmicity. This sunlight-driven circadian entrainment circuit requires no learning or practice to activate. It is present from birth and functions the first time and every time to control, among other things, your sleep-wake cycle.

The Sun

Which brings us to the final piece of the puzzle—the part that is not hard-wired but that you control with your behavior—and that is the call to action of this protocol: exposing your eyes to morning sunlight (or in its absence, to the next best alternative, which I describe below). Doing so activates the ipRGC-to-SCN circadian entrainment circuit and initiates dozens of beneficial processes that get and keep your brain and body going throughout the day. And toward the evening, it puts you in the best possible position to get a great night’s sleep.

In fact, we have copious evidence that this is foundational to both our daytime well-being and nighttime sleep. As one small example, a group of architects and neurologists came together in 2014 to investigate how windowless offices

affected the people who worked in them.[41] They asked twenty-seven office workers who had either no windows, or work areas situated far from windows, and twenty-two people working in offices with windows, about their well-being and sleep quality. The questionnaire scores were decidedly poorer among the windowless workers. They reported more sleep disturbances and an overall unsatisfactory quality of sleep.

In the early 2000s, a group of chronobiologists (scientists who study biological timing) at the Institute for Medical Psychology, in Munich, Germany, surveyed five hundred people about their sleep times, light exposure, and chronotype—as in, morning person, night owl, et cetera (you'll find out how to determine yours in Sleep Protocol 4).[42] They found that the amount of light exposure during the day correlated with what time people went to bed: More exposure during the day meant earlier bedtimes. This link can be partly explained by external factors, like employment: People whose livelihoods depend on being outdoors are more likely to be early-to-bed types, especially if their work requires them to rise early or is physically demanding. But

[41] Mohamed Boubekri et al., "Impact of Windows and Daylight Exposure on Overall Health and Sleep Quality of Office Workers: A Case-Control Pilot Study," Journal of Clinical Sleep Medicine 10, no. 6 (2014): 603–11, https://doi.org/10.5664/jcsm.3780.

[42] Till Roenneberg et al., "Life Between Clocks: Daily Temporal Patterns of Human Chronotypes," Journal of Biological Rhythms 18, no. 1 (2003): 80–90, https://doi.org/10.1177/0748730402239679.

that doesn't explain the connection entirely. It turns out that it's also due, in part, to the fact that exposure to daylight has a regulating effect on our physiology—causing elevated levels of alertness by a host of different pathways, including hormonal and neural pathways. As one review put it, "In sum, research seems to agree that daylight (at high intensities) is beneficial for sleep."[43]

There are also numerous studies showing that bright-light exposure early in the day—at the intensities that one would get from being outdoors in the morning—is beneficial for everything from mood to hormone health, daytime energy levels, and on and on.[44] But before you say, "Okay, got it!," there's just a bit more to understand about the underlying mechanisms in this protocol, and that's the relationship between sleep and wakefulness. And it will help you get the most out of this protocol, regardless of where you live and the time of year.

Our daytime levels of alertness reflect the quality and duration of our nighttime sleep. That's a giant "duh." We all know that if we sleep poorly, we drag both mentally and

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[43] Christine Blume et al., "Effects of Light on Human Circadian Rhythms, Sleep and Mood," Somnologie 23, no. 3 (2019): 147–56, https://doi.org/10.1007/s11818-019-00215-x.

[44] Blume et al., "Effects of Light"; Manuel Spitschan and Johannes Zauner, "Light Exposure as a Modifiable Determinant of Mental Health," Current Psychiatry Reports 28, no. 1 (April 2026): 24, https://doi.org/10.1007/s11920-026-01671-7.

physically the next day. We also all know that if we sleep great, we're focused and motivated to do all kinds of things that require energy… and that we have the energy to do them.

But what most people do not know is that the relationship between sleep and wakefulness works in the other direction too. When we are alert during the morning and daytime, we are more likely to expend mental and physical energy that in turn helps us sleep at night. Even those of us who sleep well typically find that increasing our daytime physical activity—especially in the hours between sunrise and 2:00 p.m.—leads to significantly better sleep,[45] which in turn leads to more energy the next day, which improves sleep, and so on. Again, you're probably thinking, "This is common knowledge. Do we really need science to explain this?"

Here is the lesser-known fact: Morning sunlight viewing is the best way to increase our morning and daytime energy levels.[46] Not food (although fine to eat early in the day), not caffeine (also fine). It's morning sunlight.

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[45] Majd A. Alnawwar et al., "The Effect of Physical Activity on Sleep Quality and Sleep Disorder: A Systematic Review," Cureus 15, no. 8 (2023): e43595, https://doi.org/10.7759/cureus.43595; Masahiro Banno et al., "Exercise Can Improve Sleep Quality: A Systematic Review and Meta-Analysis," PeerJ 6 (July 2018): e5172, https://doi.org/10.7717/peerj.5172.

[46] Christian Cajochen, "Alerting Effects of Light," Sleep Medicine Reviews 11, no. 6 (2007): 453–64, https://doi.org/10.1016/j.smrv.2007.07.009.

Morning sunlight increases your energy levels through at least three different mechanisms. First, it suppresses any residual melatonin you may have circulating in your brain (Sleep Protocol 3 includes an extensive look at melatonin, a hormone that helps promote sleepiness).[47] As discussed above, when the ipRGCs are activated by sunlight viewing, they send electrical signals to the SCN. The SCN, in turn, stimulates a collection of neurons in the nearby paraventricular nucleus, or PVN. The PVN then sets off a cascade of signals that go through the spinal cord and back up to the pineal gland in the brain, shutting off melatonin production.[48] Voila! You've suppressed the primary hormone that promotes sleepiness by viewing morning sunlight.

Second, bright light reaching your eyes helps suppress the action of adenosine, a chemical that also leads to feelings of sleepiness.[49] Sunlight does this indirectly via its effects on other systems in the brain and body, but nonetheless, it has a potent adenosine-reducing effect. (We'll take a deep dive

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[47] Vitaterna et al., "Overview of Circadian Rhythms"; Joshua J. Gooley et al., "Exposure to Room Light Before Bedtime Suppresses Melatonin Onset and Shortens Melatonin Duration in Humans," Journal of Clinical Endocrinology & Metabolism 96, no. 3 (2011): E463–72, https://doi.org/10.1210/jc.2010-2098.

[48] Shan Jiang et al., "Role of the Paraventricular Nucleus of the Hypothalamus in Sleep–Wake Regulation," Brain Science Advances 8, no. 3 (2022): 197–209, https://doi.org/10.26599/BSA.2022.9050017.

[49] Christophe Ribelayga and Stuart C. Mangel, "A Circadian Clock and Light/Dark Adaptation Differentially Regulate Adenosine in the Mammalian Retina," Journal of Neuroscience 25, no. 1 (2005): 215–22,

into adenosine in Sleep Protocol 7). After a full night's sleep, adenosine levels tend to be low, but if you happen to not get enough sleep (as so many of us do), morning light helps reduce any residual adenosine. That both increases your alertness and prevents the residual adenosine from delaying your circadian clock, which would otherwise bias you toward waking up later the next day, then going to sleep later, then waking up even later the following day, and so on (what chronobiologists call a phase delay).[50] Such circadian "drift" brought on by mild sleep deprivation is very common. It is best remedied by viewing morning sunlight to reset your circadian rhythm.[51]

Third, we need to talk about cortisol. Because it's often discussed in the context of chronic stress, which is bad, most people think cortisol is "bad" too. But it's not bad. Cortisol is essential to health and your levels of energy—you just

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https://doi.org/10.1523/jneurosci.3138-04.2005; Tarja Porkka-Heiskanen et al., "Adenosine: A Mediator of the Sleep-Inducing Effects of Prolonged Wakefulness," Science 276, no. 5316 (May 1997): 1265–68, https://doi.org/10.1126/science.276.5316.1265; Radhika Basheer et al., "Adenosine and Sleep-Wake Regulation," Progress in Neurobiology 73, no. 6 (August 2004): 379–96, https://doi.org/10.1016/j.pneurobio.2004.06.004.

[50] Aarti Jagannath et al., "Adenosine Integrates Light and Sleep Signalling for the Regulation of Circadian Timing in Mice," Nature Communications 12, article no. 2113 (2021), https://doi.org/10.1038/s41467-021-22179-z; Carolin Franziska Reichert et al., "Adenosine, Caffeine, and Sleep–Wake Regulation: State of the Science and Perspectives," Journal of Sleep Research 31, no. 4 (2022): e13579, https://doi.org/10.1111/jsr.13597.

[51] Kenneth P. Wright Jr. et al., "Entrainment of the Human Circadian Clock to the Natural Light-Dark Cycle," Current Biology 23, no. 16 (2013): 1554–58, https://doi.org/10.1016/j.cub.2013.06.039.

don’t want your cortisol levels to be elevated at the wrong times.[52]

Your SCN is located in your hypothalamus (H), which is part of a biological circuit that also includes your pituitary gland (P), a small, stalk-like protrusion at the base of your brain that releases hormones into your bloodstream. The circuit also includes your adrenal glands (A), which reside atop your kidneys. This circuit is known as the HPA axis.[53] The HPA axis is one of the main systems that responds to stress (which is why you’ll hear more about it in chapter 3). Here’s how it works: It starts with signals from the hypothalamus, which then stimulate the release of specific hormones from the pituitary, which in turn stimulate adrenaline and cortisol release from your adrenals.

The HPA axis is also involved in the sleep-wake cycle. When your ipRGCs are activated by low solar angle morning sunlight and your SCN receives the ensuing signal, it alerts your adrenal glands to release adrenaline and cortisol via the

[52] Jasleen Kaur et al., “Physiology, Cortisol,” StatPearls, last updated December 1, 2025, https://www.ncbi.nlm.nih.gov/books/NBK538239/.

[53] Catherine J. Dunlavey, “Introduction to the Hypothalamic-Pituitary-Adrenal Axis: Healthy and Dysregulated Stress Responses, Developmental Stress and Neurodegeneration,” Journal of Undergraduate Neuroscience Education 16, no. 2 (2018): R59–60, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6057754/.

HPA axis.[54] Cortisol and adrenaline dramatically increase levels of alertness, the latter likely by activating receptors on your vagus nerve,[55] a large portion of which extends from the organs in your body up into your brain and stimulates areas there such as the locus coeruleus to make norepinephrine, which has the net effect of "waking" you up.[56] Cortisol does a lot of things. It also decreases inflammation and regulates our immune systems to protect against viral and bacterial infections.[57] And it helps regulate our energy levels. Think about cortisol as a wakefulness-promoting hormone, not a stress hormone, if you want to understand what the cortisol in your body is doing most of the time. It's also useful to consider how cortisol ebbs and flows across time due to signals from your circadian clock. By measuring cortisol levels every twenty

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[54] Kaur et al., "Physiology, Cortisol"; Dunlavey, "Introduction to the Hypothalamic-Pituitary-Adrenal Axis"; Sarah Leach and Kazuhiro Suzuki, "Adrenergic Signaling in Circadian Control of Immunity," Frontiers in Immunology 11 (June 2020): 1235, https://doi.org/10.3389/fimmu.2020.01235.

[55] Christoph Szeska et al., "A Vagal Route to Memory: Evidence from Invasive and Non-Invasive Electrical Vagus Nerve Stimulation Studies and Areas for Future Clinical Application," Frontiers in Human Neuroscience 19 (2025): 1595737, https://doi.org/10.3389/fnhum.2025.1595737; T. Miyashita and C. L. Williams, "Epinephrine Administration Increases Neural Impulses Propagates along the Vagus Nerve: Role of Peripheral Beta-adrenergic Receptors," Neurobiology of Learning and Memory 85, no. 2 (March 2006): 116–124, https://doi.org/10.1016/j.nlm.2005.08.013.

[56] Jennifer A. Ross and Elisabeth J. Van Bockstaele, "The Locus Coeruleus-Norepinephrine System in Stress and Arousal: Unraveling Historical, Current, and Future Perspectives," Frontiers in Psychiatry 11 (January 2021): 601519, https://doi.org/10.3389/fpsyt.2020.601519.

[57] Kaur et al., "Physiology, Cortisol."

minutes in human subjects, researchers found cortisol reaches its apex at about sixty minutes or so after you wake up and that it is lowest about four to five hours after you fall asleep.[58] Then it begins to rise sharply across the final two to three hours of your sleep until—boom! You wake up due in large part to what is called the cortisol awakening response (CAR). Yes, reaching a threshold level of cortisol is a major factor in what ultimately rouses you from sleep. As mentioned above, your cortisol continues to rise and peaks about sixty minutes after waking, then slowly drops down as you head into the early and late afternoon. Cortisol levels generally stay low and continue to drop in the evening and at nighttime (assuming it doesn't spike from stress or intense exercise—more on that below), and then it falls to its nadir in those first four to five hours of sleep before starting to rise again toward morning. We return to this circadian cortisol curve in chapter 3, and the role it plays in promoting health in women and men of all ages. For now, the key thing to know is that for cortisol to support our health, it needs to peak in the first hour or so after waking; it then needs to drop

[58] Sharon Chan and Miguel Debono, "Replication of Cortisol Circadian Rhythm: New Advances in Hydrocortisone Replacement Therapy," Therapeutic Advances in Endocrinology and Metabolism 1, no. 3 (2010): 129–38, https://doi.org/10.1177/2042018810380214; Elliot D. Weitzman et al., "Twenty-Four Hour Pattern of the Episodic Secretion of Cortisol in Normal Subjects," Journal of Clinical Endocrinology and Metabolism 33, no. 1 (July 1971): 14–22, https://doi.org/10.1210/jcem-33-1-14.

precipitously in the evening and night and continue falling to reach its nadir in those initial four to five hours of sleep.[59]

Viewing morning sunlight doesn't directly cause the morning peak in cortisol, but it can dramatically amplify its rise by as much as 50 percent.[60] This also helps ensure that this peak comes early in the day and isn't shifted to the afternoon. Late-day elevations in cortisol have negative consequences over time such as sleeplessness, anxiety, and metabolic disease.[61]

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[59] Weitzman et al., "Twenty-Four Hour Pattern"; Katja Petrowski et al., "The Effects of Post-Awakening Light Exposure on the Cortisol Awakening Response in Healthy Male Individuals," Psychoneuroendocrinology 108 (2019): 28–34, https://doi.org/10.1016/j.psyneuen.2019.05.016; E. Van Cauter et al., "Effects of Gender and Age on the Levels of Circadian Rhythmicity of Plasma Cortisol," Journal of Clinical Endocrinology and Metabolism 81, no. 7 (July 1996): 2468–73, https://doi.org/10.1210/jcem.81.7.8675562; Ines Wilhelm et al., "Is the Cortisol Awakening Rise a Response to Awakening?," Psychoneuroendocrinology 32, no. 4 (May 2007): 358–66, https://doi.org/10.1016/j.psyneuen.2007.01.008; Timothy M. Brown et al., "Recommendations for Daytime, Evening, and Nighttime Indoor Light Exposure to Best Support Physiology, Sleep, and Wakefulness in Healthy Adults," PLOS Biology 20, no. 3 (2022): e3001571, https://doi.org/10.1371/journal.pbio.3001571.

[60] Rachel Leproult et al., "Transition from Dim to Bright Light in the Morning Induces an Immediate Elevation of Cortisol Levels," Journal of Clinical Endocrinology & Metabolism 86, no. 1 (2001): 151–57, https://doi.org/10.1210/jcem.86.1.7102; Weitzman, "Twenty-Four Hour Pattern"; Katja Petrowski et al., "The Effects of Post-Awakening Light Exposure on the Cortisol Awakening Response in Healthy Male Individuals," Psychoneuroendocrinology 108 (October 2019): 28–34, https://doi.org/10.1016/j.psyneuen.2019.05.016.

[61] Nora A. O'Byrne et al., "Sleep and Circadian Regulation of Cortisol: A Short Review," Current Opinion in Endocrine and Metabolic Research 18 (June 2021): 178–86, https://doi.org/10.1016/j.coemr.2021.03.011; Petrowski et al., "Effects of Post-Awakening Light"; Van Cauter et al., "Effects of Gender and

A quick note: A morning workout, morning caffeine, and breakfast (provided it isn't a huge and/or carbohydrate-laden one) will all promote wakefulness, too, and thereby contribute to a well-aligned circadian rhythm. But morning sunlight is the most potent lever for increasing morning cortisol and thus energy during your day, and for improving your sleep at night. Don't underestimate its power.

Keep your circadian rhythm attuned to the sun, and all of these mechanisms will work as they should.

COMMON QUESTIONS

It's often dark when I wake up. What do I do if I have no access to morning sunlight but I want to start my day

If you wake up before sunrise, and you want to start your day, I suggest turning on as many bright indoor lights in your home—or hotel room, or wherever you may be—as possible. Sunlight cannot be replaced by standard indoor lighting, but turning on indoor lighting first thing won't hurt. Then get outside to view morning sunlight once the sun is up.

If you wake before sunrise every day, or if you live in a part of the world where it is very dark in winter, you may want to invest in a 10,000 lux light to help ensure you get enough light in the morning (lux is a measure of brightness). Most

Age"; Wilhelm, "Is the Cortisol Awakening"; Brown, "Recommendations for Daytime;" Kaur, "Physiology, Cortisol."

cost about $100, and some are portable for travel. I often wake up before sunrise, and because the ipRGCs can be activated by very bright artificial lights if those bright lights include sufficient short wavelengths, I keep a 10,000 lux lamp in my kitchen and bathroom.[62] It lacks long wavelength light, so I also have a red incandescent light next to it and I turn that on too. I use the red incandescent light from SaunaSpace (no affiliation). I turn them both on first thing in the morning and then I get outside to view morning sunlight once the sun is up. I am well aware these lights are not a replacement for sunlight, but they help bridge the gap until the sun is up. Once the sun is up, you can be sure I go outside to view it!

And I will sometimes keep the 10,000 lux lamp and a red incandescent light on and facing me for an hour or so once I'm back working at my desk or getting ready for the day. Chapter 5 further explains the logic and benefits of this extended morning light exposure. There I also mention options that can be of real help for sleep as well as other aspects of health too, such as metabolic health.

But remember, morning sunlight is the best stimulus for your ipRGCs, SCN, and circadian rhythm, and it costs nothing!

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[62] Berson, "Phototransduction"; Xiwu Zhao et al., "Prolonged Inner Retinal Photoreception Depends on the Visual Retinoid Cycle," Journal of Neuroscience 36, no. 15 (April 2016): 4209–17, https://doi.org/10.1523/JNEUROSCI.2629-14.2016.

Figure out how to view it each morning. Again, you don't need to see it cross the horizon as it rises. If you do, great, but if you wake up after sunrise, that's fine. Do your best to view sunlight within the first thirty to sixty minutes after waking up. Bring your coffee or tea outside with you. Bring your spouse or kids if they're awake. Bring your dog if you have one. Remember, dogs, cats, cows, and other animals have ipRGCs and an SCN, too, and they benefit from morning sunlight in many of the same ways and via the same mechanisms as we do.[63]

What if it's cloudy outside

To activate your ipRGCs fastest, go outside and face east toward the sun on a nice, clear morning with no clouds in the sky. Obviously, this isn't possible for everyone every day.

Again, fortunately, sparse clouds do not interfere with the interaction between sunlight and our circadian clock. However, if the sky is densely overcast and gray, a 10,000 lux light can be a real help, but still aim for getting more sunlight in your eyes to compensate for the lower intensity of sunlight and reduced yellow-blue contrast exposure you're likely to get on very overcast days. In fact, it's

[63] Vincent M. Cassone, "Avian Circadian Organization: A Chorus of Clocks," Frontiers in Neuroendocrinology 35, no. 1 (January 2014): 76–88, https://doi.org/10.1016/j.yfrne.2013.10.002; Deng Pan et al., "Melanopsin-Mediated Optical Entrainment Regulates Circadian Rhythms in Vertebrates," Communications Biology 6, no. 1054 (October 2023), https://doi.org/10.1038/s42003-023-05432-7.

especially important that you get sunlight in your eyes on overcast days. This is for three reasons: (1) dense clouds reduce the number of photons reaching your eyes in a given amount of time; (2) they filter out much of the longer (yellow, orange, and red) wavelengths of light, and (3) you are likely to be indoors more on cloudy days, which can cause the SCN to fall out of sync with the solar cycle, which leads to later nights, more artificial light exposure at night, more circadian disruption, and so on.

Aim to extend the sunny, clear day recommendation of a minimum of five to ten minutes, to at least fifteen minutes on cloudy days, and thirty minutes (minimum) on densely overcast days. That may seem like a long time, but remember, you don’t need to stand in one place gazing at the sky. Take a walk. Make a call. Listen to a book or podcast or talk to a friend, partner, or other family member. And even if you can’t be outside for the full time recommended above, some sunlight hitting your eyes in the morning is better than none. If you miss a day completely due to travel or sickness or other reasons, don’t despair—your energy levels that day may be lower because you didn’t get the cortisol spike bright morning light causes, but your circadian rhythm is the reflection of your previous three to four days of sunlight viewing activity.

Do I need to worry about sunlight damaging my eyes

Excessive UV light exposure can cause cataracts and, as we have all heard, skin damage, accelerated skin aging, and some forms of skin cancer, topics I cover in more detail later in the book. But viewing morning sunlight is much safer than viewing overhead midday sunlight—because that Rayleigh scattering of short wavelength light when the sun is low in the sky includes UV and thus reduces your UV exposure significantly. It is when the sun is overhead that the UV index is high. UV light isn't bad, per se, but you don't want too much of it reaching your eyes or skin. Chapter 5 goes into the benefits and drawbacks of UV.

NEVER, EVER force yourself to stare at the sun—or any light source for that matter, at any time of day. Your natural blink and "look away" reflex is a built-in circuit (yes, there is a dedicated circuit for this) to protect your eyes. People with macular degeneration or other eye conditions ought to be especially careful about viewing midday sunlight, and if very sensitive to sunlight, they may want to consult with their ophthalmologist before doing any deliberate sunlight viewing.

What about eyeglasses and contact lenses

Eyeglasses and contact lenses are both perfectly fine to wear while viewing morning sunlight. Neither one blocks light

from reaching your eyes, even if your eyeglasses or contacts have UV filters.

Sunglasses and transition lenses, however, should be avoided while viewing morning sunlight because they block out a lot of the relevant wavelengths that trigger the valuable mechanisms covered in this protocol. Moreover, they reduce the number of photons reaching your eyes. Of course it's fine to wear sunglasses as needed for comfort and safety at other times of day.

What about viewing sunlight through a window or windshield? Is that the same

People relentlessly ask me this and my response is equally persistent: No, it's not the same. It's far less effective, though better than nothing. But if there is any way to get outside, do. Windows and windshields, especially nowadays, filter the intensities and wavelengths you want for this protocol to work. "How come viewing morning sunlight through eyeglasses or contacts is fine but not windows or windshields?" you may wonder. Because windows do something called spectral filtering: They reduce the amount of light—especially short wavelength light—that would otherwise reach your eyes. Corrective lenses don't; they let far more light through. That's their job. Well, that and to correct any optical errors in the structure of your eyeballs so

you can see things clearly. Windows and windshields do neither.



SLEEP PROTOCOL

View Sunlight in the Late Afternoon

WHAT TO DO

- Get sunlight in your eyes in the late afternoon or early evening. Ideally, go outside and spend at least ten to fifteen minutes there before the sun sets. If at all possible, face or walk west toward the sun. You do not need to see the sun drop below the horizon. That's a wonderful thing if you can—who doesn't love a sunset? Alas, most of us don't live or work in the right place to do so. Luckily, just going outside and getting some ambient light in your eyes in the late afternoon or early evening is beneficial. Seeing the sunset is a bonus.
- A few things to keep in mind. As with the previous protocol, it is best to face the sun directly, not through a window or windshield, and without wearing sunglasses or a brimmed hat. Perhaps you can get an unobscured view by going up to a rooftop or by walking a few blocks from where you work or live. From a health perspective, that is well worth it. Again it's especially important to do this protocol on overcast days.

HOW IT WORKS

We've already unpacked the basic physiological processes linking your circadian clock, your eyes, and specific qualities of light that drive the sleep-wake cycle. Afternoon sunlight exposure taps into some of the same basic underlying cells, circuits, and mechanisms, but has both distinct and added benefits, including partially protecting you from some of the negative health effects of viewing artificial lights late at night.

Here again we start with the SCN. Earlier I described the SCN as a "cluster of neurons," but that doesn't really do it justice. A key feature of the SCN is that it contains two oscillators, each following a roughly twenty-four-hour rhythm of electrical activity, but offset from one another.[64] It's as if they are singing in a round, with peaks happening at different yet specific moments within the twenty-four-hour cycle. One peak roughly corresponds to morning and the other to evening, and each can be entrained to sunrise and sunset. And they communicate with one another to be sure neither the morning nor the evening oscillator runs too fast

[64] Lily Yan et al., "Two Antiphase Oscillations Occur in Each Suprachiasmatic Nucleus of Behaviorally Split Hamsters," Journal of Neuroscience 25, no. 39 (2005): 9017–26, https://doi.org/10.1523/JNEUROSCI.2538-05.2005; Jagota et al., "Morning and Evening"; De La Iglesia et al., "Antiphase Oscillation"; Colin S. Pittendrigh and Serge Daan, "A Functional Analysis of Circadian Pacemakers in Nocturnal Rodents. V. Pacemaker Structure: A Clock for All Seasons," Journal of Comparative Physiology A 103, no. 3 (1976): 333–55, https://doi.org/10.1007/BF01417856.

or too slow. That helps generate the SCN's coherent daily rhythm that all other cells in the body follow. It also allows us to shift our clock forward or back as the lengths of days and nights change across the seasons.

Here's the best way to think about it: These two oscillators are like bookends buttressing a row of vertically stacked books, making the books less likely to fall over. This is the right analogy given that viewing morning sunlight shifts the phase of our circadian clock forward, making us want to go to sleep earlier and wake up earlier. Viewing late afternoon or early evening sunlight delays the phase of our clock, which, because it shifts the timing of the release of hormones like melatonin, makes us want to go to bed later and wake up later. So viewing both morning and afternoon sunlight thus helps keep us on a regular sleep-wake schedule, because the phase shift effects of morning and evening sunlight offset each other.[65]

Another great reason to view sunlight in the late afternoon or early evening is that it helps buffer us against the negative

[65] Ashna Ramkisoensing et al., "Enhanced Phase Resetting in the Synchronized Suprachiasmatic Nucleus Network," Journal of Biological Rhythms 29, no. 1 (2014): 4–15, https://doi.org/10.1177/0748730413516750; Andries Kalsbeek et al., "Circadian Disruption and SCN Control of Energy Metabolism," FEBS Letters 585, no. 10 (2011): 1412–26, https://doi.org/10.1016/j.febslet.2011.03.021; Pittendrigh and Daan, "Functional Analysis."

effects of viewing bright artificial lights at night.[66] In the morning, we need to see a lot of bright light to train our SCN with the solar cycle, and to elevate our levels of alertness. Unless it's a 10,000 lux lamp, most artificial lights and certainly our phone or computer screens just can't do the job at that time. After sundown, however, it takes extremely little bright light to quash our melatonin levels, and shift our circadian clock. And even if we can still fall asleep at night, that sleep is not going to be as high quality.

In fact, studies show that as little as fifteen seconds of bright light can significantly shift our circadian clock.[67] And while most screens are not as bright as the lights used in these studies, our ipRGCs and SCN work to sum photons—that is, add up all the light we see from various sources. And let's face it, most people are looking at their phone and other screens for a lot longer than fifteen seconds after sundown.

To recap: In the morning, you need more bright light in your eyes than a screen or conventional home lighting can provide, and at night you want far less light reaching your

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[66] Jamie M. Zeitzer et al., "Effectiveness of Evening Phototherapy for Insomnia Is Reduced by Bright Daytime Light Exposure," Sleep Medicine 12, no. 8 (2011): 805–7, https://doi.org/10.1016/j.sleep.2011.02.005; Marije te Kulve et al., "Early Evening Light Mitigates Sleep Compromising Physiological and Alerting Responses to Subsequent Late Evening Light," Scientific Reports 9, no. 16064 (2019), https://doi.org/10.1038/s41598-019-52352-w.

[67] Shadab A. Rahman et al., "Circadian Phase Resetting by a Single Short-Duration Light Exposure," JCI Insight 2, no. 7 (2017): e89494, https://doi.org/10.1172/jci.insight.89494.

eyes than typically comes from a phone or computer screen or conventional home lighting. It all has to do with how our eyes are more sensitive to detecting light at different times of day or night. If that seems inconvenient, you're right; nowadays it is! But the circuits underlying retinal and circadian sensitivity across the twenty-four-hour cycle evolved long before humans had artificial lighting, much less the bright, short-wavelength-enriched LED screens and LED bulbs that we use today.

I, like most people, often wake up once in the middle of the night to use the bathroom. If I had things my way, our circadian system would be buffered from the phase-shifting effects of light at that time, but I wasn't consulted at the design phase, and our ipRGCs are exquisitely sensitive to bright light in the middle of the night and wee hours of morning. In fact, our ipRGCs and SCN become increasingly responsive to light with every hour after sundown.[68] If you're wondering if fireplaces, campfires, candlelight, or

[68] Rahman et al., "Circadian Phase Resetting"; Sat Bir S. Khalsa et al., "A Phase Response Curve to Single Bright Light Pulses in Human Subjects," Journal of Physiology 549, no. 3 (June 2003): 945–52, https://doi.org/10.1113/jphysiol.2003.040477; Andrew J. K. Phillips et al., "High Sensitivity and Interindividual Variability in the Response of the Human Circadian System to Evening Light," Proceedings of the National Academy of Sciences U. S. A. 116, no. 24 (2019): 12019–24, https://doi.org/10.1073/pnas.1901824116; Jamie M. Zeitzer et al., "Sensitivity of the Human Circadian Pacemaker to Nocturnal Light: Melatonin Phase Resetting and Suppression," Journal of Physiology 526, no. 3 (August 2000): 695–702, https://doi.org/10.1111/j.1469-7793.2000.00695.

moonlight are too bright to view after dark, the answer is no. Firelight sources emit very long wavelengths, and moonlight is low lux. Candlelight is typically just 1 lux. A bright, full moon is less than 1 lux! Compare those numbers to a typical table lamp, which is 200 to 400 lux, or commonly used overhead indoor LED lighting, which is 500 to 1,000 lux.

So are we to all avoid artificial lighting and opt for candlelight or moonlight? Maybe, but more on that later. In the meantime, let's acknowledge that most people just won't do that, and even if we opt to at home, we can't control the lighting of most other environments like office cubicles, restaurants, hotels, and the rest.

That is where this protocol can help. Viewing sunlight in the afternoon reduces the sensitivity of neural circuits in your retinas, so that any artificial light you see after dark does not have as strong an effect in disrupting melatonin and other circadian-related molecules and processes in your eyes and brain. This protocol is especially important if your mornings and days are mostly spent under artificial indoor lights, as that will cause any artificial lights you view at night to have an outsized negative effect on your sleep and circadian rhythms.[69] Think of viewing the sunset as your (partial!)

[69] te Kulve et al., "Early Evening Light"; Tomoaki Kozaki et al., "Effects of Day-Time Exposure to Different Light Intensities on Light-Induced Melatonin Suppression at Night," Journal of Physiological Anthropology 34, no. 27 (2015), https://doi.org/10.1186/s40101-015-0067-1.

nighttime TV-watching or phone-scrolling inoculation. That said, you should still strive to dim the lights in the evening, and make it as dark as possible while you sleep, as discussed in the next protocol.

COMMON QUESTIONS

What if I can't get myself to a place where I can see sunlight in the late afternoon or early evening

I understand. Most of us are working, driving, or otherwise occupied indoors when the sun is setting—especially in the fall and winter months. And while sunsets are both beautiful and beneficial, telling your boss "I need to leave to go watch the sunset" is not going to go over well. (Although saying "Let's go watch the sunset together because it's good for our health" might be a good idea, depending on how you frame the question.) If you can't watch the sunset, just try to go outside at some point in the afternoon to get some ambient daylight in your eyes before dark, even if it's just for a stroll around the block, or a few moments on a balcony. Do whatever you can. The ambient light will still serve your SCN, and thus your circadian rhythm, sleep, and health, in numerous beneficial ways.

3

SLEEP PROTOCOL

Avoid Viewing Bright Lights in the Final Hour Before Bedtime

WHAT TO DO

- After sundown, limit your nighttime light exposure. You have a few options for how to do that. Which you use will depend on where you are and what activities you need to carry out. Option 1: Dim the lights. After sundown, dim the lights in your home or wherever you are in the hours before sleep.
- Option 2: Use red lights at night. I am a fan of using soft amber or red incandescent (or even amber or red LED lights, which are easier to find nowadays) for indoor lighting after dark as they don't spike cortisol or suppress melatonin as much as many bright white LED lights do.[70] Remember, you want to try to keep your cortisol low in the late evening and especially the final few hours before your bedtime at night. You also want your melatonin

[70] Mariana G. Figueiro and Mark S. Rea, "The Effects of Red and Blue Lights on Circadian Variations in Cortisol, Alpha Amylase, and Melatonin," International Journal of Endocrinology 2010 (June 2010): 829351, https://doi.org/10.1155/2010/829351; Sebastian Babilon et al., "Study Protocol for Measuring the Impact of (Quasi-)Monochromatic Light on Post-Awakening Cortisol Secretion Under Controlled Laboratory Conditions," PLOS One 17, no. 5 (2022): e0267659, https://doi.org/10.1371/journal.pone.0267659; Sonia Ancoli-Israel et al., "Effect of Light Treatment on Sleep and Circadian Rhythms in Demented Nursing Home Patients," Journal of the American Geriatrics Society 50, no. 2 (2002): 282–89, https://doi.org/10.1046/j.1532-5415.2002.50060.x.

elevated then. While this protocol might seem extreme, shifting to amber or red lighting in the final hours before sleep can make a marked difference for some people's ability to fall and stay deeply asleep, which extends to every aspect of next-day energy, mood, and focus.[71] Consider buying a few high-quality red light bulbs. Bon Charge makes great LED bulbs that are just the right level of brightness to see, but not so bright as to disrupt circadian rhythms and Ra Optics has LED bulbs that lack the usual flicker of LEDs and can shift between daytime and "candlelight" mode (I have no affiliation with Bon Charge or Ra Optics, but I use these brands at home, and I travel with one or two of their bulbs when on the road and staying in hotels.) SaunaSpace makes a large, red incandescent bulb that is about the size of a shoebox that can illuminate a medium-sized room. I have one in my living room and another in my bedroom. Incandescent bulbs give off a softer illumination than LED ones, and have some known advantages for mitochondrial health, which I cover in chapter 5.

[71] Jiexiu Zhao et al., "Red Light and the Sleep Quality and Endurance of Chinese Female Basketball Players," Journal of Athletic Training 47, no. 6 (November–December 2012): 673–78, http://doi.org/10.4085/1062-6050-47.6.08; Emiliano Terán et al., "Home lighting, Blue-Light Filtering, and Their Effects on Melatonin Suppression," Scientific Reports 16 (2026): 2850, https://doi.org/10.1038/s41598-025-29882-7.

- Consider small, portable side table LED lamps or bulbs that can be flipped from bright white to yellow to red, depending on the time of day or night. Bon Charge makes these. They are great if you need to navigate to the bathroom at night. They charge off your computer or a wall outlet and (hooray) don't require Wi-Fi or an app. I know many people who have solved their "I wake up in the middle of the night to go to the bathroom and then can't fall back asleep" issues by switching to these.

Note: I am not talking about "red light therapy," and you don't need a fancy red-light panel for this protocol. In fact, red-light panels tend to be very bright—exactly the opposite of what you want to use at night! They have their place, which I will discuss later in this book.

- Option 3: Use combination UV, blue, and green light blocking glasses after sundown. This is the most versatile and cost-effective option. Consider buying a pair of blue and green light blocking glasses if you need to be on a computer, look at a phone, or watch TV at night. They are also helpful when you can't control the lighting because you're out at a restaurant, in an airport, et cetera, at night. I helped design ROKA Wind Down glasses, which block the full range of blue and green light, are lightweight, and come in a variety of frames. They are excellent quality.

- Option 4: Use good ol' candlelight. Just make sure to blow it out before you go to sleep!

HOW IT WORKS

Humans are not the strongest or fastest creatures on earth, but we are the best at building technologies. Every technology—the automobile, the smartphone, airplanes, antibiotics—brings the potential for unprecedented power over nature in our daily lives. Artificial lights have vanquished darkness. But in the case of LED lights, in particular, this technology also carries a risk to our circadian cycle, our mitochondria, and thus, our health.

Until about ten years ago, most artificial lighting used incandescent bulbs, which contain more long-wavelength light and a more even distribution of different wavelengths than LED bulbs. Incandescent bulbs have been largely phased out because they are less energy efficient than LED bulbs. Nowadays, most indoor lighting comes from LEDs, which emit a disproportionate amount of short-wavelength light that can potently reduce melatonin levels.[72] Emerging evidence also points to negative effects of LEDs on

[72] Ana Sanchez-Cano et al., "Comparative Effects of Red and Blue LED Light on Melatonin Levels During Three-Hour Exposure in Healthy Adults," Life 15, no. 5 (April 2025): 715, https://doi.org/10.3390/life15050715.

mitochondrial health (which we'll go into later, in chapter 5).[73]

Those are some of the risks. Now let me explain why bright artificial light at night may also be affecting our mood. Each of us has a brain structure called the habenula that sits at the middle and top part of an egg-shaped brain structure called the thalamus.[74] Your thalamus is a station for routing different types of sensory and motor information to the correct parts of your brain so you can detect and integrate that information into your thoughts and actions. When I was an undergraduate and graduate student taking neuroanatomy, the habenula was listed as having an "unknown function." Recent studies in rodents, fish, and humans finally revealed what it does.[75]

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[73] Lynne Peeples, "The Surprising Science Behind Red-Light Therapy—and How It Really Works," Nature 651 (March 2026): 871–74, https://doi.org/10.1038/d41586-026-00878-1; Emiliano Terán et al., "Home Lighting, Blue-Light Filtering, and Their Effects on Melatonin Suppression," Scientific Reports 16, no. 2850 (January 2026), https://doi.org/10.1038/s41598-025-29882-7; Edward M. Barrett and Glen Jeffery, "LED Lighting (350–650nm) Undermines Human Visual Performance Unless Supplemented by Wider Spectra (400–1500nm+) like Daylight," Scientific Reports 16, no. 1 (January 2026): 3061, https://doi.org/10.1038/s41598-026-35389-6.

[74] Diego Carlos Fernandez et al., "Light Affects Mood and Learning Through Distinct Retina-Brain Pathways," Cell 175, no. 1 (2018): 71–84.e18, https://doi.org/10.1016/j.cell.2018.08.004.

[75] Laura-Joy Boulos et al., "Translating the Habenula—from Rodents to Humans," Biological Psychiatry 81, no. 4 (2016): 296–305, https://doi.org/10.1016/j.biopsych.2016.06.003.

We now know the habenula is part of a network that controls depressive feelings and disappointment.[76] Habenula activation creates those states by altering dopamine and serotonin dynamics in brain areas associated with reward.[77] Why would we harbor such a structure and circuit? The answer is that not all behaviors are meant to be rewarded. Some need to be punished so we are less likely to repeat them. And it turns out that viewing bright light in the "you really should be asleep now" phase of your circadian cycle is one of them. Animal studies and emerging evidence in humans show that viewing bright lights at night causes your ipRGCs to activate a specific region called the perihabenula.[78] That leads to elevated levels of cortisol,

[76] Jessica L. Ables et al., "Understanding the Habenula: A Major Node in Circuits Regulating Emotion and Motivation," Pharmacological Research 190 (April 2023): 106734, https://doi.org/10.1016/j.phrs.2023.106734; Sarah Cameron et al., "The Disappointment Centre of the Brain Gets Exciting: A Systematic Review of Habenula Dysfunction in Depression," Translational Psychiatry 14, no. 1 (2024): 499, https://doi.org/10.1038/s41398-024-03199-x; Liang Yang et al., "Stress Dynamically Modulates Neuronal Autophagy to Gate Depression Onset," Nature 641 (2025): 427–37, https://doi.org/10.1038/s41586-025-08807-4.

[77] Christophe D. Proulx et al., "Reward Processing by the Lateral Habenula in Normal and Depressive Behaviors," Nature Neuroscience 17, no. 9 (2014): 1146–52, https://doi.org/10.1038/nn.3779.

[78] Tara A. LeGates et al., "Aberrant Light Directly Impairs Mood and Learning Through Melanopsin-Expressing Neurons," Nature 491 (November 2012): 594–98; Fernandez et al., "Light Affects Mood"; Tenley Weil et al., "Daily Changes in Light Influence Mood via Inhibitory Networks Within the Thalamic Perihabenular Nucleus," Science Advances 8, no. 23 (June 2022), https://doi.org/10.1126/sciadv.abn3567; Christian Kaiser et al., "The Human Habenula Is Responsive to Changes in Luminance and Circadian Rhythm," NeuroImage 189 (2019): 581–88, https://doi.org/10.1016/j.neuroimage.2019.01.064.

reductions in dopamine, next-day anxiety, and deficits in learning—all consequences that appear to be separate from any reductions in total sleep time.[79]

For most people, the "you really should be asleep now" time is somewhere between the hours of 11:00 p.m. and 4:00 a.m. But of course, neither your SCN nor your habenula knows clock time; they only know circadian time, which follows the circadian body temperature, cortisol, and other twenty-four-hour rhythms I described briefly earlier in this chapter. Your body temperature starts to rise in the final two hours before waking, and continues to rise throughout your morning and day until you reach the mid to late afternoon.[80] At that point, it starts to decline. That decline continues as we sleep, reaching its lowest point, called, unimaginatively, your temperature minimum time about two to three hours

[79] Grace E. Guindon et al., "Turn Off That Night Light! Light-at-Night as a Stressor for Adolescents," Frontiers in Neuroscience 18, no. 1 (July 2024): 1451219, https://doi.org/10.3389/fnins.2024.1451219; Christopher J. Young et al., "Circadian Influences on the Habenula and Their Potential Contribution to Neuropsychiatric Disorders," Frontiers in Behavioral Neuroscience 15, no. 1 (January 2022): 815700, https://doi.org/10.3389/fnbeh.2021.815700.

[80] Kryger et al., Kryger's Principles and Practices of Sleep Medicine; Mats Gillberg and Torbjörn Åkerstedt, "Body Temperature and Sleep at Different Times of Day," Sleep 5, no. 4 (1982): 378–88, https://doi.org/10.1093/sleep/5.4.378; Victor E. Del Bene, "Temperature," in Clinical Methods: The History, Physical, and Laboratory Examinations, 3rd ed., ed. H. Kenneth Walker et al. (Butterworths, 1990), https://www.ncbi.nlm.nih.gov/books/NBK331/.

prior to waking.[81] So when I say avoid viewing bright lights from LEDs or other sources at night, I am referring to the time about six to eight hours before your temperature minimum time. This may seem overly technical, but knowing your temperature minimum time can be very valuable. And no, you don't have to measure your body temperature; your temperature minimum is the approximate time in your twenty-four-hour circadian cycle that roughly corresponds to two hours before you typically wake up, alarm clock or no. The exact clock time of course will vary depending on whether you are a morning lark or a night owl, or somewhere in between (more on how to determine that soon). But since most people wake up between 6:00 and 8:00 a.m., their temperature minimum time is between 4:00 and 6:00 a.m. If that's you, assuming you'd like to fall asleep anytime between 10:00 p.m. and midnight, it means you should avoid viewing bright artificial lights between 11:00 p.m. and 4:00 a.m. You would do even better to dim the lights starting around 8:00 or 9:00 p.m. Put simply, to avoid habenula activation and shifting your circadian clock, avoid bright artificial lights in the four to six hours prior to your temperature minimum time.

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[81] Leon Lack et al., "Chronotype Differences in Circadian Rhythms of Temperature, Melatonin, and Sleepiness as Measured in a Modified Constant Routine Protocol," Nature and Science of Sleep 1 (November 2009): 1–8, https://doi.org/10.2147/nss.s6234.

If you want to figure out your exact temperature minimum time, simply take the average of your wake-up times over the course of three to four days and subtract two hours. Then strive to avoid viewing bright artificial light in the full six-hour period prior to that. I generally wake up at about 6:30 a.m., which means my temperature minimum time is 4:30 a.m., so I avoid viewing bright artificial light between 10:30 p.m. and 4:30 a.m.

I know this all sounds technical, but it's very simple. By doing it, you'll likely enjoy improvements in your sleep and thus daytime mood, focus, and alertness. I acknowledge that many people can fall asleep after being under bright artificial lights in the hours before bedtime, but knowing several such folks, I can assure you even they benefit by adopting this simple (and incidentally cost-saving) change. Bonus: your home feels cozier at night too.

A bit more on those mood improvements: In 2018, a team of researchers, including noted UCLA psychiatrist Peter Whybrow, reviewed the data on the potential link between LED lighting and mental health. They concluded that exposure to blue light, in particular, posed a risk to sleep and other processes governed by the circadian system.[82] There is also evidence suggesting that the way in which artificial

[82] Michael Bauer et al., "The Potential Influence of LED Lighting on Mental Illness," World Journal of Biological Psychiatry 19, no. 1 (2018): 59–73, https://doi.org/10.1080/15622975.2017.1417639.

lighting at night interferes with this system may lead to depression.[83]

These findings do not mean we should all throw away our LED lights. However, if you can find incandescent lights for your home, they are preferable because they're less disruptive to the circadian system. In a study from 2020, researchers found that incandescent lighting interfered with the production of melatonin far less often than LED or fluorescent lights.[84] (I cover why this is in chapter 5.) If you can't find incandescent bulbs, consider the Bon Charge or Ra Optics bulbs or ROKA Wind Down or other short wavelength (not just "blue") blocking glasses.

Experiment with the options in this protocol to see what works best for you. And no matter what bulbs you choose for your home, still dim them before bed, and keep it dark in the six to eight hours prior to your temperature minimum time.

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[83] Yu-Sheng Shen, "Multiple Pathways and Mediation Effects of Built Environment on Kidney Disease Rate Via Mitigation of Atmospheric Threats," Science of the Total Environment 833 (August 2022): 155177, https://doi.org/10.1016/j.scitotenv.2022.155177; Manman Chen et al., "Artificial Light at Night and Risk of Depression: A Systematic Review and Meta-Analysis," Environmental Health and Preventive Medicine 29 (December 2024): 73, https://doi.org/10.1265/ehpm.24-00257; Xiaomeng Li et al., "Light at Night Exposure and Risk of Depression: A Meta-Analysis of Observational Studies," Journal of Global Health 15 (2025): 04304, https://doi.org/10.7189/jogh.15.04304.

[84] Sean W. Cain et al., "Evening Home Lighting Adversely Impacts the Circadian System and Sleep," Scientific Reports 10, no. 1 (2020): 19110, https://doi.org/10.1038/s41598-020-75622-4.

If a light is bright enough, it doesn't matter if it is an LED or incandescent, it will still alter your circadian cycle.[85]

COMMON QUESTIONS

What if I want to go to a game, concert, party, or movie and view bright lights? Will that destroy my sleep that night

This is an important question, as we need to live a life that is not just focused on health. From a circadian perspective, no, it will not destroy your sleep. Your natural wake-up and to-sleep times are determined by your genetics, your social activity, your exercise routine, your stress levels, and the average timing and amount of your morning and evening light viewing—especially morning sunlight—in the prior days. As I mentioned earlier, the idea is to strive to get great sleep about 90 percent of the nights of your life, and reserve the remaining about 10 percent for when challenges or fun keep you up late, or even until dawn. Yes, wearing glasses that block UV, blue, AND green wavelength light can help keep your sleep on target, even when you're going out a lot, but who wants to wear those lenses on a first date or on your wedding night? They can be fun to wear to concerts and dinner, but I hate to think that anyone would miss out on the joys of life to avoid bright light exposure to their eyes. That said, if you're getting a lot of screen LED light in your eyes

[85] Blume et al., "Effects of Light."

at night because you're scrolling on your phone or computer or watching a big flat-screen monitor, that will take a negative toll on your health. Please adjust this protocol so that it works for you. Also, you shouldn't wear these glasses while driving, for obvious reasons; there are a lot of road signs and lights that you would not be able to see accurately—or at all depending on their color, and you could get into an accident.

What I describe here is what to strive for. Everyone diverges from that now and again. Enjoy your life and meet its moments by prioritizing your health most of the time and not stressing about the occasional night out.

What about kids? Do the same rules about lighting apply

Yes. They need great sleep even more than we do, given the stage of brain and bodily development they are in. Teens, though, tend to stay up late to socialize online or otherwise. Keep in mind that kids and teens often need eight to ten hours of sleep.[86] But their socialization is vital, too, so you'll have to strike a balance between encouraging a healthy relationship to artificial light, screens, and their social life. That said, if your teen or child is depressed or anxious, it's important to assess their sleep patterns and relationship to

[86] S. Paruthi et al., "Consensus Statement of the American Academy of Sleep Medicine on the Recommended Amount of Sleep for Healthy Children: Methodology and Discussion," Journal of Clinical Sleep Medicine 12 (November 2016): 1549–61, https://doi.org/10.5664/jcsm.6288.

light, especially the amount and timing of screen use. I have heard from many parents whose kids had a significant improvement in their mental health with the protocols listed here. Some were only able to convince their teens to use their phones outside in the yard instead of in their rooms in the morning. Others have told me their kids love the morning and evening sunlight viewing for the immediate mental and physical benefits it provides. I won't make this a parenting protocol, but the phrase "be a channel, not a dam" comes to mind as possibly useful in this context.

My infant wakes up frequently throughout the night. Can dim lighting help me cope with interrupted sleep

Yes. Even if your infant is waking frequently through the night, try to keep the lights as dim as possible during those stretches, and/or use red light bulbs (again, not the same as red light therapy panels). Dimming lights will help you fall back asleep with your infant and will also help your baby's circadian system start to become attuned to the solar cycle. Babies have very different REM and other sleep cycles compared to adults—they naturally get much more REM, but the general effects of light and darkness on their cortisol and melatonin levels are very similar to those of healthy adults. That is one reason why the practice of dividing breastmilk refrigerated for later use into "collected in morning" and "collected at night" is becoming more

commonly discussed in health circles. The former contains relatively more cortisol and the latter relatively more melatonin.[87]



SLEEP PROTOCOL

Find Your Ideal To-Sleep and Wake-Up Times and Strive to Adhere to Them

WHAT TO DO

- Find your chronotype to set your schedule. Some people sleep better when they stay up late and sleep in late; others feel best when they go to bed early and wake up early; and most people fall somewhere in between, preferring to go to sleep somewhere between 10:00 p.m. and midnight and to wake up between 6:00 and 8:00 a.m.
- Try to stick to your schedule. As long as you sleep for at least six to eight hours a night, pick a sleep-wake schedule that works for you. Even shifting slightly closer to your chronotype will afford you much more restorative sleep.

[87] Melissa A. Woortman et al., "Day/Night Fluctuations of Breast Milk Bioactive Factors and Microbiome," Frontiers in Nutrition 12 (2025), https://doi.org/10.3389/fnut.2025.1618784.

A systematic approach to figuring out your ideal sleep-wake schedule

Let's get something out of the way. If you've been reading this chapter and thinking, "Hmm, my circadian rhythm is nothing like the one you describe," then maybe you are a night owl. Or maybe you just think you are. Kenneth Wright, whose lab at the University of Colorado Boulder studies circadian rhythms, wanted to figure out how much of people's preference to go to bed long after sundown and sleep in many hours past sunrise was due to a natural inclination, and how much was due to the influence of artificial indoor lighting.[88] He and several colleagues gave participants wrist monitors that measured the intensity and timing of their light exposure throughout the day, as well as their activity patterns, so that they could know when the participants were sleeping. Next, they measured their melatonin levels every hour to see whether the participants' bedtimes matched up with their circadian-driven increases in melatonin—which is a reliable measure of circadian clock output. Finally, the participants were sent packing. They went camping for a week, with only the sun and campfires as their light sources; no flashlights, screens, or other electronic devices permitted.

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[88] Ellen R. Stothard et al., "Circadian Entrainment to the Natural Light-Dark Cycle Across Seasons and the Weekend," Current Biology 27, no. 4 (2017): 508–13, https://doi.org/10.1016/j.cub.2016.12.041.

That one week was all it took for participants to sync sleep-wake cycles to the rising and setting of the sun. Before the outing, some of the participants were convinced they were morning people and others were convinced they were night owls. After a week in the wilderness, those differences faded. They all went to sleep around sunset. The researchers noted that the intensity of the light the participants experienced while camping was four times greater than the intensity of light in their normal lives—a testament to how much time they normally spent indoors. Interestingly, just one weekend spent camping had a similar effect of shifting people's circadian rhythm toward being early risers and people who prefer to go to sleep early. Mind you, those shifts were not just while camping. They persisted after they came back. This study was small and in no way does it mean there is no such thing as a night owl. But the rapidity and degree of the change suggest that artificial lighting masks the preference of many people to have a solar-cycle-aligned rhythm.

There are potential health implications tied to shifting toward (or away) from your innate sleep-wake timing. And they suggest that except for bona fide night owls, a more solar-cycle-aligned rhythm is beneficial. A large-scale 2019 study of nearly 700,000 subjects showed a causal link between being a morning person and better mental

health—that's right: a causal link.[89] A different study found a higher risk for cardiovascular disease among people who went to sleep later and slept for a long time.[90] Another analysis using data from 6,258 people in Finland found a link between routine late nights and type 2 diabetes.[91] The patterns a night owl keeps have also been linked to depression.[92]

And there are new data showing that even if you prefer to go to bed and sleep late, shifting to becoming an early riser can improve your health.[93] But telling a night owl to become a morning person can be dangerous—people cling to their natural sleep cycles like a pit bull on a rope toy.

Whether you think you're a night owl or not, by systematically experimenting with your to-sleep and

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[89] Samuel E. Jones et al., "Genome-Wide Association Analyses of Chronotype in 697,828 Individuals Provides Insights into Circadian Rhythms," Nature Communications 10, article no. 343 (2019), https://doi.org/10.1038/s41467-018-08259-7.

[90] Freda Patterson et al., "Interactive Effects of Sleep Duration and Morning/Evening Preference on Cardiovascular Risk Factors," European Journal of Public Health 28, no. 1 (2018): 155–61, https://doi.org/10.1093/eurpub/ckx029.

[91] Ilona Merikanto et al., "Associations of Chronotype and Sleep with Cardiovascular Diseases and Type 2 Diabetes," Chronobiology International 30, no. 4 (2013): 470–77, https://doi.org/10.3109/07420528.2012.741171.

[92] Rosa Levandovski et al., "Depression Scores Associate with Chronotype and Social Jetlag in a Rural Population," Chronobiology International 28, no. 9 (2011): 771–78, https://doi.org/10.3109/07420528.2011.602445.

[93] Elise R. Facer-Childs et al., "Resetting the Late Timing of 'Night Owls' Has a Positive Impact on Mental Health and Performance," Sleep Medicine 60 (August 2019): 236–47, https://doi.org/10.1016/j.sleep.2019.05.001.

wake-up times a bit and by limiting bright artificial light exposure in the final hour or two before sleep, you can determine the ideal sleep-wake schedule for you. Few people can do genetic testing or sleep lab analysis to determine their true chronotype, but there is a simpler way and it's worth exploring.

Here's how:

Unless you are perfectly happy with your sleep, for one week, try getting into bed by 8:30 to 9:00 p.m. If you're not ready for bed at that time, read a book or do another non-screen activity, but turn all the lights out by 9:00 to 9:30… maybe 10:00 p.m. at the latest.

Many people who do this wake up at 6:00 a.m. or even 5:00 a.m. the next day feeling more refreshed than ever before. Those who had previously been going to sleep at 11:00 p.m. or 12:00 a.m. and assumed they needed eight or more hours of sleep to feel rested often find they can feel great with just six or seven hours of sleep if they just get to sleep a bit earlier.

This process aligns your sleep more accurately with your genetically driven cortisol and melatonin rhythms. The genes that determine chronotype impact what are called clock genes, which are located in your SCN, and in almost every other cell of your body too. Your SCN entrains every cell in your body by changing its patterns of clock gene expression,

either directly or indirectly, by altering the timing and type of peptides, hormones, and neural signals those cells receive. It's a multipronged and highly distributed network that allows your SCN to do that. That is one reason why once people find their correct sleep schedule, they often notice their digestion and immune systems improve, as well as key biomarkers of metabolism, not just their sleep and mood. It's not unusual to get more deep and REM sleep and awaken feeling substantially more refreshed when you obey your chronotype. Nocturnal growth hormone release[94] is strongly regulated by your circadian clock,[95] and is released in the first hours of sleep. If you are meant to go to sleep between 8:30 and 10:00 p.m. and you stay up late, you miss out on getting some of that growth hormone. And growth hormone is vital for bodily and brain repair, metabolism, and myriad other health functions.[96]

A few things to keep in mind as you try this protocol. If you typically have been going to sleep after 10:30 p.m., dim the lights as described in Sleep Protocol 3. Also, remember that

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[94] Y. Takahashi et al., "Growth Hormone Secretion During Sleep," Journal of Clinical Investigation 47, no. 9 (1968): 2079–90, https://doi.org/10.1172/JCI105893.

[95] Weihao Wang et al., "The GH-IGF-1 Axis in Circadian Rhythm," Frontiers in Molecular Neuroscience 14 (September 2021): 742294, https://doi.org/10.3389/fnmol.2021.742294.

[96] Joshua E. Brinkman et al., "Physiology, Growth Hormone (Archived)," StatPearls, last updated May 1, 2023, https://www.ncbi.nlm.nih.gov/books/NBK482141/.

it is perfectly normal to wake up once to go to the bathroom in the middle of the night. If you do, I highly recommend that you do not look at a clock or turn on bright lights. Chances are you'll fall back asleep within thirty minutes or less. No matter when you typically have been going to sleep, eating too close to bedtime and/or drinking caffeine late in the day will mess up this whole experiment—both lead to more frequent nighttime wakings, so don't do either.[97] In general, eating your last bite of food at least two to three hours before bedtime is best, provided you are not experiencing gnawing hunger near bedtime, in which case a small snack one hour before is helpful. Yes, eating too close to bedtime may disrupt sleep, but gnawing hunger will keep you awake, so we need to be practical about this whole "don't eat too close to bedtime" thing.[98] And try not to drink caffeine after 1:00 or 2:00 p.m. (more on the reasons why in Sleep Protocol 11).

If you are not used to going to sleep this early, and you find yourself unable to fall asleep even after several nights, you may just be genetically wired to go to sleep between 10:30 p.m. and midnight. True night owls do exist! There is

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[97] Nikola Chung et al., "Does the Proximity of Meals to Bedtime Influence the Sleep of Young Adults? A Cross-Sectional Survey of University Students," International Journal of Environmental Research and Public Health 17, no. 8 (April 2020): 2677, https://doi.org/10.3390/ijerph17082677.

[98] Li-Ming Yan et al., "Chronobiological Perspectives: Association Between Meal Timing and Sleep Quality," PLOS One 19, no. 8 (August 2024): e0308172, https://doi.org/10.1371/journal.pone.0308172.

nothing wrong with that, provided your schedule lets you sleep long enough for you to feel rested the next day. Again, this assumes you are not drinking caffeine after 1:00 or 2:00 p.m., stressed out, or otherwise generating elevated levels of arousal late in the day, all things that can affect your sleep-wake schedule.

Indeed, people prefer different sleep schedules. Approximately 33 percent of people feel great going to sleep between 10:30 p.m. and midnight and waking up between 6:30 and 7:30 a.m.[99] The other 33 percent do best going to sleep between 8:30 and 10:00 p.m. and waking up between 4:30 and 6:30 a.m. The remaining third are bona fide night owls—in the absence of caffeine or work demands, their genetically driven melatonin rhythms and circadian rhythms cause them to be wide awake until 12:30 a.m., 1:00 a.m., or even 2:00 to 3:00 a.m., and they do best waking up between 9:00 and noon.[100] The point is to determine which group you are in, and to do so while not letting other variables screw up your experiment.

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[99] Arcady A. Putilov et al., "Single-Item Chronotyping (SIC), a Method to Self-Assess Diurnal Types by Using 6 Simple Charts," Personality and Individual Differences 168 (2021): 110353, https://doi.org/10.1016/j.paid.2020.110353.

[100] Christer Hublin and Jaakko Kaprio, "Chronotype and Mortality—a 37-Year Follow-Up Study in Finnish Adults," Chronobiology International 40, no. 7 (2023): 841–49, https://doi.org/10.1080/07420528.2023.2215342.

It is oh-so-valuable to know what your body and brain need, and thrive on. If you're truly an early or late chronotype, perhaps you can delegate tasks to family members or roommates to meet life's demands. You can select courses and workout times that better match your rhythms. I will even go so far as to say that, because of the way that getting your sleep-wake times right generates outsized positive effects on your health and cognition, you might even be wise to try to build your life around it. Tell your boss you'll have fewer sick days (you will), tell your partner you'll be a better partner (you will), and if you have kids, a better parent (you will). Feel free to cite me during the negotiations. There are aspects of our biology that we can neglect for a while before it causes issues. Chronotype is not one of them. Of course, we can shift our circadian clock for child-rearing, work, or travel, and this book teaches you how to do that more effectively, but we can't disobey our chronotype for more than a few years before it catches up with us, and often it only takes a few weeks of disobeying it before we start to notice mild to severe malaise, irritability, or poor digestion. Fortunately, the reverse is also true. Align better with your chronotype—even by just an hour or two and you'll often feel much better within days.

That said, our ideal sleep timing and total sleep needs do change as we age. Some of that is socially driven, but other aspects are related to changes in our pineal gland—the gland

that makes melatonin and other hormones and neurotransmitters.[101]

Also, when we are sick or recovering from an injury or bout of chronic stress, we need more sleep. The point here is to figure out the best sleep schedule for you, this year, and know that it may drift a bit over time, or in specific circumstances.

Key point 1:

Keep in mind that regardless of your ideal sleep-wake schedule, you benefit from being consistent with any sleep-wake schedule. That means going to bed and waking up at the same time, plus or minus thirty minutes, six to seven nights out of the week. Consistent does not mean neurotic. Strive, don't strain, to get things right. Nothing is worse than trying to force yourself to sleep.

Key point 2:

It is very normal to feel very awake about one to three hours before your bedtime. That increase in arousal, sometimes

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[101] Rikie M. Scholtens et al., "Physiological Melatonin Levels in Healthy Older People: A Systematic Review," Journal of Psychosomatic Research 86 (July 2016): 20–27, https://doi.org/10.1016/j.jpsychores.2016.05.005; L. Thiélbot, "Physiology of the Pineal Body," Progress in Brain Research 10 (1965): 479–88, https://doi.org/10.1016/S0079-6123(08)63466-4; K. P. Minneman and R. J. Wurtman, "The Pharmacology of the Pineal Gland," Annual Reviews of Pharmacology and Toxicology 16 (April 1976): 33–51, https://doi.org/10.1146/annurev.pa.16.040176.000341; Josephine Arendt et al., "Physiology of the Pineal Gland and Melatonin," in Endotext [internet], ed. K. R. Feingold et al. (MDText.com, 2022).

referred to as the "wake maintenance zone," is thought to be an adaptive mechanism for allowing us to secure our physical surroundings and safety prior to sleep, but that is just speculation. A transient increase in alertness—that for some occurs around sunset and for others around one to three hours before their typical bedtime—also may happen; it's well documented in sleep studies, although to my knowledge its physiological basis has not yet been explored in detail.[102] Remember, this is a transient increase, so don't sweat it. Let it pass and embrace the drop in body temperature, the rise in melatonin, and sleepiness that follows.

HOW IT WORKS

The intricacies of what happens when we sleep—the so-called architecture of sleep—deserves a book of its own, and indeed many have been written about it.

Rarely discussed, however, is how the architecture of each sleep-wake cycle influences our sleeping and our waking states. We touched on this topic a bit earlier in this chapter. But now, by exploring the architecture of sleep itself, we can start to understand how the placement of six to eight hours of

[102] Jennifer A. Shekleton et al., "Improved Neurobehavioral Performance During the Wake Maintenance Zone," Journal of Clinical Sleep Medicine 9, no. 4 (April 2013): 353–62, https://doi.org/10.5664/jcsm.2588; Jan de Zeeuw et al., "The Alerting Effect of the Wake Maintenance Zone During 40 Hours of Sleep Deprivation," Scientific Reports 8, no. 1 (July 2018): 11012, https://doi.org/10.1038/s41598-018-29380-z.

sleep in each twenty-four-hour cycle, and the consistency of that placement from one day to the next, has a profound effect on exactly what happens while we sleep, and how that impacts whether our sleep is maximally restorative.

The Stages of Sleep

Many people are familiar with the two major types of sleep: rapid eye movement, or REM, sleep, and non-REM sleep. Non-REM sleep is often referred to as deep or slow-wave sleep, but that's oversimplified. Non-REM sleep consists of three distinct phases that we enter, one after the other. These are referred to as N1, N2, and N3 sleep, and only in the N3 stage does actual deep or slow-wave sleep occur. Cycling through N1, to N2, to N3, to REM sleep takes, on average, about ninety minutes. So a person who sleeps for three hours might make it through two cycles, a person who sleeps for six hours experiences four cycles, and a person who sleeps 7.5 hours gets five cycles, and so on.

We enter N1, or light sleep, first and it typically lasts five to ten minutes. It is easy to rouse us from N1 sleep. But N1 is not merely a transition stage; during N1, our body starts to generate additional adenosine triphosphate, or ATP, the chemical that fuels all our biological processes. What's more, proteins that have become misshapen or broken during our waking hours get repaired during N1, and proteins that

help build memories by reconfiguring our neural circuits are created.[103]

Next we enter N2 sleep, which lasts about twenty minutes. During N2 our body temperature drops, our heart rate slows down, and our brain produces sleep spindles—surges of transient brain activity that help consolidate information we have learned in the days prior.[104] Sleep spindles are vital for learning new cognitive information and motor skills, which both represent types of neuroplasticity.[105] We explore this in greater detail in chapter 6.

N2 has another intriguing feature: PGO waves, which are surges of large-scale brain activity that travel from the pons (P) of the brainstem, to the geniculate (G), a structure in our thalamus (the sensory waystation we discussed earlier), and then to our occipital (O) cortex, which resides in the back of our brain and is involved in, among other things, visual processing. PGO waves trigger the release of massive amounts of glutamate, an excitatory neurotransmitter that is

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[103] Gina Poe, in discussion with the author; Dante Picchioni et al., "Increased Rates of Brain Protein Synthesis During [N1,N2] Sleep: L-[1-11C]leucine PET Studies in Human Subjects," Journal of Cerebral Blood Flow & Metabolism 43, no. 1 (2023): 59–71, https://doi.org/10.1177/0271678X221121873.

[104] Matthew P. Walker et al., "Sleep-Dependent Learning and Memory Consolidation," Neuron 44, no. 1 (2004): 121–33, https://doi.org/10.1016/j.neuron.2004.08.031.

[105] Matthew P. Walker and Robert Stickgold, "Sleep, Memory, and Plasticity," Annual Review of Psychology 57 (January 2006): 139–66, https://doi.org/10.1146/annurev.psych.56.091103.070307.

a requisite component of the multistage process of learning, such as in the formation and elimination of synapses.[106]

After N1 and N2 we enter deep sleep, what sleep researchers call slow-wave sleep, or N3. This is an extremely restorative stage of sleep. During N3, our brain waves shift to very high amplitude and low frequency, aka slow; our heart rate slows way down, and growth hormone is released from our pituitary gland into our bloodstream.[107] What's more, the active repair of our muscle, brain, and other bodily tissues occurs—including our tendons, skin, heart, et cetera.[108] That process involves a lot of protein synthesis, or the repair and regeneration of cells and tissues by use of specific amino

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[106] Mia Michaela Pal, "Glutamate: The Master Neurotransmitter and Its Implications in Chronic Stress and Mood Disorders," Frontiers in Human Neuroscience 15 (October 28, 2021): 722323, https://doi.org/10.3389/fnhum.2021.722323; Roberto Malinow and Robert C. Malenka, "AMPA Receptor Trafficking and Synaptic Plasticity," Annual Review of Neuroscience 25 (2002): 103–26, https://doi.org/10.1146/annurev.neuro.25.112701.142758.

[107] Eve Van Cauter and Laurence Plat, "Physiology of Growth Hormone Secretion During Sleep," Journal of Pediatrics 128, no. 5 (1996): S32–37, https://doi.org/10.1016/s0022-3476(96)70008-2; Lauren M. Hablitz and Maiken Nedergaard, "The Glymphatic System: A Novel Component of Fundamental Neurobiology," Journal of Neuroscience 41, no. 37 (2021): 7698–711, https://doi.org/10.1523/JNEUROSCI.0619-21.2021.

[108] Andy R. Eugene and Jolanta Masiak, "The Neuroprotective Aspects of Sleep," MEDtube Science 3, no. 1 (2015): 35–40, https://pmc.ncbi.nlm.nih.gov/articles/PMC4651462/; Brinkman et al., "Physiology, Growth Hormone (Archived)"; Joshua E. Brinkman et al., "Physiology of Sleep," StatPearls, last updated April 3, 2023, https://www.ncbi.nlm.nih.gov/books/NBK482512/.

acids.[109] Some of that repair and regeneration occurs in response to tissue damage from exercise, but most of it is simple "housekeeping"-based cellular turnover and repair, which is also vital to our health and longevity. N3 is also when glymphatic washout of our brain occurs.[110] This process was discovered just in the past fifteen years or so and involves the pulsing of lymph-like fluid through our brain, thereby removing the waste products caused by normal brain metabolism and any cellular debris that accumulated the previous day.[111] This glymphatic clearance is actively governed by multiple brain regions—including the locus coeruleus—that we will talk about more below and in later chapters.

REM sleep, or rapid eye movement sleep, follows N3, but often we cycle from N3 back to N2 for a few minutes or more before advancing into REM sleep. REM sleep is crucial for sorting through all that waking life brings our way by eliminating or strengthening specific neural connections in accordance with what information we need to

[109] Aakash K. Patel et al., "Physiology, Sleep Stages," StatPearls, last updated January 26, 2024, https://www.ncbi.nlm.nih.gov/books/NBK526132/.

[110] Oliver Cameron Reddy and Ysbrand D. van der Werf, "The Sleeping Brain: Harnessing the Power of the Glymphatic System Through Lifestyle Choices," Brain Sciences 10, no. 11 (2020): 868, https://doi.org/10.3390/brainsci10110868.

[111] Helene Benveniste et al., "The Glymphatic System and Waste Clearance with Brain Aging," Gerontology 65, no. 2 (2018): 106–19, https://doi.org/10.1159/000490349; Hablitz and Nedergaard, "The Glymphatic System."

keep and what we should discard.[112] Thus, think of REM sleep as an experience audit, with only the meaningful experiences surviving the cut.

One more interesting thing about REM: Our body becomes mostly paralyzed during this stage, a phenomenon known as sleep atonia.[113] The most prevalent hypothesis to explain sleep atonia is that it keeps us from acting out our dreams during REM sleep.[114] Our brains are extremely active during REM and the dreams we experience during this stage tend to be vivid and emotionally laden—often marked by flying, swimming, being filled with anxiety or awe, and highly dynamic in how we move through time and space (in the dream of course).[115] The idea is that REM-associated atonia prevents us from acting in accordance with what are

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[112] John Peever and Patrick M. Fuller, "The Biology of REM Sleep," Current Biology 27, no. 22 (2017): R1237–48, https://doi.org/10.1016/j.cub.2017.10.026.

[113] Patricia L. Brooks and John H. Peever, "Unraveling the Mechanisms of REM Sleep Atonia," Sleep 31, no. 11 (2008): 1492–97, https://doi.org/10.1093/sleep/31.11.1492.

[114] Mark S. Blumberg and Alan M. Plumeau, "A New View of 'Dream Enactment' in REM Sleep Behavior Disorder," Sleep Medicine Reviews 30 (December 2016): 34–42, https://doi.org/10.1016/j.smrv.2015.12.002; Patricia L. Brooks and John H. Peever, "Unraveling the Mechanisms of REM Sleep Atonia," Sleep 31, no. 11 (2008): 1492–97, https://doi.org/10.1093/sleep/31.11.1492.

[115] Jerome C. J. Fryer, "Is a Purpose of REM Sleep Atonia to Help Regenerate Intervertebral Disc Volumetric Loss?," Journal of Circadian Rhythms 7, article no. 1 (January 2009), https://doi.org/10.1186/1740-3391-7-1.

essentially in-sleep hallucinations, which would put us in danger.

The relative amount of time we spend in REM sleep increases, and the amount spent in deep N3 sleep decreases, as we complete more sleep cycles across a night of sleep. This means our most vivid dreams tend to occur as we approach morning. A baby's sleep is largely REM sleep, and the amount of REM we experience decreases with age starting in adolescence.[116] If we are sleep-deprived, or we do something that reduces the amount of REM sleep we get on a given night—using cannabis, for instance—the next night we experience REM rebound by entering REM sleep sooner in the sleep cycle, and experiencing more REM overall. Chronic cannabis users who stop using cannabis often report vivid dreams in the first weeks and months after quitting.[117]

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[116] Hai-Lin Chen et al., "Rapid Eye Movement Sleep During Early Life: A Comprehensive Narrative Review," International Journal of Environmental Research and Public Health 19, no. 20 (2022): 13101, https://doi.org/10.3390/ijerph192013101.

[117] Carissa Gardiner et al., "The Effect of Alcohol on Subsequent Sleep in Healthy Adults: A Systematic Review and Meta-Analysis," Sleep Medicine Reviews 80 (April 2025): 102030, https://doi.org/10.1016/j.smrv.2024.102030; Leela McKinnon et al., "Sound Reasons for Unsound Sleep: Comparative Support for the Sentinel Hypothesis in Industrial and Nonindustrial Groups," Evolution, Medicine, and Public Health 11, no. 1 (November 2022): 53–66, https://doi.org/10.1093/emph/eoac039; Rob Velzeboer et al., "Cannabis and Sleep Architecture: A Systematic Review and Meta-Analysis," Sleep Medicine Reviews 84 (December 2025): 102164, https://doi.org/10.1016/j.smrv.2025.102164; Irwin Feinberg et al., "Effects of Marijuana Extract and Tetrahydrocannabinol on Electroencephalographic

Alcohol and many prescription drugs reduce the fraction of our sleep that is REM as well. Once someone stops using these, they also experience an REM rebound.[118]

REM sleep is fascinating. As stated before, it is characterized by more brain activity than we ever experience in our waking life, and yet we are paralyzed while our brain is in that hyperactive state. And there's more. REM sleep is also the only time in our lives when our locus coeruleus—the brainstem region that releases norepinephrine and increases our focus and alertness—goes largely dormant. Also, we don't readily release adrenaline from our adrenals during REM.

Researchers believe that this paradoxical state, during which our brain is highly active but our body is paralyzed, our locus coeruleus is almost silent, and our body can't easily release adrenaline, serves as a sort of naturally occurring therapy, unloading the emotional burdens from our daytime experiences. Indeed, this general set of conditions parallels many established trauma therapies, in which patients recount a traumatic event while in an intentionally or progressively lowered state of anxiety, so as to detach the body's and

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Sleep Patterns," Clinical Pharmacology and Therapeutics 19, no. 6 (June 1976): 782–94, https://doi.org/10.1002/cpt1976196782.

118 Elahhe Afkhamnejad and Sandhya Kumar, "The Role of Prescription Drugs in REM Sleep (P1.109)," Neurology 90, no. S15 (April 2018), https://doi.org/10.1212/WNL.90.15_supplement.P1.109; Gardiner et al., "Effect of Alcohol."

brain's emotional responses from the details of the memory. People selectively deprived of REM sleep have a harder time unloading emotional burdens from past experiences, including traumas.[119]

REM, like some of the other sleep stages, is also a key time for the consolidation of new skills and information that are of a nonemotional nature. We will cover more on that in chapter 6.

Health risks of sleep deprivation

Given all this, it becomes clearer and clearer how lack of sleep, but also altered sleep architecture, affects our mind and body in a variety of ways. Sleep deprivation is associated with diminished response time, alertness, attention, memory, and executive functioning.[120]

Additionally, as described at the chapter outset, lack of sleep can bring about longer-term problems as well. Insufficient REM sleep may increase the risk of dementia. In 2017, a group of researchers published their investigation of links

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[119] Dimitris Repantis et al., "REM Sleep in Acutely Traumatized Individuals and Interventions for the Secondary Prevention of Post-Traumatic Stress Disorder," European Journal of Psychotraumatology 6, no. 1 (2020): 1740492, https://doi.org/10.1080/20008198.2020.1740492.

[120] William D. S. Killgore, "Effects of Sleep Deprivation on Cognition," Progress in Brain Research 185 (2010): 105–29, https://doi.org/10.1016/B978-0-444-53702-7.00007-5.

between dementia and sleep architecture.[121] They looked at three years of data on thirty-two adults over age sixty with dementia and found that the likelihood of cognitive decline fell in line with a reduction in REM sleep. In fact, with every 1 percent reduction in REM sleep, the risk of dementia rose by about 9 percent. (Non-REM sleep deprivation did not share this connection.) The role of REM in building and consolidating neural connections, thereby protecting against cognitive decline, may also explain why truncating this stage would raise the risk of dementia, but direct evidence for this mechanism in humans is still lacking.

Getting sufficient REM sleep is also connected to overall longevity. In 2020, a group of researchers from multiple institutions reported that less REM sleep is associated with an increased mortality risk. Using data from 4,050 people, they found that for every 5 percent decrease in REM sleep, the mortality rate increased by 13 percent.[122] That held true for most causes of death, including cardiovascular disease.

Ask any sleep researcher “How can I get more REM sleep?” and they will tell you “Sleep in an extra hour or two.” But most people can’t afford to sleep in an extra hour or two past

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[121] Matthew P. Pase et al., “Sleep Architecture and the Risk of Incident Dementia in the Community,” Neurology 89, no. 12 (2017): 1244–50, https://doi.org/10.1212/WNL.0000000000004373.

[122] Eileen B. Leary et al., “Association of Rapid Eye Movement Sleep with Mortality in Middle-Aged and Older Adults,” JAMA Neurology 77, no. 10 (2020): 1241–51, https://doi.org/10.1001/jamaneurol.2020.2108.

their usual wake-up time. The good news is that when we sleep is almost as important as getting enough of it. If we get to bed and wake up at a time that is in sync with our chronotype, we end up getting more growth hormone release, and we spend less time in light sleep and more time in deep sleep and REM sleep. As a consequence, we tend to need less sleep overall. Our sleep can become more efficient and restorative.[123]

Studies of shift workers tell us what happens if we sleep enough, but at the wrong times of the twenty-four-hour cycle. The short answer is: Bad things happen. Working at nighttime and sleeping in the daytime is associated with greatly elevated risk of type 2 diabetes, cardiovascular disease, depression, and suicide.[124] Shift workers are also susceptible to neurocognitive impairment and at greater risk

[123] Kelly Baron et al., "Circadian Phase Alignment and Sleep Duration in Overweight Adults: Associations with Cardiometabolic Risk Factors," Sleep 47, no. S1 (2024): A7–8, https://doi.org/10.1093/sleep/zsae067.0017; Andrew J. K. Phillips et al., "Irregular Sleep/Wake Patterns Are Associated with Poorer Academic Performance and Delayed Circadian and Sleep/Wake Timing," Scientific Reports 7, no. 1 (2017): 3216, https://doi.org/10.1038/s41598-017-03171-4.

[124] Yinyan Gao et al., "Association Between Shift Work and Risk of Type 2 Diabetes Mellitus: A Systematic Review and Dose-Response Meta-Analysis of Observational Studies," Chronobiology International 37, no. 1 (2020): 29–46, https://doi.org/10.1080/07420528.2019.1683570; Johanna Samulin Erdem et al., "Night Shift Work and Indicators of Cardiovascular Risk: A Systematic Review and Meta-Analysis," Environmental Research 276 (July 2025): 121503, https://doi.org/10.1016/j.envres.2025.121503; Qi-Jun Wu et al., "Shift Work and Health Outcomes: An Umbrella Review of Systematic Reviews and Meta-Analyses of Epidemiological Studies," Journal of Clinical Sleep Medicine 18 (2022): 653–62, https://doi.org/10.5664/jcsm.9642; Minzhi Xu et al.,

for substance abuse.[125] One large survey from Canada found that executive function is significantly worse among people whose jobs take place outside normal work hours.[126] One study found the level of performance impairment to be equal to a blood alcohol concentration of 0.10, which exceeds the legal limit for driving in many places.[127]

To be clear: Being a night owl is not the same thing as forcing yourself to be nocturnal, so if that's your chronotype, don't worry. The health detriments of shift work are likely to be multifactorial—the lack of sunlight, the altered social interactions associated with an inverted schedule, and more. We know that one of the worst types of shift work for one's health is the swing shift, where people switch back and forth from daytime to nighttime work each week or even within the same week.

"Lifestyle Factors in the Association of Shift Work and Depression and Anxiety," JAMA Network Open 6, no. 8 (2023): e2328798, https://doi.org/10.1001/jamanetworkopen.2023.28798.

[125] Thomas Vlasak et al., "Neurocognitive Impairment in Night and Shift Workers: A Meta-Analysis of Observational Studies," Occupational & Environmental Medicine 79, no. 6 (2022): 365–72, https://doi.org/10.1136/oemed-2021-107847; Kneginja Richter et al., "Shiftwork and Alcohol Consumption: A Systematic Review of the Literature," European Addiction Research 27, no. 1 (2021): 9–15, https://doi.org/10.1159/000507573.

[126] Rea Alonzo et al., "Does Shiftwork Impact Cognitive Performance? Findings from the Canadian Longitudinal Study on Aging (CLSA)," International Journal of Environmental Research and Public Health 19, no. 16 (2022): 10124, https://doi.org/10.3390/ijerph191610124.

[127] Drew Dawson and Kathryn Reid, "Fatigue, Alcohol and Performance Impairment," Nature 388 (1997): 235, https://doi.org/10.1038/40775.

How long we sleep, when we sleep, and how consistent we are with our sleep-wake schedule affect our physical and mental health. That's why I hope you try the approach outlined here. Figure out what is best for you and then stick with it.

COMMON QUESTIONS

Does changing time zones for travel or moving the clock for daylight saving time affect our sleep-wake cycle

Yes. When we turn the clocks forward an hour to move from standard time to daylight saving time, or back an hour to return to standard time, or when we change time zones during travel, our circadian clock has to shift.

Changing our clocks forward or back means we have collectively decided that 7:00 a.m. is now 8:00 a.m., or 8:00 a.m. is now 7:00 a.m., but our SCN really doesn't care about clock time, it only knows time according to the entrainment cues it receives. You already know the most powerful entrainment cue and that's viewing morning sunlight. Even if we move the clocks just by one hour, the SCN needs a few days to shift its patterns of gene expression and communication with all the cells of the body to get them on schedule. In the meantime, we are groggy at times in the morning when we were perfectly chipper the day before. And a whole host of other systems in our body and brain

have to shift before we feel like ourselves again. The dangers of such shifts extend beyond mere irritability. Traffic accidents increase in the days immediately following seasonal clock changes.[128] So do heart attacks.[129] It also makes people more susceptible to depression.[130]

Jet lag occurs when our sleep-wake cycle no longer matches the solar cycle of our location. Our internal timekeeper may be telling us it's time for bed when the locals are just eating dinner. Many people find that traveling east, into a later time, is harder than traveling west, into an earlier time, because we have to move our circadian clocks forward. It's often easier to stay up a few hours longer, or even take a brief nap, than to force ourselves to go to bed before we're ready.

Some people don't suffer much from jet lag. Others suffer tremendously. For those of you in the latter group, Sleep Protocol 15 offers a technique that can help avoid jet lag or at least dampen its impact.

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[128] Jijo Varughese and Richard P. Allen, "Fatal Accidents Following Changes in Daylight Savings Time: The American Experience," Sleep Medicine 2, no. 1 (January 2001): 31–36, https://doi.org/10.1016/S1389-9457(00)00032-0.

[129] Amneet Sandhu et al., "Daylight Savings Time and Myocardial Infarction," Open Heart 1, no. 1 (2014): e000019, https://doi.org/10.1136/openhrt-2013-000019.

[130] Birgitte T. Hansen et al., "Daylight Savings Time Transitions and the Incidence Rate of Unipolar Depressive Episodes," Epidemiology 28, no. 3 (May 2017): 346–53, https://doi.org/10.1097/EDE.0000000000000580.

Is it bad to sleep in on weekends or to stay up late now and again

We should all strive to keep a consistent sleep-wake schedule, for the reasons discussed above. That means we should try to get to bed at the same time, plus or minus thirty minutes, on most nights. Same goes for waking up; try to wake up at roughly the same time each day, plus or minus thirty minutes. Larger changes in sleep or wake times every once in a while won't be too disruptive, because the entrainment of your SCN reflects the weighted average of inputs from multiple factors, as discussed. So by all means, enjoy that birthday or wedding or New Year's celebration and don't despair about arriving home late. One late night is not something to worry about.

Many people wonder whether wake times should be later after that occasional late night. If you usually go to sleep somewhere between 8:30 and 11:00 p.m. and have a night where you're up until 1:00 or 2:00 a.m., should you sleep late or wake up at your usual time? In general, it's best to get enough sleep to feel rested but not to sleep more than one or two hours past your usual wake time. So on those "I stayed up too late last night" mornings, get an extra hour in bed asleep if you can, and try to take a twenty-minute nap, or practice non-sleep deep rest (NSDR) in the afternoon (more on that soon), and then get to bed around your usual time that night. Your SCN—and thus you—thrive on consistency.

If you go to bed much later than usual and sleep in much later, too, be sure not to shift your whole schedule forward. It's still fine (and encouraged) to view sunlight, exercise, and so on, but be especially mindful not to drink caffeine much later in the day than usual, or you will soon find yourself off your usual rhythm and it will take three to four or more days to get back to it.

What about children and teenagers

The advice is a little different for young children and teenagers. Babies need to sleep A LOT and enter REM far more frequently and for longer than any other age group. Toddlers and young children should also sleep as much or as often as necessary, or desired, to support their extraordinarily fast brain and bodily growth rates. The amount of neuroplasticity in infants and children, all the way up to adolescence and beyond, is incredible—and sleep is a key part of that process. Limiting their sleep is never recommended but of course once kids are in school, setting a regular schedule (and keeping devices out of their rooms near bedtime and throughout the night) becomes important.

Sleep needs are somewhat different for teenagers. Bedtimes shift during puberty for at least two known reasons. First is what researchers call the homeostatic sleep drive. This term refers to a brain and bodily pressure to sleep that builds up gradually throughout the day, mainly due to adenosine

accumulation.[131] The longer we are awake, the more adenosine in our bodies, and the greater the pressure to sleep. Adenosine levels drop while we sleep and begin to mount again when we wake up for the new day. This pressure to sleep is separate from the circadian clock mechanisms we have discussed a lot thus far. Teenagers (good rebels that they are) resist this pressure and force themselves to stay up late, building up more adenosine and, therefore, the desire to sleep in the next morning.[132]

And they often don't need to force themselves to stay up late; it just happens naturally, as the hormonal testosterone, estrogen, and progesterone surges of puberty shift their circadian clock to a later phase.[133] Melatonin production begins later in the teenage brain than in the brains of either adults or younger children.

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[131] Porkka-Heiskanen et al., "Adenosine"; Basheer et al., "Adenosine and Sleep-Wake Regulation."

[132] M. H. Hagenauer et al., "Adolescent Changes in the Homeostatic and Circadian Regulation of Sleep," Developmental Neuroscience 31, no. 4 (2009): 276–84, https://doi.org/10.1159/000216538.

[133] Stephanie J. Crowley et al., "Sleep, Circadian Rhythms, and Delayed Phase in Adolescence," Sleep Medicine 8, no. 6 (September 2007): 602–12, https://doi.org/10.1016/j.sleep.2006.12.002; Mary A. Carskadon et al., "Association Between Puberty and Delayed Phase Preference," Sleep 16, no. 3 (May 1993): 258–62, https://doi.org/10.1093/sleep/16.3.258; M. H. Hagenauer et al., "Adolescent Changes in the Homeostatic and Circadian Regulation of Sleep," Developmental Neuroscience 31, no. 4 (2009): 276–84, https://doi.org/10.1159/000216538.

All of these changes leave the typical sixteen-year-old awake until about 11:00 p.m.[134] And this age group should be sleeping about nine hours per night, making 8:00 a.m. the preferred wake-up time. Studies have found that moving school start times to 8:30 a.m. or later leads to better grades and other measures of academic performance. There is even evidence suggesting that this shift also reduces car crashes among teen drivers.[135] It's beneficial for teenagers to have days when they can sleep as long as they want, even if that means staying in bed until noon. That doesn't mean they should never be woken up earlier on a Saturday or Sunday, but parents and caregivers do not need to worry about prolonged sleeping on the weekends. Teenagers, yes, reference this book if you need leverage to sleep in and parents, yes, you still want to encourage them to get sunlight in their eyes once they get up, even though it is midday, and not low-solar-angle sun.

Current data indicate that teenagers in the US are in the midst of an epidemic of sleep deprivation that is likely having serious consequences for their health. Ideally, young people between the ages of six and twelve years should sleep

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[134] Adolescent Sleep Working Group, Committee on Adolescence, Council on School Health, "School Start Times for Adolescents," Pediatrics 134, no. 3 (2014): 642–49, https://doi.org/10.1542/peds.2014-1697.

[135] Robert D. Foss et al., "School Start Times and Teenage Driver Motor Vehicle Crashes," Accident Analysis & Prevention 126 (May 2019): 54–63, https://doi.org/10.1016/j.aap.2018.03.031.

at least nine hours per night, and thirteen-to-eighteen-year-olds should sleep at least eight hours per night. More than 70 percent of young people do not sleep enough, according to one large survey from the CDC.[136] Anxiety, mood disorders, obesity, high-risk behavior, poor academic performance, and substance use have all been tied to sleep deprivation among adolescents.[137] Good practices for teenagers include shutting off screens at least an hour before bedtime, keeping the phone (and the laptop, the tablet, etc.) out of the bedroom at night and the room cool and dark, having a consistent bedtime, and, ideally, sleeping for those eight to nine hours per night.

Do sleep patterns change for older adults

Parsing the changes in sleep that arise in older adults is tricky. Ailments and medications of various kinds impact sleep in many ways. Absent any such interferences, sleep can vary in our later years, but not to a great extent. Older adults sleep a bit less than middle-aged adults by about an hour or so, on average.[138] It can take longer to fall asleep as

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[136] Anne G. Wheaton et al., "Short Sleep Duration Among Middle School and High School Students—United States, 2015," Morbidity and Mortality Weekly Report 67, no. 3 (2018): 85–90, https://doi.org/10.15585/mmwr.mm6703a1.

[137] Sara Uccella et al., "Sleep Deprivation and Insomnia in Adolescence: Implications for Mental Health," Brain Sciences 13, no. 4 (2023): 569, https://doi.org/10.3390/brainsci13040569; Wheaton et al., "Short Sleep Duration."

[138] Junxin Li et al., "Sleep in Normal Aging," Sleep Medicine Clinics 13, no. 1 (2018): 1–11, https://doi.org/10.1016/j.jsmc.2017.09.001.

we move past age fifty, but again, not in a dramatic way. And older adults are able to doze back off to sleep after a middle of the night waking, just as well as teenagers and younger adults can.

The stages of sleep also shift a bit as we age. The percentage of time we spend in deep sleep and REM decreases, although studies have shown a surge in REM in people between the ages of seventy-five and eighty-five years.[139] Older people also tend to nap more.

Does counting sheep help with falling asleep

Counting sheep has long been recommended for calming a busy brain before bed. The idea is that by giving ourselves a task, we stop ruminating on the day and going over our to-do lists for tomorrow. It turns out that counting sheep isn't such a great nighttime activity. Allison Harvey, a psychologist at UC Berkeley, found that counting sheep actually made it more difficult for people to fall asleep.[140] She concluded that this basic mental imagery was not engaging enough to hold the attention of people lying in bed trying to keep anxieties at bay. Without that focus to help calm our brain,

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[139] Judith A. Floyd et al., "Changes in REM-Sleep Percentage over the Adult Lifespan," Sleep 30, no. 7 (2007): 829–36, https://doi.org/10.1093/sleep/30.7.829.

[140] Allison G. Harvey and Suzanna Payne, "The Management of Unwanted Pre-Sleep Thoughts in Insomnia: Distraction with Imagery Versus General Distraction," Behaviour Research and Therapy 40, no. 3 (2002): 267–77, https://doi.org/10.1016/s0005-7967(01)00012-2.

we're back to ruminating and reviewing to-do lists. A subsequent study in which Harvey instructed participants to deliberately suppress whatever thought was most likely to intrude into their mind before sleep found this didn't help—predictably, the thought they were supposed to be suppressing just became more preoccupying.[141]

Instead, try these somewhat weird tricks:

Take a mental walk: Imagine walking from your bed to your mailbox and then down the street and to a nearby location, then home again by the same or different familiar route. Occupying a busy brain with an easy and engaging task can help stop ruminating thoughts. Mental walks like these are occupying but not stimulating. And they have the tendency to lull people off to sleep.

Move your eyes following the pattern described below: Warning: This is going to sound a bit out there, but it works well for many people, and studies are indirectly starting to suggest why.

Lie down in bed and close your eyes. Keep your eyes closed and move your eyes slowly to the right then to the left, then up, then down, then counterclockwise and then clockwise. Then cross them somewhat by directing them gently toward

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[141] Allison G. Harvey, "The Attempted Suppression of Presleep Cognitive Activity in Insomnia," Cognitive Therapy and Research 27 (2003): 593–602, https://doi.org/10.1023/A:1026322310019.

your nose. Finally, do a big inhale and long exhale. Repeat one to three times as needed.

Told you this would sound a bit out there! This trick often helps people fall asleep, because to do so we need to stop thinking about our limb position, or proprioception. Our visual and vestibular systems, or balance, and proprioception systems interact, and the eye movements just described "confuse" those systems. A similar mechanism partially explains why gentle slow rocking and pacing can help babies, and slow rocking of a bed (yes, such beds exist) can help adults improve their sleep metrics.[142] We need more data to understand the mechanism behind this pattern, which is why it's not its own protocol, but it is reasonable to assume it is related to the eye-to-vestibular circuits involved in, among other things, proprioception and sleep. New eye mask and rocking-bed technologies that trigger this eye movement pattern are designed to cause similar eye movements. I have tried a few of these nascent technologies and confess I fell asleep in less than two minutes! Meanwhile, this technique is free and easy to try. "Next thing I knew it was morning!" is not an uncommon refrain from people with whom I have shared this trick.

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[142] Nami Ohmura et al., "A Method to Soothe and Promote Sleep in Crying Infants Utilizing the Transport Response," Current Biology 32, no. 20 (October 2022): 4521–29.e4, https://doi.org/10.1016/j.cub.2022.08.041; Aurore A. Perrault et al., "Whole-Night Continuous Rocking Entrains Spontaneous Neural Oscillations with Benefits for Sleep and Memory," Current Biology 29, no. 3 (February 2019): 402–11.e3, https://doi.org/10.1016/j.cub.2018.12.028.

5

SLEEP PROTOCOL

Lower Your Core Body Temperature at Night

WHAT TO DO

- Lower the temperature of the room you sleep in at night. Cooler ambient temperatures facilitate sleep. Keep your bedroom between 65 and 68 degrees Fahrenheit, and a maximum of 70 degrees.
- Warm your extremities before bed. A hot shower, hot bath, sauna, or even washing your face with very warm water can help induce drowsiness by warming hands and feet and by triggering a drop in core body temperature in compensation for this heating.
- Make sure your feet and hands can be easily exposed from underneath your covers. The skin of these areas is a primary way we shed heat from our body. Exposing them to cool air is one of the most effective ways to decrease core body temperature. So untuck your sheets at the foot and sides of the bed.
- Avoid wearing socks while you sleep, especially if you tend to wake in the middle of the night. Socks can stop the natural cooling process that happens through the feet. (People who tend to run very cold at night may want to skip this recommendation.)

HOW IT WORKS

Socks or no socks? Window open or closed? Blanket or just sheets? Feet exposed? We all have our sleep habits and a lot of them revolve around temperature. That makes sense. Thermoregulation—the maintenance of our core body temperature—and the management of our sleep environment to control our body temperature is crucial for a great night's sleep, both falling and staying asleep. Indeed, changes in our core body temperature both reflect and influence our circadian rhythm, how much melatonin and other hormones we release, and even the amount of deep and REM sleep we get in a given night.

As I mentioned earlier in this chapter, our core body temperature declines as the day goes on, eventually reaching its lowest point at a specific time—we call this temperature minimum time.[143] A decline in body temperature precedes us falling asleep. The rate of temperature reduction, and not just the absolute core temperature value, seems to matter. In one experiment, researchers made the case that people tended to head off to bed at exactly the time that their temperature was declining fastest.[144] Body temperature

[143] Patricia J. Murphy and Scott S. Campbell, "Nighttime Drop in Body Temperature: A Physiological Trigger for Sleep Onset?," Sleep 20, no. 7 (1997): 505–11, https://doi.org/10.1093/sleep/20.7.505.

[144] Scott S. Campbell and Roger J. Broughton, "Rapid Decline in Body Temperature Before Sleep: Fluffing the Physiological Pillow?," Chronobiology

starts to increase shortly before we wake up. The relationship between core body temperature and our level of autonomic arousal—which is what sleepiness and alertness reflect—is causal but is best thought of as "necessary but not sufficient," meaning we need the change in temperature, but other factors are key too.[145]

Research has found that the ideal ambient temperature while we sleep is between approximately 66 and 70 degrees Fahrenheit, which typically brings the microclimate immediately around our skin down to somewhere between 88 and 95 degrees Fahrenheit.[146] To fall asleep our core body temperature needs to decrease by about 1 to 2 degrees.

A lower core body temperature facilitates various sleep-related biological processes you are now familiar with. First, research indicates that this reduction may help initiate non-REM sleep,[147] and our lowest body temperature is associated with deep sleep. Melatonin, a hormone that

International 11, no. 2 (1994): 126–31, https://doi.org/10.3109/07420529409055899.

[145] Yaqian Fan et al., "How Temperature Influences Sleep," International Journal of Molecular Sciences 23, no. 20 (2022): 12191, https://doi.org/10.3390/ijms232012191; Shreyas S. Joshi et al., "The Importance of Temperature and Thermoregulation for Optimal Human Sleep," Energy and Buildings 131 (November 1, 2016): 153–57, https://doi.org/10.1016/j.enbuild.2016.09.020; Kryger et al., Kryger's Principles and Practices of Sleep Medicine.

[146] Edward C. Harding et al., "The Temperature Dependence of Sleep," Frontiers in Neuroscience 13 (April 23, 2019): 336, https://doi.org/10.3389/fnins.2019.00336.

[147] Harding et al., "Temperature Dependence of Sleep."

facilitates falling asleep, is released in sync with the nightly drop in body temperature. In one small study of twelve women who were entrained to the solar cycle, the secretion of natural melatonin at night occurred with a simultaneous decline in core body temperature.[148] Other studies have observed a similar coupling.[149]

The stillness of our body during early sleep also plays a role in further dropping our core body temperature. In sleep, our autonomic nervous system shifts from more sympathetic activity (or tone, as it's often referred to), which is related to being alert and able to generate movement, to a far more parasympathetic tone, which is associated with calmness, lower heart rate, and less or no bodily movement.[150] All that combined causes our blood to flow more to our periphery, to

[148] A. Cagnacci et al., "Melatonin: A Major Regulator of the Circadian Rhythm of Core Temperature in Humans," Journal of Clinical Endocrinology & Metabolism 75, no. 2 (1992): 447–52, https://doi.org/10.1210/jcem.75.2.1639946.

[149] James K. Wyatt et al., "Circadian Temperature and Melatonin Rhythms, Sleep, and Neurobehavioral Function in Humans Living on a 20-h Day," American Journal of Physiology 277, no. 4 (1999): R1152–63, https://doi.org/10.1152/ajpregu.1999.277.4.r1152; Kurt Kräuchi et al., "Thermoregulatory Effects of Melatonin in Relation to Sleepiness," Chronobiology International 23, no. 1–2 (2006): 475–84, https://doi.org/10.1080/07420520500545854; Stephen Deacon and Josephine Arendt, "Melatonin-Induced Temperature Suppression and Its Acute Phase-Shifting Effects Correlate in a Dose-Dependent Manner in Humans," Brain Research 688, no. 1–2 (1995): 77–85, https://doi.org/10.1016/0006-8993(95)96872-i.

[150] Ronald Szymusiak, "Body Temperature and Sleep," Handbook of Clinical Neurology 156 (2018): 341–51, https://doi.org/10.1016/B978-0-444-63912-7.00020-5.

our hands and feet, where it “dumps heat,” which in turn leads to a lower core body temperature.

This means that warming up our hands and feet a bit before getting into bed could facilitate falling asleep. That may seem paradoxical: Why would warming up your periphery help at a time when cooling down your core body temperature is the goal? Because heating your periphery acts as a trigger. A hot shower or bath, or a sauna, warms the surface of your body. That causes the preoptic area of your hypothalamus—a brain area that works sort of like a thermostat—to trigger a compensatory response, lowering the core temperature in your body.[151] This process makes it easier to fall asleep, a phenomenon known colloquially as the warm bath effect.[152]

Once you get in bed, having your sleeping environment cool is also helpful. Many people sleep best in a cool room but with their whole body under loads of warm blankets. Other people run hot and like to sleep under just a sheet or a light blanket. Try out a few of these recommendations and pick what suits you best.

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[151] Takafumi Maeda et al., “Effects of Bathing-Induced Changes in Body Temperature on Sleep,” Journal of Physiological Anthropology 42, article no. 20 (2023), https://doi.org/10.1186/s40101-023-00337-0.

[152] Harding, “Temperature Dependence of Sleep.”

In My Experience

I prefer a comforter and a weighted blanket, but stick my bare feet out from underneath them, which we know helps cool core body temperature. And I keep the room about 68 degrees Fahrenheit. Closer to morning, I find that I often bring my feet under the blankets (in my sleep!), which we know can increase the amount of time spent in REM in the final few hours before waking.[153] Temperature-controlled mattress covers like Eight Sleep make controlling your bed temperature across the night easy, and I use one. (They are a podcast sponsor, and I am a scientific advisor to Eight Sleep). But with or without Eight Sleep, it is a good idea to use programmable room temperature controls to help create that cooler-at-start and warmer-toward-morning sleep environment. Experiment and find what works for you.

COMMON QUESTIONS

Do daytime temperature changes affect nighttime sleep

Some data from animal studies suggest that a slightly elevated body temperature for a few hours in the afternoon may lead to more time spent in N3 sleep at night, when waste products from the brain and spinal cord are cleared

[153] Jeong-Whun Kim et al., "Polysomnographic Evidence of Enhanced Sleep Quality with Adaptive Thermal Regulation," Healthcare (Basel) 13, no. 19 (October 2025): 2521, https://doi.org/10.3390/healthcare13192521.

away (among other processes).[154] That's one reason why many people find that doing some moderate to intense exercise in the early or midafternoon improves their sleep, provided they don't consume a lot of caffeine before their workout.

So should I avoid exercising at night

It's best to avoid intense exercise late at night, within three to four hours of sleep. High-intensity exercise raises our core body temperature, which is counterproductive to falling asleep. That said, some people can only exercise at this time, and many learn how to down-shift afterward with a hot bath or shower, and by using some of the tools covered in chapter 3, like deliberate long-exhale breathing.

If you consistently find it difficult to fall and stay asleep after exercising hard at night, consider moving your more intense workouts to mornings and doing something lighter in the evening. A review of several randomized clinical trials found that meditative movement like tai chi and qigong improved sleep quality.[155]

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[154] Stephen R. Morairty et al., "Selective Increases in Non-Rapid Eye Movement Sleep Following Whole Body Heating in Rats," Brain Research 617, no. 1 (1993): 10–16, https://doi.org/10.1016/0006-8993(93)90606-n.

[155] Fang Wang et al., "The Effect of Meditative Movement on Sleep Quality: A Systematic Review," Sleep Medicine Reviews 30 (December 2016): 43–52, https://doi.org/10.1016/j.smrv.2015.12.001.

However, if night is the only time of day you can exercise, then it's more important to get the workout in than to skip it just because it's late in the day (more on that in chapter 2). In this case, it may help to take some additional steps for ensuring a solid night of sleep: no caffeine after 1:00 or 2:00 p.m., a nighttime wind-down routine, a hot shower, some breathing exercises to reduce stress, no screens, and dim lighting will all help counteract the energizing effects of late evening exercise.



SLEEP PROTOCOL

Increase Your Core Body Temperature in the Final One to Two Hours of Sleep Before You Wake Up

WHAT TO DO

- Program your heating system for the room you sleep in at night. If you have one that can be set to different times, program it to rise one to three degrees above your room temperature, starting one to two hours before your wake-up time.
- Keep clothing by your bed. If you wake up one to two hours before your preferred wake-up time or in the middle of the night to use the bathroom, try putting on some socks, sweats, or even a knit cap and go back to

sleep until your preferred wake-up time. This can increase your core body temperature, which increases the amount of REM sleep you get and will help you wake up feeling energized.

HOW IT WORKS

We touched on this in the previous protocol but given the importance of getting sufficient REM sleep for mental and physical health, and for performance, it's worth delving into this topic as its own protocol. So here it goes: Just as cooling our core body temperature facilitates falling and staying asleep, warming our core body temperature starting about one to two hours before we wake up can lead us to feel far more refreshed and ready to lean into the day.

Hot temperatures are disruptive to falling and staying asleep.[156] Interestingly the body tends not to adapt; even after several days of trying to sleep well in hot temperatures, sleep remains disrupted.[157] One theory holds that wakefulness itself is triggered by the need for

[156] Kazue Okamoto-Mizuno and Koh Mizuno, "Effects of Thermal Environment on Sleep and Circadian Rhythm," Journal of Physiological Anthropology 31, article no. 14 (2012), https://doi.org/10.1186/1880-6805-31-14.

[157] J. P. Libert et al., "Effect of Continuous Heat Exposure on Sleep Stages in Humans," Sleep 11, no. 2 (1988): 195–209, https://doi.org/10.1093/sleep/11.2.195.

thermoregulation; in other words, waking up is a way to bring our body temperature back to normal.[158]

We have less data on the use of temperature in this part of the sleep cycle than we do for the previous protocol. However, we know that our lowest body temperature, or temperature minimum time, occurs at about two hours before we wake up and begins to rise after that.[159] A return to our normal core body temperature accompanies our return to wakefulness, so it stands to reason that supporting the general process of warming up after we reach our temperature minimum helps us wake up feeling energized.[160] (It likely reflects having had at least two REM-enriched sleep cycles.)

Admittedly, there's a small problem here. It's hard to get out of a warm bed into a cold room. Heating the room you sleep in on a timer that increases the temperature one to two hours prior to waking solves that problem. Otherwise you just have to get up and get to the thermostat. As mentioned before,

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158 Pier Luigi Parmeggiani, "Interaction Between Sleep and Thermoregulation: An Aspect of the Control of Behavioral States," Sleep 10, no. 5 (1987): 426–35, https://doi.org/10.1093/sleep/10.5.426.

159 Okamoto-Mizuno and Mizuno, "Effects of Thermal Environment."

160 Mats Gillberg and Torbjörn Åkerstedt, "Body Temperature and Sleep at Different Times of Day," Sleep 5, no. 4 (1982): 378–88, https://doi.org/10.1093/sleep/5.4.378; Institute of Medicine Committee on Military Nutrition Research, "Physiological Responses to Exercise in the Heat," in Nutritional Needs in Hot Environments: Applications for Military Personnel in Field Operations (Washington, DC: National Academies Press, 1993), https://www.ncbi.nlm.nih.gov/books/NBK236240/.

devices such as Eight Sleep mattress covers that can cool at first and then heat your sleep environment in the final hours of sleep are very effective. These can also measure your deep and REM sleep, and use AI to make real-time adjustments to your sleep temperature to help you get the best possible sleep. But these carry a cost that for many could be prohibitive, and most versions don't travel easily. Anyone can adjust their bedroom temperature and/or use clothes to try to achieve some of the ideal temperature ranges described here.



SLEEP PROTOCOL

Nap (Correctly) to Offset Minor Sleep Deficits

WHAT TO DO

- If you need a nap, make it short. Keep the duration to no more than about twenty minutes, and avoid naps longer than ninety minutes.
- Avoid late-day naps that interfere with your nighttime sleep. Each person can determine what time constitutes "too late" with a bit of experimentation. See how groggy you are in the hours following your nap. Most importantly, note if you can still easily fall asleep at your regular bedtime afterward.

- Know your goal. Nap strategies depend on what you are trying to accomplish. If you need an energy boost to sustain your focus and drive for the rest of the day: Aim for a fifteen-to-twenty-minute nap. Five to ten minutes will give you a mild energy boost and help your focus but it doesn't last long. After a fifteen to twenty minute nap, concentration, mood, and energy will improve in a way that will sustain you for the remainder of the day.
- If you are dealing with heavy emotions, say from a loss or a rejection or an argument: Aim for a longer nap, but no longer than ninety minutes. You'll recall from Sleep Protocol 4 that the REM stage is when emotions are processed, and that stage arrives about sixty to seventy minutes into the sleep cycle. So know you might feel groggy after a nap of forty-five minutes or so because sleep is cut off before the REM cycle is completed. And know too that when people's nighttime sleep is altered due to heavy emotions, and they nap in the day, they often go into REM more quickly; that's the REM rebound effect discussed in that same protocol.
- If you are sleep-deprived for any reason: It is fine to nap for thirty, sixty, or even ninety minutes but avoid longer naps as these can alter your circadian rhythm. It can often "split" your circadian rhythm so that you're asleep from 10:00 p.m. to 2:00 a.m., then awake for a few hours, then asleep again.

- Consider other forms of daytime rest besides sleep to replenish mental and physical vigor. Particularly if you have trouble falling asleep at night, napping may not be ideal. Sleep Protocol 8 describes non-sleep deep rest (NSDR), a highly effective alternative to napping that can help you feel refreshed without grogginess or impairment to your nighttime sleep. It has other benefits as well.

HOW IT WORKS

There are conflicting views on napping among sleep experts.[161] Some say if you are sleepy enough to nap in the daytime, you must be getting less than optimal sleep at night, and you should focus on improving that instead. Others take the stance that napping is fine, beneficial even, and not always reflective of a sleep deficit. Provided you adhere to a few simple constraints, napping is great. But if you don't crave naps, there is no need to nap.

And yes, there is a proper way to nap.

[161] Janna Mantua and Rebecca M. C. Spencer, "Exploring the Nap Paradox: Are Mid-Day Sleep Bouts a Friend or Foe?," Sleep Medicine 37, no. 1 (September 2017): 88–97, https://doi.org/10.1016/j.sleep.2017.01.019; Pengqiang Du et al., "Multiple Health Outcomes of Daytime Napping: A Comprehensive Umbrella Review," Public Health Reviews 47, no. 1 (February 2026): 1609013, https://doi.org/10.3389/phrs.2026.1609013; Yves Dauvilliers and Alain Buguet, "Hypersomnia," Dialogues in Clinical Neuroscience 7, no. 4 (December 2005): 347–56, https://doi.org/10.31887/DCNS.2005.7.4/ydauvilliers.

First off, it's normal to feel a bit sleepy in the afternoon as your body temperature shifts from rising to falling—remember that the rate of core temperature dropping is what initiates the feeling of drowsiness. For people on a conventional schedule of being asleep at night and being awake in the daytime, this dip in early afternoon energy passes. I fall asleep at about 10:30 p.m. most nights and wake up at 6:30 a.m., and I always get a bit sleepy between 2:00 and 3:00 p.m. irrespective of what I eat, how much caffeine I drink, and so on. After 3:30 p.m. or so, I feel wide awake and energetic again. The point is, you might get a bit tired in the afternoon, but the feeling is temporary. Not everyone experiences this dip. Some people just barge right through the day, morning until night.

We know that napping can increase our ability to focus and learn. Matthew Walker, a professor of neuroscience and biomedical engineering at the University of Texas in Dallas, conducted a study in which a group of participants learned a new set of facts at midday.[162] Half of them napped for ninety minutes afterward; half of them relaxed without sleeping. At 5:00 p.m., they learned a second new set of facts. The group that did not nap experienced a decline in learning capacity. The nap group sustained or even improved

[162] Matthew P. Walker, "Sleep, Memory and Emotion," Progress in Brain Research 185 (2010): 49–68, https://doi.org/10.1016/B978-0-444-53702-7.00004-X.

their capacity to learn. The difference reached about 25 percent, which is meaningful.

The finding goes back to the role of sleep spindles, which we covered in Sleep Protocol 4. These are the types of brain waves present during N2 sleep, which happens early in the sleep cycle. Even a brief snooze of twenty minutes takes us into N2.[163] Sleep spindles help us process new learning, but also prepare our brain for more learning after we wake up.

Napping also influences our mood. In another study, Walker and his group asked participants to rate the level of emotion they observed in a photograph.[164] How angry were the angry faces? How happy were the happy faces? Again, half the subjects napped for about an hour and half did not. Toward the end of the day, the non-nappers were rating the emotions as more extreme; angry people were angrier, and so on. The napping group was more likely to rate faces as happy; the rest time seemed to blunt the normal increase in anger sensitivity across the day. As Walker sometimes puts it, sleep gives us a "rose tint" to our worldview. In this study, the naps were ninety minutes, long enough to get some REM sleep, so the benefit is likely due to that, which as we discussed, is when we unload the emotional burdens from

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[163] Patel et al., "Physiology, Sleep Stages."

[164] Ninad Gujar et al., "A Role for REM Sleep in Recalibrating the Sensitivity of the Human Brain to Specific Emotions," Cerebral Cortex 21, no. 1 (2011): 115–23, https://doi.org/10.1093/cercor/bhq064.

our daytime experiences. Although, in this case, it wasn't about the nappers' emotions; it was about how they perceived the emotions of others.

Finally, there is the simple benefit of restoring energy. Many people experience what sleep researchers call a postprandial drop, a dip in energy after eating, especially large meals rich in starchy carbohydrates. Aside from modifying your dietary pattern, naps can help get you to the other side of a postprandial drop if and when you do experience the phenomenon. That said, for anyone who feels tired in the middle of the day and would do better without that sleepiness, I suggest making sure you don't overeat, and possibly even reduce starchy carbohydrate intake, at lunch. Take a short ten-minute walk afterward; that can help for reasons related to blood sugar, or glucose, regulation (covered in chapter 2). Many people find that just reducing the amount of starchy carbs they eat at lunch allows them to go straight through the rest of their day feeling energized and focused. I'll talk more about the relationship between food and brain states in chapter 4.

When Not to Nap (and Why)

A short midday sleep isn't a good idea for everyone. To understand why, we need to return to adenosine. As you recall, adenosine acts as a neurotransmitter that inhibits neural activity in brain regions involved in strategic

planning.[165] Sleep actually results from two processes: the circadian clock and sleep pressure from adenosine. Most of the protocols thus far depend on the former. This one has to do with the latter, which you might remember from Sleep Protocol 4. Adenosine is so vital to this process that we ought to explore it a bit further, so here goes.

ATP, or adenosine triphosphate, is the basic unit of energy for all our cellular processes. It is generated during the early stages of sleep. Adenosine is the downstream product of the burning of ATP.[166] It's a product of metabolism that accrues as we engage in activity over the course of a day. As we move from morning to afternoon to evening, more energy has been spent and therefore more adenosine builds up in our brains, gathering in the space between our cells. As it does, we register a need to sleep. This is, again, called homeostatic sleep drive.

The longer we are awake and the more physical activity and mental activity we engage in over the course of a day, the higher the levels of adenosine we accumulate. On days when we get more exercise, for example, our brains accrue more adenosine. And the more they do, the easier it is to fall and

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[165] Thomas V. Dunwiddie and Susan A. Masino, "The Role and Regulation of Adenosine in the Central Nervous System," Annual Review of Neuroscience 24 (March 2001): 31–55, https://doi.org/10.1146/annurev.neuro.24.1.31.

[166] Theresa E. Bjorness and Robert W. Greene, "Adenosine and Sleep," Current Neuropharmacology 7, no. 3 (2009): 238–45, https://doi.org/10.2174/157015909789152182.

stay asleep. Additionally, greater amounts of adenosine are associated with an increase in deeper slow-wave sleep and a decrease in shallower sleep.

Napping—which is, after all, a form of sleep—reduces the amount of adenosine in our brains.[167] It won't zero it out like a solid eight hours of sleep will, but a forty-five-to-ninety-minute nap will clear away a substantial amount of adenosine. For this reason, people who have trouble falling asleep at night should not nap. You want the buildup of adenosine to create enough sleep pressure so that you easily fall and stay asleep at night.

How to Nap

Research at NASA back in 1995 found that napping for twenty-six minutes maximizes the benefits for alertness, focus, and motivation.[168] These became known as power naps, and are frequently used to improve memory, reduce blood pressure, and increase concentration. I don't believe there is anything magic about twenty-six minutes except that it is less than forty-five minutes and more than ten, so set your alarm for a fifteen-to-twenty-minute nap and perhaps hit the five-minute snooze button if you don't feel refreshed

[167] Dunwiddie and Masino, "Role and Regulation of Adenosine"; Porkka-Heiskanen et al., "Adenosine."

[168] Mark R. Rosekind et al., "Alertness Management: Strategic Naps in Operational Settings," Journal of Sleep Research 4, no. S2 (1995): 62–66, https://doi.org/10.1111/j.1365-2869.1995.tb00229.x.

the first time your alarm goes off. Just don't hit it a second time!

A twenty-minute nap shouldn't make you groggy. That feeling, known as sleep inertia, happens when we wake up in the middle of the deeper stages of non-REM sleep. By keeping the nap to about twenty minutes, you get the benefits of N1 and N2 sleep without suffering from sleep inertia.

Perhaps surprisingly, some people find that resting without sleeping is sufficient to restore mental and physical energy. When you do, you can enter a liminal state in which you are still awake but some parts of your brain are in a sleeplike state. Many people find that even if they just lie down, elevate their feet on a few pillows or a chair, close their eyes, and rest for ten to fifteen minutes, they get some of the restorative effects of a real nap. Elevating your legs and feet slightly in this way promotes a parasympathetic response, increases venous blood flow to the heart, may lower blood pressure slightly, and—provided you don't ruminate on stressful topics—can shift your brain's state toward one of more "relaxation." No, it's not a good idea to spend this time tapping or scrolling on your phone, as it will offset the brain state shift you're seeking. You need to get your mind into a sort of idling mode. Idling is clearly not neuroscientific language. But it's a good way to describe how our brains

shift to a very different pattern of activity under these resting conditions, moving from higher frequency beta waves (13–30 Hz) to lower frequency alpha waves (8–13 Hz). Napping, on the other hand, lets our brain enter N1 and N2 sleep, allowing even lower frequency theta waves (4–8 Hz) to emerge.[169] Experienced meditators can get themselves into a state where theta waves emerge quickly, although that takes practice. Relaxation techniques like those offered in the very next protocol can help you rest during the day without interfering with your ability to fall asleep at night.

COMMON QUESTIONS

I'm not a napper but I don't have trouble falling asleep at night. Should I nap

If you are not tired in the daytime, don't force yourself to nap. Some people get through a day just fine without naps.

Can naps help with insomnia

[169] Erin J. Wamsley, "Memory Consolidation During Waking Rest," Trends in Cognitive Sciences 23, no. 3 (March 2019):171–73, https://doi.org/10.1016/j.tics.2018.12.007; Olivia King and Jessica Nicosia, "The Effects of Wakeful Rest on Memory Consolidation in an Online Study," Frontiers in Psychology 13 (October 2022): 932592, https://doi.org/10.3389/fpsyg.2022.932592; Anika T. Löwe et al., "N2 Sleep Promotes the Occurrence of 'Aha' Moments in a Perceptual Insight Task," PLOS Biology 23, no. 6 (2025): e3003185, https://doi.org/10.1371/journal.pbio.3003185.

People with insomnia should generally avoid naps.[170] The temptation to sleep during the day can be powerful when coping with poor sleep at night. But all evidence points to the idea that resisting the urge to nap is the way to go for insomniacs. If you are suffering from chronic insomnia, try the protocols in this book, and if those don't solve the issue then talk to a doctor who specializes in sleep medicine.

For older adults, is there an added health benefit to napping

Studies of napping among older adults have shown mixed results.[171] A concern with prolonged midday rests among older adults is that the need to nap may be indicative of a problem with nighttime sleep. Older adults who always feel tired during the day should consider changes that can benefit sleep, such as making sure to get plenty of natural daylight in their eyes in the morning and dimming indoor lighting at night before relying on naps (Sleep Protocols 1–6 all have useful tools for this situation).

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[170] Kryger et al., Kryger's Principles.

[171] Maher Souabni et al., "Benefits and Risks of Napping in Older Adults: A Systematic Review," Frontiers in Aging Neuroscience 14 (October 2022): 1000707, https://doi.org/10.3389/fnagi.2022.1000707.

8

SLEEP PROTOCOL

Use Non-Sleep Deep Rest (NSDR) to Offset Sleep Deficits (and to Get Better at Falling and Staying Asleep at Night)

WHAT TO DO

- Do non-sleep deep rest (NSDR) if you feel tired and need to restore your mental and physical vigor. NSDR is fast, safe, effective, and costs nothing. It has both immediate and long-term benefits for your brain state, and there is evidence it can improve focus and learning. It takes as little as ten minutes, although longer versions of twenty, thirty, or even sixty minutes can be even more effective. It's nearly impossible to do NSDR incorrectly (unless you are texting in the middle of a session). Do NSDR first thing in the morning if you failed to get enough sleep the night before and want to feel more rested as you enter your day. If you don't have time just then, I suggest you do NSDR in the midmorning or afternoon. It will still help.
- Do NSDR at night if you have trouble falling asleep, or if you wake up in the middle of the night and can't fall back asleep. NSDR is a far better option than tossing and turning and worrying about the sleep you're not getting.

- There is a link below that will take you to a list of NSDR recordings available at zero cost in downloadable formats (no sign-up required) done by me, and some by Kelly Boys—an expert in NSDR—if you prefer a female voice.

https://www.hubermanlab.com/nsdr

https://www.youtube.com/@kellyboys5313

https://www.kellyboys.org

HOW IT WORKS

First off, it's important to emphasize that NSDR is not anything like traditional meditation. NSDR is not breathwork either. Rather, NSDR belongs to a category of techniques that can be broadly thought of as self-directed relaxation. NSDR is also not anything like a nap. A central feature of NSDR is to stay awake (but not focused on planning or action), while simultaneously bringing the body into an extremely relaxed state. It puts you in a unique brain state.[172] Meditation often involves holding one's attention on a specific location inside or near one's body (such as the spot just behind our forehead), or on an external object (a candle flame, for example), or even on a sensory experience (e.g., the taste of a single almond). These kinds of directed perceptual focus exercises have been shown to improve

[172] Omar Boukhris et al., "The Acute Effects of Nonsleep Deep Rest on Perceptual Responses, Physical, and Cognitive Performance in Physically Active Participants," Applied Psychology: Health and Well-Being 16, no. 4

concentration, reduce stress, strengthen memory and improve mental health.[173] When done consistently, meditation techniques can lead to profound shifts in our awareness of our internal thought patterns. And over time, meditation can provide a rich understanding of how our minds work and can even change our temperament.

Self-directed relaxation protocols like NSDR are different. NSDR involves shifting our autonomic nervous system into a more parasympathetic, or relaxed, state by breathing in a manner in which the duration of our exhales is deliberately

(November 2024): 1967–87, https://doi.org/10.1111/aphw.12571; Shashank Ghai et al., "Effects of Yoga Nidra on Stress, Anxiety, and Depression: A Systematic Review and Meta-Analysis," Annals of the New York Academy of Sciences 1556, no. 1 (February 2026): e70149, https://doi.org/10.1111/nyas.70149; V. Kachera et al., "The Effects of Yoga Nidra Practice on EEG: A Systematic Review," medRxiv (June 2025), https://doi.org/10.1101/2025.06.16.25329676; Karuna Datta et al., "Improved Sleep, Cognitive Processing and Enhanced Learning and Memory Task Accuracy with Yoga Nidra Practice in Novices," PLOS One 18, no. 12 (December 2023): e0294678, https://doi.org/10.1371/journal.pone.0294678.

[173] Julia C. Basso et al., "Brief, Daily Meditation Enhances Attention, Memory, Mood, and Emotional Regulation in Non-Experienced Meditators," Behavioural Brain Research 356 (January 2019): 208–20, https://doi.org/10.1016/j.bbr.2018.08.023; Daniel Goleman and Richard J. Davidson, Altered Traits: Science Reveals How Meditation Changes Your Mind, Brain, and Body (Avery, 2017); Simon B. Goldberg et al., "Sitting and Active Meditation Practice: Utilization and Associations with Outcomes in Naturalistic and Clinical Trial Data," Mindfulness 17 (2026): 1120–37, https://doi.org/10.1007/s12671-026-02787-w; Simon B. Goldberg et al., "Mindfulness-Based Interventions for Psychiatric Disorders: A Systematic Review and Meta-Analysis," Clinical Psychology Review 59 (February 2018): 52–60, https://doi.org/10.1016/j.cpr.2017.10.011.

emphasized.[174] NSDR also involves deliberately and systematically moving our focus of attention from one part of our body to another, then to locations far from our body, then closer to us, then away again. The focusing, then defocusing, and moving and resetting of attention over and over keeps the mind active. And because it is done while the body is deeply relaxed, it brings about a somewhat dreamlike state, and yet you are completely aware and alert.

NSDR is a modification of an ancient practice known as yoga nidra, which roughly translates to "yoga sleep."[175] Yoga nidra has been described as creating the experience of moving our mind away "from thinking and doing toward a state of being and feeling"—which is an admittedly vague statement. There is a small but growing body of modern science on the brain and bodily states that yoga nidra and NSDR bring about and their ability to restore mental and physical vigor, improve consolidation of newly learned information, offset sleep loss, and more.

When I first tried yoga nidra in 2017, I found it to be incredibly restorative. Doing this technique for twenty to

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[174] Camila Ferreira-Vorkapic et al., "The Impact of Yoga Nidra and Seated Meditation on the Mental Health of College Professors," International Journal of Yoga 11, no. 3 (2018): 215–23, https://doi.org/10.4103/ijoy.IJOY_57_17; Seithikurippu R. Pandi-Perumal et al., "The Origin and Clinical Relevance of Yoga Nidra," Sleep and Vigilance 6, no. 1 (2022): 61–84, https://doi.org/10.1007/s41782-022-00202-7.

[175] Pandi-Perumal et al., "Origin and Clinical Relevance."

thirty minutes first thing in the morning after a restless or less-than-full night's sleep left me feeling fully rested, like I'd had the best sleep of my life. I also noticed that incorporating this practice in the morning improved my ability to fall back asleep quickly if I happened to wake up in the middle of the night. That's right—even though NSDR stands for non-sleep deep rest, and even though it is typically done in the morning or afternoon, something about doing it at that time significantly improved my ability to self-direct my brain and body into a relaxed state and drift back to sleep at night if ever I needed to. What's more, if I ever woke up in the middle of the night, listening to a guided NSDR recording beat tossing and turning, and often sent me back to sleep. Challenges with falling asleep and staying asleep are very common, so I was intrigued. I also noticed that replacing an afternoon nap with an NSDR session left me feeling especially reinvigorated and did not alter my nighttime sleep in any negative way.

I wondered how this simple practice really worked. So I dove into the literature and found there wasn't much known. But the few studies I did find were interesting. A 2002 study from Denmark found that doing yoga nidra dramatically increases levels of dopamine in a brain region called the basal ganglia, which are involved in taking, and withholding,

action.[176] Yoga nidra also reduced people's blood pressure and heart rate. Although this and other studies of yoga nidra were small in terms of the number of subjects studied, other investigations indicated that this practice could improve various measures of mental health and help consolidate certain forms of learning.[177]

At one point, I decided to devote a portion of my lab to studying yoga nidra as a tool for stress reduction. To do so we needed to keep its core elements—breathing that emphasizes long exhales to slow the heart rate and place us into a more parasympathetic state, the deliberate and systematic movement of one's perceptual attention to the surface and then the interior of the body, then away from the body, and so on—but we opted to remove the "intentions" and any language pertaining to more spiritual aspects of

[176] Troels W. Kjaer et al., "Increased Dopamine Tone During Meditation-Induced Change of Consciousness," Cognitive Brain Research 13, no. 2 (2002): 255–59, https://doi.org/10.1016/s0926-6410(01)00106-9.

[177] Ferreira-Vorkapic et al., "Impact of Yoga Nidra"; Heather Eastman-Mueller et al., "iRest Yoga-Nidra on the College Campus: Changes in Stress, Depression, Worry, and Mindfulness," International Journal of Yoga Therapy 23, no. 2 (2013): 15–24, https://doi.org/10.17761/ijyt.23.2.r8735770101m8277; Roberta Anderson et al., "Using Yoga Nidra to Improve Stress in Psychiatric Nurses in a Pilot Study," Journal of Integrative and Complementary Medicine 23, no. 6 (2017): 494–95, https://doi.org/10.1089/acm.2017.0046; Khushboo Nayak and Kedarmal Verma, "Yoga-Nidra as a Mental Health Booster: A Narrative Review," Journal of Ayurveda and Integrative Medicine 14, no. 6 (2023): 100842, https://doi.org/10.1016/j.jaim.2023.100842; Datta et al., "Improved Sleep, Cognitive Processing."

yoga nidra.[178] This parsing was not an attempt at cultural appropriation of an ancient tradition. Rather, it was a way to examine this powerful tool within the context of Western science, where we seek to understand what works and why, and to then use this knowledge to build more efficient and effective tools. Any aspect of the practice that couldn't be reliably measured was discarded.

Later, as I moved into public education and spoke about stress mitigation tools, I coined the term non-sleep deep rest, which keeps the key elements related to the physiology of yoga nidra, but removes the subjective aspects that are hard to keep constant from one person to the next.

And eventually, given the wide interest in stress mitigation tools, my podcast team produced various NSDR scripts of short (ten minute), medium (twenty minute), and long (thirty minute) durations and made them freely available online.[179]

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[178] Melis Yilmaz Balban et al., "Brief Structured Respiration Practices Enhance Mood and Reduce Physiological Arousal," Cell Reports Medicine 4, no. 1 (2023): 100895, https://doi.org/10.1016/j.xcrm.2022.100895; Andrea Zaccaro et al., "How Breath-Control Can Change Your Life: A Systematic Review on Psycho-Physiological Correlates of Slow Breathing," Frontiers in Human Neuroscience 12 (September 2018): 353, https://doi.org/10.3389/fnhum.2018.00353; Melis Yilmaz Balban et al., "Human Responses to Visually Evoked Threat," Current Biology 31, no. 3 (February 2021): 601–12, https://doi.org/10.1016/j.cub.2020.11.035.

[179] "#NSDR (Non-Sleep Deep Rest) with Dr. Andrew Huberman," Virtusan, YouTube, September 11, 2022, https://www.youtube.com/watch?v=AKGrmY8OSHM; "20 Minute Non-Sleep Deep Rest (NSDR) to Restore Mental & Physical Energy | Dr. Andrew Huberman," Huberman Lab Clips, YouTube, April 26, 2024, https://www.youtube.com/watch?v=hEypv90GzDE.

Public feedback has confirmed the positive effects of NSDR, but of course it just follows the thousands of years of people across the world doing this time-tested practice. (Later in chapter 3, I'll get back to what my lab's studies on stress discovered about the benefits of NSDR).

Among all the protocols that I have shared on the Huberman Lab podcast, NSDR is in the top ten of those that people tell me have changed their lives, especially individuals who struggle to sleep enough, to maintain high levels of daytime energy and focus, and also among those struggling with excessive work, school, or other demands. I am the last person in the world to suggest people sleep less than they need to at night, but I will be the first to acknowledge how challenging it can often be to get a full seven to nine hours of sleep, especially for parents of young children and those working very long hours. NSDR is a remarkable tool for anyone who simply can't get enough sleep because they have to wake up and meet the demands of the day. Or for those who don't like naps or who can't nap for whatever reason. We will revisit NSDR several times in this book, but in the meantime, if you're seeking tools for improving sleep, or even replacing missed sleep, I highly suggest doing NSDR.

COMMON QUESTIONS

What if I fall asleep during NSDR

That's just fine. Chances are, if done during the daytime, the sleep will be relatively shallow and won't create much, if any, sleep inertia. On many occasions, I have fallen asleep during a morning or afternoon NSDR session and then woken up at the last instruction to move my limbs, which meant I wasn't in a deep sleep. If you are using NSDR to reset during the day, or first thing in the morning to offset the effects of not getting enough sleep, just be sure you set an alarm in case you fall asleep.

How long should my NSDR session be

That's up to you. A ten-minute NSDR is often enough for a quick afternoon reset, but I suggest doing a twenty- to thirty-minute NSDR if you have time. If you slept poorly the night before, you may find that a longer session in the morning or afternoon leaves you feeling more rested, compensating for the sleep you missed in the night. Find what works for you.

Can NSDR truly replace sleep or reduce the amount of time I need to sleep

Claims that "thirty minutes of yoga nidra is worth two hours of sleep" abound. I've also heard that "twenty minutes of meditation twice daily reduces your nighttime sleep needs by

several hours.” I am not aware of much laboratory data supporting those statements. It can be hard to get enough sleep, and most of us would love to need less of it to feel fully rested. NSDR won’t necessarily reduce the amount of nighttime sleep you need, but it may very well improve the quality of the sleep you get. Combined with proper placement of your nighttime sleep in the twenty-four-hour cycle (see Sleep Protocol 4 for more), and other protocols in this chapter, it may, however, reduce your total sleep needs by improving your sleep architecture, as we discussed earlier in this chapter.

In one study, researchers randomized forty-one people with chronic insomnia to participate in either cognitive behavioral therapy for insomnia or twice-daily yoga nidra for twenty minutes per session.[180] In both groups, participants slept a few minutes longer during the study than at the start of it. However, the yoga nidra group fell asleep faster, fell back asleep faster after night wakings, and had a more efficient sleep (an increase in the number of times they went through N1, N2, N3, and REM) compared to their sleep prior to the study. So while they weren’t sleeping less, the sleep they were getting was better than before.

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[180] Karuna Datta et al., “Yoga Nidra Practice Shows Improvement in Sleep in Patients with Chronic Insomnia: A Randomized Controlled Trial,” National Medical Journal of India 34, no. 3 (2021): 143–50, https://doi.org/10.25259/NMJI_63_19.

The study also found a significant reduction in cortisol following yoga nidra.[181] That may be one factor that reduces the total amount of sleep you need because, in theory, lowering your evening cortisol should improve sleep. If you practice relaxation techniques during the day and happen to need less sleep that night, that's an added bonus.[182]



SLEEP PROTOCOL

Consider Supplements to Support Sleep

WHAT TO DO

- Find a combination of supplements that support you in getting the best possible sleep. Learn a bit about how each of these works. Consider your specific needs and consult with your physician to ensure that any supplements you take are safe for you and will not interfere with any medications you are currently taking. As stated in the introduction, supplements can have their place but ideally, try behavioral tools first. There is simply no pill to replace poor sleep habits, or "sleep hygiene." Below is a list of common supplements sold over the counter that can assist in getting the best possible

[181] Prashant Kaul et al., "Meditation Acutely Improves Psychomotor Vigilance, and May Decrease Sleep Need," Behavioral and Brain Functions 6, article no. 47 (July 2010), https://doi.org/10.1186/1744-9081-6-47.

[182] Boukhris et al., "Acute Effects of Nonsleep Deep Rest."

night's sleep: Magnesium L-threonate or magnesium bisglycinate.Dosage: 145–250 mg These two forms of magnesium are interchangeable for improving sleep.

- Apigenin, a derivative of chamomile. Dosage: 50 mg
- Theanine. Dosage: 100–200 mg
- GABA. Dosage: 250–500 mg
- Glycine. Dosage: 1–2 g (1,000–2,000 mg)
- Myoinositol. Dosage: 900 mg
- Kiwi extract. Dosage: 500–1,000 mg
- Tart cherry extract. Dosage: 500 mg
- Saffron. Dosage: 30 mg
- Valerian. Dosage: 50–75 mg.
- Melatonin. Dosage: 0.5–1.5 mg of rapid release; 0.5–1.5 mg of sustained release (Note the decimal points here; these are not typos. These are MUCH lower doses of melatonin than you commonly see recommended.)
- Build a sleep stack. The best approach is to experiment with one compound at a time in order to become familiar with the effects of each individually. Start with the lowest amount listed and adjust the dosage if necessary until you find the lowest dose that works for you. The dosages provided here are averages. All the compounds are considered safe for healthy individuals but results vary,

so I suggest taking one at a time for a few nights each and then adding an additional supplement. That will be the most cost-effective and safest approach, and it will help you understand what works for you. Th/at said, if you want to start with a small stack of two to three sleep supplements right out of the gate, here is one that has worked for a lot of people. Take it about thirty minutes before going to sleep. 145–250 mg magnesium L-threonate (or magnesium bisglycinate)

- 50 mg apigenin
- 100–200 mg theanine. (Don’t take theanine if you tend to have intense dreams or nightmares, or if you sleepwalk, as it can enhance the vividness of your dreams.)
- Pay attention to potential side effects. The supplements listed above are generally very safe but not all of them will sit well for all people. Here’s what to watch for: Gut discomfort: A small percentage of people (about 5 percent in my experience) who take magnesium threonate as a sleep aid experience digestive issues or other gut discomfort. Magnesium L-threonate and bisglycinate are far less likely to do this than other forms of magnesium like citrate or malate, et cetera. Still, discontinue use of magnesium L-threonate or bisglycinate if gut discomfort persists. The dosage listed above for magnesium is

already pretty low so you don't have a much lower limit to work with.

- Vivid or disturbing dreams: A small percentage (about 5 percent) of people who take theanine report having vivid dreams that wake them at night.
- Proceed with caution. Again, it is important to consult with a physician before you take supplements. You need to be sure that you are not risking a dangerous interaction, and that a medical professional has information about your sleep issues that may be symptoms of an underlying health problem. And always check that any product you decide to use is reliable.

HOW IT WORKS

When it comes to getting optimal sleep, it's always best to start with behavioral tools and changes to your sleep environment. (Sun)light exposure, adjusting your core body temperature, and NSDR can all be employed to improve sleep. So, too, can exercise and proper nutrition (more on both later). These carry no cost, cause no side effects, and quite often resolve persistent issues.

Many people turn to supplements to help improve their sleep. Parsing the scientific literature on the efficacy of over-the-counter, supplement-based sleep aids can be difficult. The studies are often small, they have conflicting

results, and they are too few to provide a significant body of evidence.[183] That being said, enough data exist to indicate that some supplements may be beneficial and carry a very low risk of harm for most people trying them. In my opinion, these are much safer than the majority of prescription sleep medications.

A major issue when it comes to supplements is that the dose of the pills inside the bottle often does not match the dosage stated on the label.[184] In addition, just because a supplement contains a "natural" ingredient does not mean it is safe. And these ingredients may react badly with a prescribed medication. So a reminder here, again, to talk to your doctor before proceeding, and to only buy supplements from reputable brands. I discuss a few things to look for on the packaging to ensure this below.

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[183] Ji Won Yeom and Chul-Hyun Cho, "Herbal and Natural Supplements for Improving Sleep: A Literature Review," Psychiatry Investigation 21, no. 8 (2024): 810–21, https://doi.org/10.30773/pi.2024.0121; Monica Kazlausky Esquivel and Brittany Ghosn, "Current Evidence on Common Dietary Supplements for Sleep Quality," American Journal of Lifestyle Medicine 18, no. 3 (2024): 323–27, https://doi.org/10.1177/15598276241227915; Vicky Chan and Kenneth Lo, "Efficacy of Dietary Supplements on Improving Sleep Quality: A Systematic Review and Meta-Analysis," Postgraduate Medical Journal 98, no. 1158 (2022): 285–93, https://doi.org/10.1136/postgradmedj-2020-139319.

[184] Victor Navarro et al., "The Contents of Herbal and Dietary Supplements Implicated in Liver Injury in the United States Are Frequently Mislabeled," Hepatology Communications 3, no. 6 (June 2019): 792–94, https://doi.org/10.1002/hep4.1346; Pieter A. Cohen et al., "Quantity of Melatonin and CBD in Melatonin Gummies Sold in the US," JAMA 329, no. 16 (April 2023): 1401–2, https://doi.org/10.1001/jama.2023.2296.

- Magnesium L-threonate: Increasing brain magnesium with this compound may improve learning and memory.[185] A randomized study of magnesium L-threonate versus placebo found that this supplement improved the quality of sleep and also cognitive performance during the day.[186] There are more data on magnesium bisglycinate than threonate for sleep but these seem to work interchangeably for this purpose.
- Apigenin: This compound is among the most widely studied flavonoids (chemicals found in plants) and is present in chamomile, parsley, oranges, celery, and thyme, among herbs and vegetables.[187] A growing body of evidence supports apigenin as a sleep aid.[188] Animal studies have found a mild sedative effect from

[185] Inna Slutsky et al., "Enhancement of Learning and Memory by Elevating Brain Magnesium," Neuron 65, no. 2 (2010): 165–77, https://doi.org/10.1016/j.neuron.2009.12.026.

[186] Heather A. Hausenblas et al., "Magnesium-L-Threonate Improves Sleep Quality and Daytime Functioning in Adults with Self-Reported Sleep Problems: A Randomized Controlled Trial," Sleep Medicine: X 8 (December 15, 2024): 100121, https://doi.org/10.1016/j.sleepx.2024.100121.

[187] A. N. Panche et al., "Flavonoids: An Overview," Journal of Nutritional Science 5 (December 29, 2016): e47, https://doi.org/10.1017/jns.2016.41; A. N. Panche et al., "Flavonoids: An Overview—CORRIGENDUM in Figure 1," Journal of Nutritional Science 14 (January 29, 2025): e11, https://doi.org/10.1017/jns.2024.73; Bahare Salehi et al., "The Therapeutic Potential of Apigenin," International Journal of Molecular Sciences 20, no. 6 (2019): 1305, https://doi.org/10.3390/ijms20061305.

[188] Daniel J. Kramer and Adiv A. Johnson, "Apigenin: A Natural Molecule at the Intersection of Sleep and Aging," Frontiers in Nutrition 11 (February 26, 2024): 1359176, https://doi.org/10.3389/fnut.2024.1359176.

apigenin.[189] Studies in humans also indicate a sleep benefit. In one investigation, apigenin mildly improved daytime function among people with insomnia.[190] Among a group of new mothers suffering from poor sleep and depression, apigenin was found to improve both.

- Theanine: Evidence suggests that theanine increases levels of GABA, an amino acid that, when broadly increased in the brain, may reduce anxiety.[191]
- Glycine: Research suggests glycine works by elevating serotonin, which helps regulate sleep, and activating NMDA receptors in the SCN, where the circadian clock is headquartered.[192]

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[189] P. Zanoli et al., "Behavioral Characterisation of the Flavonoids Apigenin and Chrysin," Fitoterapia 71, suppl. 1 (2000): S117–23, https://doi.org/10.1016/s0367-326x(00)00186-6; Nicholas K. Chow et al., "Telemetry as a Tool to Measure Sedative Effects of a Valerian Root Extract and Its Single Constituents in Mice," Planta Medica 77, no. 8 (2011): 795–803, https://doi.org/10.1055/s-0030-1250589.

[190] Suzanna M. Zick et al., "Preliminary Examination of the Efficacy and Safety of a Standardized Chamomile Extract for Chronic Primary Insomnia: A Randomized Placebo-Controlled Pilot Study," BMC Complementary and Alternative Medicine 11, article no. 78 (2011), https://doi.org/10.1186/1472-6882-11-78; Pandi-Perumal et al., "Origin and Clinical Relevance."

[191] Rebeka Olga Mátyus et al., "Promising, but Not Completely Conclusive—The Effect of L-Theanine on Cognitive Performance Based on the Systematic Review and Meta-Analysis of Randomized Placebo-Controlled Clinical Trials," Journal of Clinical Medicine 14, no. 21 (October 2025): 7710, https://doi.org/10.3390/jcm14217710.

[192] Chiara M. Portas et al., "Serotonin and the Sleep/Wake Cycle: Special Emphasis on Microdialysis Studies," Progress in Neurobiology 60, no. 1 (2000): 13–35, https://doi.org/10.1016/s0301-0082(98)00097-5; Nobuhiro Kawai et al., "The Sleep-Promoting and Hypothermic Effects of Glycine Are

- GABA: As noted earlier, GABA is an amino acid associated with reducing brain activity. It promotes relaxation and is therefore conducive to sleep.[193]
- Melatonin: As discussed in Sleep Protocol 3, we produce natural melatonin, a hormone that causes us to feel sleepy. This production surges at night, leading to feelings of sleepiness. This hormone works well all on its own. However, melatonin supplements have been a staple of vitamin stores for years. Clinical trials and meta-analyses of multiple studies indicate that melatonin supplements may benefit older adults, who experience a lower level of natural melatonin production.[194] It may also be beneficial in combination with morning light

Mediated by NMDA Receptors in the Suprachiasmatic Nucleus," Neuropsychopharmacology 40, no. 6 (2015): 1405–16, https://doi.org/10.1038/npp.2014.326.

[193] Claude Gottesmann, "GABA Mechanisms and Sleep," Neuroscience 111, no. 2 (2002): 231–39, https://doi.org/10.1016/s0306-4522(02)00034-9.

[194] Rebecca B. Costello et al., "The Effectiveness of Melatonin for Promoting Healthy Sleep: A Rapid Evidence Assessment of the Literature," Nutrition Journal 13, article no. 106 (November 7, 2014), https://doi.org/10.1186/1475-2891-13-106; Eduardo Ferracioli-Oda et al., "Meta-Analysis: Melatonin for the Treatment of Primary Sleep Disorders," PLOS One 8, no. 5 (2013): e63773, https://doi.org/10.1371/journal.pone.0063773; Lucreția Anghel et al., "Benefits and Adverse Events of Melatonin Use in the Elderly (Review)," Experimental and Therapeutic Medicine 23, no. 3 (2022): 219, https://doi.org/10.3892/etm.2022.11142; Daniel P. Cardinali et al., "Melatonin's Benefits and Risks as a Therapy for Sleep Disturbances in the Elderly: Current Insights," Nature and Science of Sleep 14 (October 2022): 1843–55, https://doi.org/10.2147/NSS.S380465.

exposure for people with Alzheimer's disease,[195] which tends to disrupt sleep patterns. However, there is no clear evidence of benefit for all other populations. Once again, be aware of the dosages listed here. The amount of melatonin contained in the typical single, over-the-counter pill is far greater than what our bodies produce. I often see dosages of 3 milligrams, 5 milligrams, or even 8 to 10 milligrams of melatonin, which are incredibly high and not advised. That's a problem, because melatonin does more than make us sleepy; it can also interact with other hormones and biological pathways in the brain and body unrelated to sleep. Therefore, taking melatonin chronically can be problematic, especially in children and adolescents who are still growing. If you decide to take melatonin, make sure that you are choosing a reputable brand with third-party certification and use the minimal effective dose. I would suggest taking no more than the amounts listed above (0.5–1.5 mg of rapid or sustained release, although it's fine, and useful, to take both).

- Tart cherry extract: A small, randomized, double-blind, placebo-controlled study found elevated natural melatonin levels and improved sleep quality among ten

[195] Glenna A. Dowling et al., "Melatonin and Bright-Light Treatment for Rest-Activity Disruption in Institutionalized Patients with Alzheimer's Disease," Journal of the American Geriatrics Society 56, no. 2 (2008): 239–46, https://doi.org/10.1111/j.1532-5415.2007.01543.x.

people who drank tart cherry juice for seven days.[196] This preliminary evidence suggests a potential benefit, with larger and longer-term studies needed. Tart cherry juice is safe to try as a sleep aid taken shortly before bedtime. Tart cherry extract capsules are preferable if you don't want the added calories from the juice.

- Kiwi extract: A small study of twenty-five people who self-reported their sleep patterns concluded that kiwi may modestly improve sleep duration and efficiency.[197] Consuming kiwi fruit with the skin on an hour before bed may be a mild sleep aid. More evidence is needed to be certain of this benefit, but kiwi is safe to try. Kiwi extract is a good noncaloric, capsule-based approach to get the same effect. (And you don't have to eat kiwi skins!)

COMMON QUESTIONS

When should I take supplements for sleep

It's best to take any sleep aid supplements about thirty minutes before you intend to be asleep. Magnesium supplementation during the daytime can be useful for other reasons, but if you're taking it specifically for sleep, best to

[196] Glyn Howatson et al., "Effect of Tart Cherry Juice (Prunus cerasus) on Melatonin Levels and Enhanced Sleep Quality," European Journal of Nutrition 51, no. 8 (2012): 909–16, https://doi.org/10.1007/s00394-011-0263-7.

[197] Hsiao-Han Lin et al., "Effect of Kiwifruit Consumption on Sleep Quality in Adults with Sleep Problems," Asia Pacific Journal of Clinical Nutrition 20, no. 2 (2011): 169–74, https://pubmed.ncbi.nlm.nih.gov/21669584/.

take it before bed. The one exception is theanine, which some people like to take when they drink caffeine, in order to offset the jitters that they experience from caffeine. In fact, that is one reason energy drink manufacturers often include theanine in energy drinks.

I take AGZ, a powdered sleep supplement that you mix with water and drink. (Note: I helped design AGZ.) It contains many of the compounds listed above. It remains the best sleep supplement I have ever taken, and yes, I am incentivized to say that, but I am measuring my sleep stats most nights. I am able to take half a packet (one full packet is the recommended dose). It has outperformed all the other sleep pill/powder combinations I have tried over the years and even the “sleep peptide” I have tried called pinealon. There are, of course, other great sleep supplements with multiple compounds out there too. Compare and contrast ingredient lists and costs, and once again: Only by trying individual single-ingredient supplements can you really determine what does or doesn’t work for you.

What brand of supplements should I use

Always look for third-party testing. The NSF (founded as the National Sanitation Foundation) or US Pharmacopeia (USP)

offer rigorous testing for supplements.[198] "NSF Certified for Sport" indicates that a product met the review standards of the NSF.

10

SLEEP PROTOCOL

Approach Prescription Sleep Drugs with Caution

WHAT TO DO

- Talk to your doctor. If you're still struggling with sleep after working through the prior sleep protocols, and you have tried the stress-management protocols, too, talk to your doctor. There may be an underlying health condition that needs addressing. Or it may be that your sleep issues warrant more intensive treatment. Cognitive behavioral therapy has shown efficacy for sustained sleep disturbance.
- Stay away from prescription sleep aids if you can. These are best avoided, unless your physician prescribes them as a last resort. Most can carry side effects, like confusion and daytime drowsiness. Some, like Ambien, may harm

198 "Supplement and Vitamin Certification," NSF, n.d., https://www.nsf.org/consumer-resources/articles/supplement-vitamin-certification; "Dietary Supplement Verification Program," United States Pharmacopeia (USP), n.d., https://www.usp.org/verification-services/dietary-supplements-verification-program.

sleep architecture and glymphatic function.[199] That means even if you have more time in bed, sleep might not be doing the "cleaning" job it's designed to do. Some, like trazodone, and the DORAs (dual orexin receptor antagonists), appear safer than the others, yet it is easy to become dependent on them for sleep once you begin. None are without side effects, and withdrawal then carries its own set of side effects.

HOW IT WORKS

Below I've described the different and more common classes of sleep aids, so you can understand how each works, as well as the potential risks:

- Benzodiazepines: These drugs belong to a class often referred to as hypnotics or sedatives, although they have other effects too. They are often prescribed to treat insomnia. They work by increasing the effects of GABA.[200] They are older medications that can be addictive, and can have other serious side effects including confusion, memory problems, unsteadiness,

[199] Natalie L. Hauglund et al., "Norepinephrine-Mediated Slow Vasomotion Drives Glymphatic Clearance During Sleep," Cell 188, no. 3 (February 2025): 606–22, https://doi.org/10.1016/j.cell.2024.11.027.

[200] Alan N. Bateson, "The Benzodiazepine Site of the GABAA Receptor: An Old Target with New Potential?," Sleep Medicine 5, suppl. 1 (2004): S9–15, https://doi.org/10.1016/s1389-9457(04)90002-0.

and slurred speech.[201] The withdrawal from benzodiazepines is notoriously hard.

- Non-benzodiazepine hypnotics: These short-term insomnia medications tend to be less disruptive than benzodiazepines.[202] The most well-known is Ambien. However, the FDA has issued a "black box" warning for non-benzodiazepines following reports of rare but dangerous behavior during sleep, including wandering, driving, and using a stove, and amnesia and other forms of memory alterations after the drug has worn off.[203] And again, Ambien may harm sleep architecture and glymphatic function, the clearance of waste products from the brain at night.[204]
- Antidepressants: Drowsiness is a common side effect of certain antidepressants, leading to their use as a sleep aid. The American Academy of Sleep Medicine has

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[201] Kalyanakrishnan Ramakrishnan and Dewey C. Scheid, "Treatment Options for Insomnia," American Family Physician 76, no. 4 (2007): 517–26, https://pubmed.ncbi.nlm.nih.gov/17853625/; Connor G. Bounds and Preeti Patel, "Benzodiazepines," StatPearls, last updated January 30, 2024, https://www.ncbi.nlm.nih.gov/books/NBK470159/.

[202] Judy Wagner and Mary L. Wagner, "Non-Benzodiazepines for the Treatment of Insomnia," Sleep Medicine Reviews 4, no. 6 (2000): 551–81, https://doi.org/10.1053/smrv.2000.0126.

[203] "FDA Requires Stronger Warnings About Rare but Serious Incidents Related to Certain Prescription Insomnia Medicines," US Food & Drug Administration, April 30, 2019, https://www.prnewswire.com/news-releases/fda-requires-stronger-warnings-about-rare-but-serious-incidents-related-to-certain-prescription-insomnia-medicines-300840960.html.

[204] Hauglund et al., "Norepinephrine-Mediated Slow Vasomotion."

recommended the use of certain antidepressants as a treatment for insomnia among people with depression, but clinical evidence for this benefit is limited.[205] Trazodone, a serotonin receptor agonist and reuptake inhibitor (SARI) seems to preserve sleep architecture better than other antidepressants when taken specifically for sleep.[206] Please note that many people who take selective serotonin reuptake inhibitors (SSRIs) like Prozac, SARIs like trazodone, or norepinephrine-dopamine reuptake inhibitors (NDRIs) like Wellbutrin/bupropion during the daytime for depression, anxiety, or other reasons do not experience altered sleep, and some of these, like bupropion, when taken early in the daytime, can indirectly improve sleep later that night.[207] Psychiatrists now will prescribe

[205] Molly S. Clark et al., "FPIN's Clinical Inquiries: Antidepressants for the Treatment of Insomnia in Patients with Depression," American Family Physician 84, no. 9 (2011): 1–2, https://pubmed.ncbi.nlm.nih.gov/22164891/.

[206] Andrea Fagiolini et al., "Rediscovering Trazodone for the Treatment of Major Depressive Disorder," CNS Drugs 26, no. 12 (December 2012): 1033–49, https://doi.org/10.1007/s40263-012-0010-5.

[207] Adam Wichniak et al., "Effects of Antidepressants on Sleep," Current Psychiatry Reports 19, no. 9 (August 2017): 63, https://doi.org/10.1007/s11920-017-0816-4; Takashi Maruki et al., "Efficacy and Safety of Each Class of Sleep Medication for Major Depressive Disorder with Insomnia Symptoms: A Systematic Review and Meta-Analysis of Double-Blind Randomized Controlled Trials," Psychiatry and Clinical Neurosciences 79, no. 6 (June 2025): 306–18, https://doi.org/10.1111/pcn.13811; Stephen Wilson and Svetoslav Argyropoulos, "Antidepressants and Sleep: A Qualitative Review of the Literature," Drugs 65, no. 7 (2005): 927–47, https://doi.org/10.2165/00003495-200565070-00003.

antidepressants specifically for sleep because some can improve sleep.

- Melatonin receptor agonists: This category of medications mimics the action of melatonin and may help with falling asleep faster and improving sleep quality. They do not appear to cause dependency or serious side effects.[208] However, the efficacy of melatonin receptor agonists is somewhat unclear.
- Anticonvulsants and dopamine agonist drugs: These may be prescribed to treat restless legs syndrome and other conditions involving nighttime limb movement that can make it difficult to fall and stay asleep,[209] as well as to treat pain. Examples include gabapentin and cabergoline.

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[208] Costello et al., "Effectiveness of Melatonin"; Akira Kuriyama et al., "Ramelteon for the Treatment of Insomnia in Adults: A Systematic Review and Meta-Analysis," Sleep Medicine 15, no. 4 (2014): 385–92, https://doi.org/10.1016/j.sleep.2013.11.788.

[209] Guan Jian Liu et al., "Efficacy and Tolerability of Gabapentin in Adults with Sleep Disturbance in Medical Illness: A Systematic Review and Meta-Analysis," Frontiers in Neurology 8 (July 2017): 316, https://doi.org/10.3389/fneur.2017.00316; Russell P. Rosenberg et al., "A Randomized, Double-Blind, Single-Dose, Placebo-Controlled, Multicenter, Polysomnographic Study of Gabapentin in Transient Insomnia Induced by Sleep Phase Advance," Journal of Clinical Sleep Medicine 10, no. 10 (October 2014): 1093–100, https://doi.org/10.5664/jcsm.4108; Heike Benes et al., "Long-Term Safety and Efficacy of Cabergoline for the Treatment of Idiopathic Restless Legs Syndrome: Results from an Open-Label 6-Month Clinical Trial," Sleep 27, no. 4 (June 2004): 674–82, https://doi.org/10.1093/sleep/27.4.674; Birgit Högl et al., "The Effect of Cabergoline on Sleep, Periodic Leg Movements in Sleep, and Early Morning Motor Function in Patients with Parkinson's Disease," Neuropsychopharmacology 28 (May 2003): 1866–70, https://doi.org/10.1038/sj.npp.1300250.

- Dual orexin receptor antagonists (DORAs): Insomnia is characterized by abnormal wakefulness, and orexin, a neuropeptide that helps regulate the sleep-wake cycle and that is altered in narcolepsy, has been found to cause a state of arousal.[210] Thus it makes sense to investigate orexin as a potential target for insomnia drugs. That led to the development of the DORAs, which work by reducing the action of orexin and thus wakefulness circuits in the brain rather than by direct sedation.[211] Early evidence of such drugs shows that they do effectively block orexin, and indicate a potential role for them in safely treating insomnia.[212] A meta-analysis of ten randomized clinical trials that combined a sample of 7,086 individuals with insomnia found these drugs to be generally safe, with a comparably lower incidence of mild side effects and with improved sleep compared to a placebo.[213] These new medications are expensive, typically costing about $300

[210] S. Nishino et al., "Hypocretin/Orexin and Narcolepsy: New Basic and Clinical Insights," Acta Physiologica 198, no. 3 (March 2010): 209–222.

[211] Rebeka Bustamante Rocha et al., "Dual Orexin Receptor Antagonists for the Treatment of Insomnia: Systematic Review and Network Meta-Analysis," Arquivos de Neuro-Psiquiatria 81, no. 5 (2023): 475–83, https://doi.org/10.1055/s-0043-1768667.

[212] Clemens Muehlan et al., "The Orexin Story and Orexin Receptor Antagonists for the Treatment of Insomnia," Journal of Sleep Research 32, no. 6 (2023): e13902, https://doi.org/10.1111/jsr.13902; Cecilio Álamo et al., "Orexinergic Receptor Antagonists as a New Therapeutic Target to Overcome Limitations of Current Pharmacological Treatment of Insomnia Disorder," Actas Españolas de Psiquiatría 52, no. 2 (2024): 172–82, https://doi.org/10.62641/aep.v52i2.1659.

[213] Bustamante Rocha et al., "Dual Orexin Receptor Antagonists."

or more per month.[214] Generic lower-cost versions are not yet available.



SLEEP PROTOCOL

Adjust Your Caffeine Schedule to Improve Your Sleep

WHAT TO DO

- Avoid caffeine in the eight to ten hours before bed. As it's a stimulant with a long half-life, caffeine should be avoided late in the day, especially if you have trouble falling asleep. Some people who are especially caffeine sensitive may have to avoid caffeine in the twelve hours before bedtime. Even those who can "fall asleep after drinking coffee" are likely not getting the best sleep they could. As someone who loves caffeine, I relate to anyone reluctant to stop drinking caffeine in the afternoon. As painful as it can be, I personally avoid it after 2:00 p.m. most days.
- If you tend to experience an energy crash in the afternoon despite sleeping well, consider delaying drinking caffeine until thirty to ninety minutes after waking up. This slight adjustment to your caffeine schedule may help you have

[214] Morgan Adams Rhodes, "Lemborexant (Dayvigo) for the Treatment of Insomnia," American Family Physician 103, no. 4 (2021): 241–42, https://pubmed.ncbi.nlm.nih.gov/33587572/.

more sustained energy levels throughout the day and may help you avoid the need to drink caffeine after 2:00 p.m.

HOW IT WORKS

Remember adenosine, from Sleep Protocol 7? It's a chemical that builds up in our brains as we go throughout our day, eventually making us want to sleep through a process called homeostatic sleep drive.[215]

Our scientific understanding of adenosine is deeply intertwined with that of caffeine.[216] The research history dates back to 1911, when Coca-Cola asked a psychologist named Harry Levi Hollingworth to investigate the effects of caffeine on human behavior because the US government was suing the company for including what it considered to be a drug with a "poisonous tendency" in the soda. (The cocaine that had once been a major ingredient of Coca-Cola had been removed years earlier.)[217] The company wanted reliable evidence on exactly how caffeine altered human behavior. Hollingworth agreed as long as the company did not include his study results in its advertisements. He also insisted that the results be published regardless of what they revealed.

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[215] Dunwiddie and Masino, "Role and Regulation of Adenosine."

[216] Reichert et al., "Adenosine, Caffeine, and Sleep–Wake Regulation."

[217] Deborah Blum, "How a Lawsuit Against Coca-Cola Convinced Americans to Love Caffeine," Time, September 25, 2018, https://time.com/5405132/coca-cola-trial-caffeine-history/.

Hollingworth's investigation is often cited as the first time that the double-blind, placebo-controlled study design entered psychological research.[218] Although some may argue whether it was indeed a double-blind study, it is considered a landmark for its introduction of rigor into the process, and for its findings. Sixteen participants consumed caffeine daily at a range of doses (at amounts reflecting drinking a daily soda) or a placebo. They then completed various tasks that examined their alertness, coordination, and other mental skills. Caffeine undeniably improved their capacity in all these domains. The findings were favorable toward Coca-Cola because the chemical in question seemed to only help people, not hurt them, at least in the context of what was measured. But Coca-Cola wasn't interested in sleep. (If you were wondering, the US government's case was dismissed on a technicality when the company argued that caffeine wasn't added to the beverage but rather was naturally occurring.)

According to Hollingworth's report, published in the American Journal of Psychology in 1912, a few grains of caffeine taken on successive days were enough to impair sleep.[219] The connection relied on self-reporting and each

[218] Ludy T. Benjamin Jr. et al., "Coca-Cola, Caffeine, and Mental Deficiency: Harry Hollingworth and the Chattanooga Trial of 1911," Journal of the History of the Behavioral Sciences 27, no. 1 (1991): 42–55, https://doi.org/10.1002/1520-6696.

[219] Benjamin Jr. et al., "Coca-Cola, Caffeine, and Mental Deficiency."

individual recording their impression about the quality of their sleep following each study day. But the findings were consistent.

Decades later we know how that happens: Caffeine leads to longer sleep latency (that is, the time it takes to fall asleep) and more nighttime waking, thus leading to reduced wakefulness the next day.[220] We also now know that caffeine can alter sleep architecture, including less time spent in REM.[221]

Hollingworth's study was the first of many to come. Today we understand a great deal about how caffeine stimulates brain activity. It's a psychoactive stimulant that increases the actions of epinephrine and dopamine, in part by increasing the availability of dopamine receptors, the sites on cells where it can attach and thereby initiate its effects.[222] Caffeine also increases the activity of certain neural circuits and brain areas associated with alertness, such as the locus

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[220] Ian Clark and Hans Peter Landolt, "Coffee, Caffeine, and Sleep: A Systematic Review of Epidemiological Studies and Randomized Controlled Trials," Sleep Medicine Reviews 31 (February 2017): 70–78, https://doi.org/10.1016/j.smrv.2016.01.006.

[221] Carissa Gardiner et al., "The Effect of Caffeine on Subsequent Sleep: A Systematic Review and Meta-Analysis," Sleep Medicine Reviews 69 (June 2023): 101764, https://doi.org/10.1016/j.smrv.2023.101764; Clark and Landolt, "Coffee, Caffeine, and Sleep."

[222] N. D. Volkow et al., "Caffeine Increases Striatal Dopamine D2/D3 Receptor Availability in the Human Brain," Translational Psychiatry 5, no. 4 (2015): e549, https://doi.org/10.1038/tp.2015.46.

coeruleus, which we discussed earlier in this chapter.[223] We also understand now how it interferes with sleep. That brings us back to adenosine.

Caffeine is what is called an adenosine antagonist.[224] It parks in adenosine receptors, which prevents adenosine from exerting its normal fatigue-inducing activity in the brain. As sleep expert Matthew Walker describes it, "Caffeine keeps pulling out the chairs from under adenosine."[225] So even though adenosine is building up with each hour we are awake and increasingly so when we are very alert or active, our brains don't register that fatigue or the number of hours we've been awake accurately.[226] Caffeine dampens the communication between adenosine and the brain.

This process explains the phenomenon many of us know of as the caffeine crash. Once caffeine's actions on those receptors wear off, which our metabolism accomplishes in a few hours, the non-receptor-bound adenosine that has

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[223] Astrid Nehlig and Sylvette Boyet, "Dose-Response Study of Caffeine Effects on Cerebral Functional Activity with a Specific Focus on Dependence," Brain Research 858, no. 1 (2000): 71–77, https://doi.org/10.1016/s0006-8993(99)02480-4.

[224] Joaquim A. Ribeiro and Ana M. Sebastião, "Caffeine and Adenosine," Journal of Alzheimer's Disease 20, no. S1 (2010): S3–15, https://doi.org/10.3233/JAD-2010-1379.

[225] "Coffee & Sleep: How Does Caffeine Work & Its Effects on Sleep | Matt Walker & Andrew Huberman," Huberman Lab Clips, YouTube, October 1, 2021, https://www.youtube.com/watch?v=bFdBJ2v6sps&t=291s.

[226] Porkka-Heiskanen et al., "Adenosine"; Basheer et al., "Adenosine and Sleep-Wake Regulation."

accumulated suddenly can bind and fatigue hits us. In addition, caffeine makes our brain more active and increases our capacity for physical effort, and that generates more adenosine as a by-product, which in turn means the crash is harder than it would be without caffeine.

This all sounds rough, but I want to reiterate: I love caffeine and I have no plans to give it up. The data say caffeine in moderation is fine… and may even be good for us. A recent large-scale study involving over 100,000 subjects found that daily intake of approximately two to three cups of caffeinated coffee or tea can reduce risks of stroke and dementia.[227]

"What is moderation?" you may be wondering. One or two cups of strong coffee, a few twelve-ounce diet sodas (we can debate artificial sweeteners later, in chapter 4, but we should all probably avoid the sugar in regular soda), or my favorite source of caffeine: hot brewed or cold brew yerba maté, which will deliver 100 to 300 milligrams of caffeine depending on the source.

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[227] Yu Zhang et al., "Coffee and Tea Intake, Dementia Risk, and Cognitive Function," JAMA 335, no. 11 (2025): 961–74, https://doi.org/10.1001/jama.2025.27259; Miaomiao Yang et al., "Association of Unsweetened and Sweetened Coffee Consumption with Risk of Cardiovascular Disease: A Prospective Cohort Study," Nutrition, Metabolism and Cardiovascular Diseases 36, No. 6 (June 2026): 104539, https://doi.org/10.1016/j.numecd.2026.104539; Robin Poole et al., "Coffee Consumption and Health: Umbrella Review of Meta-Analyses of Multiple Health Outcomes," BMJ 359, no. 1 (November 2017): j5024, https://doi.org/10.1136/bmj.j5024.

The ramifications of caffeine on sleep depend on the dose of caffeine and the time of day you consume it. The closer to your to-sleep time you drink a caffeinated beverage, and the more sensitive you are to caffeine, the greater the impact it will have on sleep.

So now you may be wondering: How close is too close to your to-sleep time? This may seem somewhat obvious but most people don't realize just how long the effects of caffeine last, and that's what I hope to help with in this protocol.

People who are very caffeine sensitive have probably already learned by experience not to drink caffeine past a certain time of day if they hope to sleep later that night. For the rest of us who enjoy caffeine in both the morning and in the afternoon—yes, there are many of us—a bit more understanding of caffeine's effects can go a long way toward consuming it while still experiencing great sleep.

Caffeine is hard to study. Would-be study participants who consume caffeine daily (and that includes about 85 percent of American adults, for example) would need to quit prior to joining a well-controlled investigation of the effects of caffeine, and many people just won't abstain, even for a short while, just to be in a study. Also, researchers don't always know when a person who is in caffeine withdrawal is

back to a baseline state of sleep.[228] And, as I've mentioned, people's sensitivity to caffeine varies.[229] That being said, the evidence shows that our sleep patterns tend to adapt to a daily dose of caffeine. Regular caffeine consumers are able to fall asleep, stay asleep, and cycle through the stages of sleep.[230] It appears that chronic caffeine users develop a sort of sleep-tolerance for the stimulant. This is good and bad. Good, because it means folks like me can enjoy a morning yerba maté and a 2:00 p.m. espresso and still fall asleep easily at 10:00 p.m. Bad, because there is a chance that anyone drinking caffeine past noon is getting less than optimal sleep. Not everything in life is about getting the best possible sleep of course, but if you are struggling to feel rested enough in the daytime or are waking after seven to eight hours of sleep and still not feeling alert, you may try avoiding caffeine in the eight to ten or even twelve hours prior to sleep. Note how you feel. Those tracking their sleep

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[228] Diane C. Mitchell et al., "Beverage Caffeine Intakes in the U.S.," Food and Chemical Toxicology 63 (January 2014): 136–42, https://doi.org/10.1016/j.fct.2013.10.042; Jack E. James, "Caffeine and Cognitive Performance: Persistent Methodological Challenges in Caffeine Research," Pharmacology Biochemistry and Behavior 124 (September 2014): 117–22, https://doi.org/10.1016/j.pbb.2014.05.019.

[229] J. V. Rétey et al., "A Genetic Variation in the Adenosine A2A Receptor Gene (ADORA2A) Contributes to Individual Sensitivity to Caffeine Effects on Sleep," Clinical Pharmacology & Therapeutics 81, no. 5 (2007): 692–98, https://doi.org/10.1038/sj.clpt.6100102.

[230] Janine Weibel et al., "The Impact of Daily Caffeine Intake on Nighttime Sleep in Young Adult Men," Scientific Reports 11, article no. 4668 (2021), https://doi.org/10.1038/s41598-021-84088-x.

with Eight Sleep, WHOOP, Oura rings, or other devices can see how this impacts their REM and non-REM sleep.

The recommendation to delay caffeine by sixty to ninety minutes after waking is just a suggestion. Indeed, this suggestion—when I first made it on the internet—made a lot of people VERY upset. (People don't like it when you try to take away their morning coffee). To be clear: You don't have to do it.

It is also not supported by a clinical trial. It is, however, supported by ample mechanistic evidence of how caffeine adjusts adenosine, and how adenosine continually reduces in the first one to two hours of the day if we don't ingest caffeine. If you are someone who tends to experience energy crashes in the afternoon and you rely on caffeine to overcome them, I suggest at least trying to delay your morning caffeine by sixty to ninety minutes for a few days. It can be a bit painful at first, so maybe start with a thirty-minute delay, then sixty minutes, then ninety minutes, and see if you still need the afternoon caffeine kick. Then try NSDR, as described in Sleep Protocol 8, if you're having trouble making it through the afternoon without it.

One key thing: If you like to exercise soon after you wake up and need caffeine to do so—as I and many others do—then by all means, drink it upon waking and do your workout. Just be sure to hydrate well too. Caffeine can be a mild

diuretic. It's much less of a diuretic than once thought but we tend to wake up slightly dehydrated as it is.[231] Before you slam back two coffees, it's probably best to hydrate first thing.

In My Experience

For my first caffeine of the day, I usually drink yerba maté, a Paraguayan (and Uruguayan and Argentine) tea that is traditionally sipped from a small gourd through a metal straw but that can also be brewed and consumed like any other tea. I'll drink it hot or a cold-brewed canned version. Mateina is my favorite. (Note: I am a partial owner of Mateina, but I love it without bias. Also, it is a good idea to avoid drinking smoked varieties of yerba. Mateina is air-dried, not smoked.) I find that the alertness effect from yerba maté is more gradual and causes fewer jitters, and results in a milder, if any, "crash" as it wears off. Sometimes I have coffee or espresso later in the morning. My absolute favorite time for coffee is in the early afternoon, after a twenty-to-thirty-minute NSDR session. Talk about a reset. My best work sessions seem to follow this protocol. However, my sleep suffers if I drink coffee after about 3:00 p.m. I can fall asleep perfectly fine, but my sleep is not as deep (I use a sleep tracker, and it confirms that I get much

[231] Yang Zhang et al., "Caffeine and Diuresis During Rest and Exercise: A Meta-Analysis," Journal of Science and Medicine in Sport 18, no. 5 (September 2015): 569–74, https://doi.org/10.1016/j.jsams.2014.07.017.

less REM sleep after late-day caffeine). So I try to limit coffee and other sources of caffeine after 3:00 p.m. That's not a hard-and-fast rule; sometimes I drink coffee late in the day, but that's rare and I always regret it when I'm still awake at midnight.

I like to exercise first thing in the morning, within an hour or two of waking up, so on workout days my target regimen is to drink water and caffeine soon after waking, view sunlight, and then exercise. On days when I cannot exercise early, I tend to delay caffeine intake until midmorning, which I find keeps my energy levels steadier into and throughout the afternoon. On these days, I often skip afternoon caffeine because I don't need it.

COMMON QUESTIONS

Does the source of caffeine matter

No. Different drinks contain different amounts of caffeine, and chocolate has small amounts too.[232] But the caffeine molecule works pretty much the same regardless of where it comes from. Some drinks like yerba maté and guayusa tea also contain non-caffeine stimulants in them like theobromine—a plant alkaloid also with antioxidant

[232] Mayo Clinic Staff, "Caffeine Content for Coffee, Tea, Soda and More," Mayo Clinic, February 6, 2025, https://www.mayoclinic.org/healthy-lifestyle/nutrition-and-healthy-eating/in-depth/caffeine/art-20049372; "Caffeine Chart," Center for Science in the Public Interest, last updated March–April 2025, https://www.cspinet.org/caffeine-chart.

properties. And of course nowadays there are hundreds of different energy drinks available that combine caffeine, neuromodulator precursors like L-tyrosine (the amino acid from which dopamine is made), taurine, and on and on. I will drink a zero-sugar energy drink now and again. I discuss these and other stimulants more in chapter 6.

Is caffeine unhealthy

Studies have consistently shown a health benefit from coffee, particularly when consumed before noon, up to a certain amount.[233] Generally, beyond four cups per day of coffee, or the equivalent of 400 milligrams, there are diminishing returns or possibly negative consequences, although larger individuals may be able to handle more than that with no problem.[234] Many of the benefits from coffee and tea are not from caffeine but from antioxidants contained in coffee beans and certain teas. That finding is supported by the fact that decaffeinated coffee confers the same benefit as caffeinated coffee, which includes a potential reduction in the risk of heart disease, prostate cancer, melanoma,

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[233] Luigi Barrea et al., "Coffee Consumption, Health Benefits and Side Effects: A Narrative Review and Update for Dietitians and Nutritionists," Critical Reviews in Food Science and Nutrition 63, no. 9 (2023): 1238–61, https://doi.org/10.1080/10408398.2021.1963207; Xuan Wang et al., "Coffee Drinking Timing and Mortality in US Adults," European Heart Journal 46, no. 8 (February 2025): 749–59, https://doi.org/10.1093/eurheartj/ehae871.

[234] Kitty Pham et al., "High Coffee Consumption, Brain Volume and Risk of Dementia and Stroke," Nutritional Neuroscience 25, no. 10 (2022): 2111–22, https://doi.org/10.1080/1028415X.2021.1945858.

endometrial cancer, liver conditions, type 2 diabetes, and neurodegenerative diseases.[235] In the case of yerba maté, research suggests that it may facilitate weight loss by reducing appetite and perhaps by its effects on digestion as well as the microbiome, a critical aspect of health we will address in chapter 4.[236]



SLEEP PROTOCOL

Avoid Alcohol Within Eight Hours Before Sleep

WHAT NOT TO DO

- Alcohol should not be used as a sleep aid. Alcohol severely disrupts the elegantly sequential architecture of sleep, and reduces both the total amount of deep sleep and REM sleep, so even if it helps you drift off and you sleep for seven to eight hours, it will negatively affect the quality of your sleep.

[235] Poole et al., "Coffee Consumption and Health"; Erratum, Poole et al., "Coffee Consumption and Health," BMJ 360 (January 12, 2018): k194, https://doi.org/10.1136/bmj.k194.

[236] Sun-Young Kim et al., "Anti-Obesity Effects of Yerba Mate (Ilex paraguariensis): A Randomized, Double-Blind, Placebo-Controlled Clinical Trial," BMC Complementary and Alternative Medicine 15 (September 25, 2015): 338, https://doi.org/10.1186/s12906-015-0859-1; Paolo Manghi et al., "Coffee Consumption Is Associated with Intestinal Lawsonibacter asaccharolyticus Abundance and Prevalence across Multiple Cohorts," Nature Microbiology 9, no. 12 (December 2024): 3120–34, https://doi.org/10.1038/s41564-024-01858-9.

- If you drink, avoid alcohol within eight hours of the time you plan to be asleep. This timing reduces the amount of alcohol in your body by the time you go to bed.

The “What Alcohol Does to Your Body, Brain & Health” episode of the Huberman Lab podcast remains one of the top downloaded and shared podcast episodes of all time, across all podcasts. It makes clear that the data show zero to two drinks per week is fine for nonalcoholic adults without confounding medical issues. However, zero is better than any, and there is growing data on the increased risk of cancer associated with alcohol consumption.[237]

I am not anti-alcohol! I know some people enjoy a drink now and again and that is my motivation for the inclusion of this protocol. I subscribe to a “do as you like, but know what you’re doing” philosophy. I personally don’t drink but I have never liked the taste or effects of alcohol.

But there’s a key question for everyone that has not been addressed: Are many of the negative effects of alcohol the consequence of the negative effects of alcohol on sleep? This has not been systematically studied, but for those that like a drink now and again (and also for those who abstain), keep that question in the back of your mind as you read the section below.

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[237] Office of the U.S. Surgeon General, Alcohol and Cancer Risk: The U.S. Surgeon General’s Advisory, 2025, https://www.hhs.gov/sites/default/files/oash-alcohol-cancer-risk.pdf.

HOW IT WORKS

Alcohol is a sedative that affects the neural pathways associated with sleep in multiple ways. Alcohol decreases the amount of time it takes us to fall asleep.[238] But what happens next is not good. During the first half of the night, the presence of alcohol in the body suppresses REM sleep and it takes longer for us to reach whatever REM sleep we do manage to achieve.[239] Remember that each sleep stage is like a gate to the next, so the reduced REM means less deep sleep, and less REM in the next cycle. As discussed earlier in this chapter, REM sleep is critical to our cognitive functioning, to consolidating new information, and to daytime emotional regulation. So missing out on any amount of REM is a problem. And alcohol is a potent REM suppressor.

Alcohol has also been found to decrease the amount of growth hormone and testosterone released during the

[238] D. L. Williams et al., "Dose-Response Effects of Ethanol on the Sleep of Young Women," Journal of Studies on Alcohol and Drugs 44, no. 3 (1983): 515–23, https://doi.org/10.15288/jsa.1983.44.515; Ian M. Colrain et al., "Alcohol and the Sleeping Brain," Handbook of Clinical Neurology 125 (2014): 415–31, https://doi.org/10.1016/B978-0-444-62619-6.00024-0.

[239] Eliza Van Reen et al., "Effects of Alcohol on Sleep and the Sleep Electroencephalogram in Healthy Young Women," Alcohol: Clinical & Experimental Research 30, no. 6 (2006): 974–81, https://doi.org/10.1111/j.1530-0277.2006.00111.x; Julie K. M. Chan et al., "The Acute Effects of Alcohol on Sleep Architecture in Late Adolescence," Alcohol: Clinical & Experimental Research 37, no. 10 (2013): 1720–28, https://doi.org/10.1111/acer.12141.

night—hormones critical for vitality and metabolism for both men and women.[240] Normally, growth hormone is released relatively soon after falling asleep, and testosterone is released before and during REM sleep.[241] The negative effects of alcohol on REM and these hormones happen acutely, meaning after even just a single night of drinking or for those who are alcohol sensitive, even one drink. Only if we metabolize the alcohol prior to bedtime do we avoid some of these consequences, which is why this protocol recommends a specific timing.

Chronic drinking is even more problematic for sleep. Insomnia and alcohol dependency frequently coexist, for example.[242] The cycle is self-perpetuating in that poor sleep leads to increased caffeine and other stimulant use during the

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[240] Patricia N. Prinz et al., "Effect of Alcohol on Sleep and Nighttime Plasma Growth Hormone and Cortisol Concentrations," Journal of Clinical Endocrinology & Metabolism 51, no. 4 (1980): 759–64, https://doi.org/10.1210/jcem-51-4-759; A. C. Ekman et al., "Ethanol Decreases Nocturnal Plasma Levels of Thyrotropin and Growth Hormone but Not Those of Thyroid Hormones or Prolactin in Man," Journal of Clinical Endocrinology & Metabolism 81, no. 7 (1996): 2627–32, https://doi.org/10.1210/jcem.81.7.8675588.

[241] Marco Zaffanello et al., "Complex Relationship Between Growth Hormone and Sleep in Children: Insights, Discrepancies, and Implications," Frontiers in Endocrinology 14 (January 24, 2024): 1332114, https://doi.org/10.3389/fendo.2023.1332114; Rafael Luboshitzky et al., "Disruption of the Nocturnal Testosterone Rhythm by Sleep Fragmentation in Normal Men," Journal of Clinical Endocrinology & Metabolism 86, no. 3 (2001): 1134–39, https://doi.org/10.1210/jcem.86.3.7296.

[242] Subhajit Chakravorty et al., "Alcohol Dependence and Its Relationship with Insomnia and Other Sleep Disorders," Alcohol: Clinical & Experimental Research 40, no. 11 (2016): 2271–82, https://doi.org/10.1111/acer.13217.

day and into the afternoon and evening, which leads to difficulty sleeping at night, which leads to alcohol intake as a sleep aid. Beyond the changes one or two drinks can make on REM sleep, people with alcohol dependency tend to have decreased amounts of both N2 and N3 sleep, the non-REM period when we sink into a deep sleep.[243] These alterations in sleep architecture can persist for some time even after a person stops drinking. Frequent drinking also interferes with our circadian rhythm, disrupting or even decreasing the amount of melatonin our brains produce.[244]

Adolescents are at particular risk of turning to alcohol intake as a sleep aid. In 2008, public health researchers at the University of Texas Health Science Center reported their findings from a large, yearlong study of insomnia among youth between the ages of eleven and seventeen years. They collected data on 3,134 people in this age range from a baseline date and then twelve months later and found that chronic insomnia heightened the risk of alcohol use.[245] That's concerning because insomnia is the most common sleep disorder among young people, and because nighttime

[243] Ian M. Colrain et al., "Impact of Alcoholism on Sleep Architecture and EEG Power Spectra in Men and Women," Sleep 32, no. 10 (2009): 1341–52, https://doi.org/10.1093/sleep/32.10.1341.

[244] Colrain et al., "Alcohol and the Sleeping Brain."

[245] Robert E. Roberts et al., "Chronic Insomnia and Its Negative Consequences for Health and Functioning of Adolescents: A 12-Month Prospective Study," Journal of Adolescent Health 42, no. 3 (2008): 294–302, https://doi.org/10.1016/j.jadohealth.2007.09.016.

phone use, which is very common in this age group, also interferes with sleep quality and duration.[246] Phone use at night has been linked to insomnia.[247]

That means there are two major risks to this age group when it comes to good sleep, not to mention the chronic stress that kids at that age face. We'll discuss phone use in Sleep Protocol 14.

COMMON QUESTIONS

It's not practical to avoid alcohol within eight hours of sleep because any drinking I do happens at night. How can I compensate for the negative effects of consuming alcohol close to bedtime

For most people, social drinking happens well within eight hours of going to sleep. What's important here is that drinking this way every night, multiple times per week, or even once per week will disrupt sleep, and thereby health, in detrimental ways. Occasional alcohol use closer to bedtime, say once a month or less, is very unlikely to cause long-term

[246] Eric O. Johnson et al., "Epidemiology of DSM-IV Insomnia in Adolescence: Lifetime Prevalence, Chronicity, and an Emergent Gender Difference," Pediatrics 117, no. 2 (2006): e247–56, https://doi.org/10.1542/peds.2004-2629; Sofia de Sá et al., "The Influence of Smartphones on Adolescent Sleep: A Systematic Literature Review," Nursing Reports 13, no. 2 (2023): 612–21, https://doi.org/10.3390/nursrep13020054.

[247] Liese Exelmans and Jan Van den Bulck, "Bedtime Mobile Phone Use and Sleep in Adults," Social Science & Medicine 148 (January 2016): 93–101, https://doi.org/10.1016/j.socscimed.2015.11.037.

problems, unless it's excessive during that once-a-month night out.

If you drink in the eight hours before sleep, make sure you also consume sufficient water before going to sleep and after waking. Take electrolytes too if you can, and emphasize other behaviors that support great sleep, as well as a healthy microbiome, to try to offset the effects of alcohol.

To be clear, it's good to enjoy life. If having a few drinks during a celebration or a night out now and again is how you do, you're not alone. The important considerations are that this isn't happening too often, that you're not driving, that you're not experiencing any kind of dependency, and that you're not too young to drink (the negative effects of alcohol are much worse for the developing brain compared to the adult brain).[248]

Does the type of alcohol matter

Not with respect to the negative effects on sleep and the microbiome. As mentioned above, no alcohol is better than any alcohol. Although many studies have been done on the potential benefits of red wine, conclusive evidence of this link does not exist.[249] The amount of red wine you would

[248] Lindsay M. Squeglia, "Alcohol and the Developing Adolescent Brain," World Psychiatry 19, no. 3 (2020): 393–94, https://doi.org/10.1002/wps.20786.

[249] Jeffrey S. Wojtowicz, "Long-Term Health Outcomes of Regular, Moderate Red Wine Consumption," Cureus 15, no. 10 (2023): e46786, https://doi.org/10.7759/cureus.46786.

need to drink to get benefits from grape skin, for example, would require consumption of enough alcohol to far outweigh any benefits. Put simply, the interaction between alcohol and our sleep processes is the same regardless of the drink.



SLEEP PROTOCOL

Avoid Cannabis as a Sleep Aid

WHAT TO DO

- Do not use cannabis specifically for sleep. Cannabis should not be used to facilitate falling asleep or returning to sleep after night wakings. It potently reduces REM sleep.
- Understand CBD's impact on sleep. Some evidence exists on the benefit of CBD for sleep but those effects are mainly due to the reduced anxiety it causes.[250] At higher dosages (e.g., 160 mg or more), CBD is more of a sedative. The dosage of CBD that makes it an effective sleep aid varies depending on weight and other factors.[251]

[250] Scott Shannon et al., "Cannabidiol in Anxiety and Sleep: A Large Case Series," Permanente Journal 23, no. 1 (2019), https://doi.org/10.7812/TPP/18-041.

[251] Lucy H. Silmore et al., "Food Effects on the Formulation, Dosing, and Administration of Cannabidiol (CBD) in Humans: A Systematic Review of Clinical Studies," Pharmacotherapy 41, no. 4 (April 2021): 405–20, https://doi.org/10.1002/phar.2512; Ehsan Moazen-Zadeh et al.,

HOW IT WORKS

Cannabis is a psychoactive drug that comes from the cannabis plant. The main chemical behind its psychoactive effects is THC (delta-9-tetrahydrocannabinol).[252] CBD, or cannabidiol, is another well-known component of the cannabis plant; this chemical technically is psychoactive, too, because it impacts the brain, but it's not responsible for the "high" that people are usually seeking when they use cannabis. THC can impair cognitive or behavioral functioning; CBD doesn't appear to do so.[253] When present with THC in the cannabis plant or in an edible, CBD can modify the way that THC works in the brain, so their ratios can determine the subjective effects of the drug.[254]

"Pharmacokinetics of Cannabidiol: A Systematic Review and Meta-Regression Analysis," Cannabis and Cannabinoid Research 9, no. 4 (August 2024): 939–66, https://doi.org/10.1089/can.2023.0025.

[252] Priyamvada Sharma et al., "Chemistry, Metabolism, and Toxicology of Cannabis: Clinical Implications," Iranian Journal of Psychiatry 7, no. 4 (2012): 149–56, https://pubmed.ncbi.nlm.nih.gov/23408483/.

[253] Andréia Pucinelli de Sousa Queiroz et al., "Cognitive Effects of Cannabis Use: A Comprehensive Review Across Domains," Neurology International 17, no. 7 (2025): 107, https://doi.org/10.3390/neurolint17070107; Ana Julia de Lima Bomfim et al., "Effects of the Acute and Chronic Administration of Cannabidiol on Cognition in Humans and Animals: A Systematic Review," Cannabis & Cannabinoid Research 8, no. 6 (2023): 955–73, https://doi.org/10.1089/can.2023.0086.

[254] C. Austin Zamarripa et al., "Assessment of Orally Administered Δ9-Tetrahydrocannabinol When Coadministered with Cannabidiol on Δ9-Tetrahydrocannabinol Pharmacokinetics and Pharmacodynamics in Healthy Adults: A Randomized Clinical Trial," JAMA Network Open 6, no. 2 (2023): e2254752, https://doi.org/10.1001/jamanetworkopen.2022.54752.

THC potently reduces the amount of REM sleep even at low doses.[255] THC is often used as an aid for falling asleep, as its psychoactive properties are often accompanied by, or followed by, drowsiness. As with alcohol, people often come to depend on THC to fall asleep, which disrupts the sleep-wake cycle, and can put them at risk for the physical and mental health issues associated with a deficiency in REM sleep, including depression and cardiovascular disease, among other issues.[256] Not everyone who uses cannabis experiences these challenges of course but the link between REM sleep and health is solid: THC reduces REM. Ask people who have used cannabis regularly and then stopped and you'll often hear about the wild dreams they experience afterward. That is the REM-rebound compensation for the lack of REM they experienced while using cannabis.[257]

A multidisciplinary team at McMaster University, in Canada, conducted a review of the scientific literature on the

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[255] Thomas Schierenbeck et al., "Effect of Illicit Recreational Drugs upon Sleep: Cocaine, Ecstasy and Marijuana," Sleep Medicine Reviews 12, no. 5 (2008): 381–89, https://doi.org/10.1016/j.smrv.2007.12.004.

[256] Anu-Katriina Pesonen et al., "REM Sleep Fragmentation Associated with Depressive Symptoms and Genetic Risk for Depression in a Community-Based Sample of Adolescents," Journal of Affective Disorders 245 (February 15, 2019): 757–63, https://doi.org/10.1016/j.jad.2018.11.077; Leary et al., "Association of Rapid Eye Movement Sleep."

[257] Joshua Feriante and Shantanu Singh, "REM Rebound Effect," StatPearls, last updated September 12, 2024, https://www.ncbi.nlm.nih.gov/books/NBK560713/.

link between cannabinoids and sleep.[258] One of their most noteworthy findings was that the direct effects of both THC and CBD on sleep are lacking data because most clinical studies of cannabis have focused on various health conditions, such as pain, post-traumatic stress disorder, or glaucoma. The researchers did identify data showing that cannabinoids could have a positive impact on sleep. But the small size of these studies and the fact that examining this connection was not the main objective of the investigations somewhat limits the usefulness of the findings. Furthermore, THC and CBD are vastly different, which makes any reference to cannabinoids confusing and unhelpful.

A survey of 4,637 ninth through twelfth graders in Colorado found that those who consumed cannabis through edibles or vaping tended to sleep fewer than seven hours per night.[259] Another study of adolescents found that the sedative effects of THC were followed by increased wakefulness at night and impaired memory the next day, which could have been

[258] N. Kuhathasan et al., "The Use of Cannabinoids for Sleep: A Critical Review on Clinical Trials," Experimental and Clinical Psychopharmacology 27, no. 4 (2019): 383–401, https://doi.org/10.1037/pha0000285.

[259] Evan Winiger et al., "Novel Methods of Cannabis Use and Lower Sleep Duration Among High School Students," Cannabis 5, no. 2 (2022): 66–73, https://doi.org/10.26828/cannabis/2022.02.006.

directly due to the THC, which is known to impair memory, lack of sleep, or both.[260]

Researchers have also found a potential link between use of cannabis among pregnant mothers and poor sleep habits among their children during adolescence.[261] You might be asking, "How many pregnant women use cannabis?" A recent report said that anywhere from 4 to 12 percent of pregnant mothers in the US said they'd used cannabis during pregnancy, usually by edible and usually in the first trimester prior to learning they were pregnant.[262] Cannabis use during pregnancy has reportedly increased in recent years, from 3.4 percent in 2002 to 7 percent in 2017.[263] Smoking or ingesting cannabis during pregnancy has been tied to lower birth weight, possibly due to THC-induced changes to the

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[260] Anthony N. Nicholson et al., "Effect of Delta-9-Tetrahydrocannabinol and Cannabidiol on Nocturnal Sleep and Early-Morning Behavior in Young Adults," Journal of Clinical Psychopharmacology 24, no. 3 (2004): 305–13, https://doi.org/10.1097/01.jcp.0000125688.05091.8f.

[261] Evan A. Winiger and John K. Hewitt, "Prenatal Cannabis Exposure and Sleep Outcomes in Children 9–10 Years of Age in the Adolescent Brain Cognitive Development SM Study," Sleep Health 6, no. 6 (2020): 787–89, https://doi.org/10.1016/j.sleh.2020.05.006.

[262] Kelly C. Young-Wolff et al., "Prenatal Cannabis Use and Maternal Pregnancy Outcomes," JAMA Internal Medicine 184, no. 9 (2024): 1083–93, https://doi.org/10.1001/jamainternmed.2024.3270.

[263] Kelly C. Young-Wolff et al., "Rates of Prenatal Cannabis Use Among Pregnant Women Before and During the COVID-19 Pandemic," JAMA 326, no. 17 (2021): 1745–47, https://doi.org/10.1001/jama.2021.16328.

placenta.[264] Rates of neonatal and infant death are higher with cannabis use during pregnancy compared to no such use (and combining cannabis with nicotine raises this risk even higher).[265]

Many people use CBD to try to induce sleep, but rigorous evidence of a direct benefit to sleep is currently insufficient. Chances are any positive effects of CBD on sleep are due to its anxiety-lowering effects. CBD may help people to stop ruminating—a prerequisite to falling asleep. As such, there are populations for whom CBD may be useful. In one study, fifty-seven of seventy-two adults with anxiety reported improved sleep with the use of CBD, although this study was retrospective, meaning the participants were reporting their recall of possible effects from memory rather than measuring them directly.[266] A 2025 review of twenty-nine studies identified a potential for CBD to reduce sleep disruptions but emphasized that research on this link has suffered from a lack of standardization.[267] There certainly is enough evidence to warrant continued study of CBD in this context

[264] Mohammad R. Hayatbakhsh et al., "Birth Outcomes Associated with Cannabis Use Before and During Pregnancy," Pediatric Research 71, no. 2 (2012): 215–19, https://doi.org/10.1038/pr.2011.25.

[265] B. Adam Crosland et al., "Risk of Adverse Neonatal Outcomes After Combined Prenatal Cannabis and Nicotine Exposure," JAMA Network Open 7, no. 5 (2024): e2410151, https://doi.org/10.1001/jamanetworkopen.2024.10151.

[266] Shannon et al., "Cannabidiol in Anxiety and Sleep."

[267] Juan G. Perez et al., "Cannabinoids for Anxiety and Sleep Disturbances: A Scoping Review," Medical Cannabis and Cannabinoids 8, no. 1 (October 2025): 219–237, https://doi.org/10.1159/000548890.

and no known negative effects of CBD that would preclude trying it for sleep if you feel stressed at night. That said, in January 2024, a team from Australia published the results of a small, randomized, placebo-controlled study in which fifteen people received 150 milligrams of CBD and fifteen people received a placebo as a sleep aid to combat insomnia.[268] The investigators observed no difference between the two groups in the time it took to fall asleep, the efficiency of their sleep, the duration of sleep, or overall severity of insomnia. There were some signs of an improved mood upon waking among the CBD group, and that group had fewer night wakings after two weeks of treatment.

One more thing: An increasing number of studies report a potential for THC (but not CBD) to induce psychosis in certain individuals.[269]

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[268] Andrea J. Narayan et al., "Cannabidiol for Moderate–Severe Insomnia: A Randomized Controlled Pilot Trial of 150 mg of Nightly Dosing," Journal of Clinical Sleep Medicine 20, no. 5 (2024): 753–63, https://doi.org/10.5664/jcsm.10998.

[269] Tabea Schoeler et al., "Rates and Correlates of Cannabis-Associated Psychotic Symptoms in Over 230,000 People Who Use Cannabis," Translational Psychiatry 12, no. 1 (2022): 369, https://doi.org/10.1038/s41398-022-02112-8; Alessio Fiorentini et al., "Substance-Induced Psychoses: An Updated Literature Review," Frontiers in Psychiatry 12 (December 22, 2021): 694863, https://doi.org/10.3389/fpsyt.2021.694863; Nadia Solowij and Robert Battisti, "The Chronic Effects of Cannabis on Memory in Humans: A Review," Current Drug Abuse Reviews 1, no. 1 (2008): 81–98, https://doi.org/10.2174/1874473710801010081.

If your goal is to improve your sleep, there are better tools covered in the other protocols of this chapter. The same goes for stress management—more reliable tools exist (and these can improve sleep too), which I'll explain in a later chapter.



SLEEP PROTOCOL

Avoid Screens Within an Hour or More Before Bedtime

WHAT TO DO

- Avoid screens in the bedroom at night. This includes phones, laptops, tablets, and TV screens.
- Avoid consuming information on an electronic device within an hour (at least) of bedtime. Light from phones and other screens can disrupt your circadian clock.
- If you opt to read or watch content on a screen in the hour before sleep, dim the screen. I suggest lowering the brightness and/or activating the "red" screen function. You can also wear blue and green (short wavelength) blocking glasses.

HOW IT WORKS

In 2011, the National Sleep Foundation conducted a poll.[270] They asked 1,508 Americans about their sleep habits, whether any technology was around them in the hour before bed, and if so, what it was and how they used it. Ninety percent of respondents said they used some kind of screen device at this time; for those under thirty, that percentage climbed to 96! Adolescent and young adult respondents were more likely to use cell phones or play video games before bed compared to adults over thirty, and the same went for computers and digital music players. I strongly suspect those percentages have either held constant or increased in the last decade and a half. In 2011, watching television was the most common activity among the entire sample. Others included doing homework, using social media, using a smartphone, sending work emails, and other activities. The researchers also asked about reading printed books and magazines before bed, which about half of respondents reported doing.

More than half the respondents—63 percent, to be exact—said they didn't get enough sleep on weeknights. And nearly all the people who reported insufficient sleep said it negatively affected both their personal and professional lives. The high rates of late-night technology use and insufficient sleep seen in the survey responses didn't

[270] Michael Gradisar et al., "The Sleep and Technology Use of Americans: Findings from the National Sleep Foundation's 2011 Sleep in America Poll," Journal of Clinical Sleep Medicine 9, no. 12 (2013): 1291–99, https://doi.org/10.5664/jcsm.3272.

automatically mean the two were related, so the researchers conducted some statistical analyses to check for such a link. They found that difficulty falling asleep rose in keeping with the amount of technology people used in the hour before bed. More passive devices like e-readers were less problematic than interactive devices like game consoles, cell phones, and laptops.

These screens are illuminated by LEDs, and as we've discussed in this chapter, this type of lighting is disruptive to our circadian rhythm. Evidence suggests that LED screens can interrupt melatonin production, which, as you now know, is essential for sleep. In 2014, a group of sleep researchers at Harvard Medical School reported that people who read on an e-book at night with a dim bedside light took longer to fall asleep, were less sleepy, had lower levels of melatonin, and were less alert the next morning compared to people who read a printed book with a dim bedside light before bed.[271]

And there's more. In 2018, a team of researchers, including UCLA psychiatrist Peter Whybrow, reviewed the data on the potential link between LED lighting and mental health.[272]

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[271] Anne-Marie Chang et al., "Evening Use of Light-Emitting eReaders Negatively Affects Sleep, Circadian Timing, and Next-Morning Alertness," Proceedings of the National Academy of Sciences 112, no. 4 (2015): 1232–37, https://doi.org/10.1073/pnas.1418490112.

[272] Michael Bauer et al., "The Potential Influence of LED Lighting on Mental Illness."

They concluded that exposure to blue light, in particular, posed a risk to sleep and other processes governed by the circadian system. There is also evidence suggesting that the way in which artificial lighting at night interferes with this system may lead to depression.[273]

The problem with devices at bedtime is not solely due to the type of lighting. Technology, particularly interactive technology, stimulates our brains in a way that counters the calm that we need in order to fall asleep. For example, the devices in the 2018 study were most strongly linked to difficulty falling asleep and to poor sleep quality. Also, people who sleep with their phones in their rooms with the ringer on contributed to the numbers reported in the study, indicating that the problem isn't just about late-night scrolling.

We have known since at least the 1970s that the bedroom should be reserved for sleep and intimacy.[274] The evidence linking various stimuli to insomnia is well established. Yet decades later, here we are, with about 1,350 out of 1,508 randomly selected Americans ignoring this basic practice.

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[273] T. A. Bedrosian and R. J. Nelson, "Influence of the Modern Light Environment on Mood," Molecular Psychiatry 18, no. 7 (2013): 751–57, https://doi.org/10.1038/mp.2013.70.

[274] Richard R. Bootzin, "Stimulus Control Treatment for Insomnia," Proceedings, 80th Annual Convention, APA, 1972.

Believe it or not (believe it!), this protocol is one of the most potent in this chapter. It's also one of the most difficult to stick to for many people. Sleep-based apps and the use of smartphones as alarm clocks don't make it any easier. Still: I highly suggest you try it.

COMMON QUESTIONS

Do blue-blocking glasses counter the negative effects of screens at night

Yes, somewhat. Better, however, would be glasses that block both blue and green light, aka short wavelengths of visible light, at that time—especially if you use screens at night. (See Sleep Protocol 3 for more information.)



SLEEP PROTOCOL

A Tip Sheet for Avoiding Jet Lag

Some people experience terrible jet lag, others far less so. This changes as we age. And it depends on where we fly—which direction and if we need to push ourselves to wake earlier or stay awake later than usual. The latter is always easier for people.

Five factors allow you to shift your circadian clock. Knowing these can help you switch to a new time zone quickly, which allows you to be more effective on work trips and enjoy vacations more fully. The factors you use to

entrain your SCN to a new local light-dark cycle are, in descending order of influence:

- Light
- Temperature
- Meal times
- Exercise
- Social factors

You might think: Can't I just hop on the local schedule? Alas, that's hard because your SCN is entrained to wherever you have been for the prior week or so before your trip. So for instance if you fly from California to London and arrive at 5:00 p.m. and you view the sunset, your SCN thinks it's morning sunlight and it wakes you up. But there is a way to overcome this. I'll explain.

WHAT TO DO

BEFORE YOU TRAVEL EAST

Eastward travel is typically more challenging for the body. Why? Most people find it easier to stay awake later than to fall asleep earlier. But you can prepare for your trip and partially adjust your circadian rhythm before you leave:

- Get early bright light exposure, but be careful when. When traveling east, you need to shift to waking up earlier than you normally would. Starting two days before

you leave, try waking up thirty to forty-five minutes prior to your usual wake-up time and view bright sunlight or bright artificial light as soon as you wake up.

- Drink caffeine early. This will help promote alertness early in the day and make it much easier to shift to your new time zone when you travel.
- Get moving. Within sixty minutes of your new, earlier wake-up time, exercise. Even just some light jogging will help if you can't do a full workout.

These three levers—light, caffeine, and exercise—will help adjust your circadian clock to your new eastward-of-typical location, fast.

BEFORE YOU TRAVEL WEST

As mentioned earlier, westward travel tends to be easier for most people. That said, it can still help to adjust your circadian rhythm before you leave by using light, exercise, and caffeine. In this case, however, you'll want to shift everything thirty to forty-five minutes later in the day as opposed to earlier. A slightly later wake-up time, a slightly later caffeine intake in the morning and early afternoon (but still avoid caffeine in the eight to ten hours prior to bedtime), and a slightly later workout—even just by forty-five minutes—will all help.

You'll also want to make sure that you view bright light from artificial lights one to two hours after sunset. Remember, you're trying to prepare to stay up later as you soon fly westward.

WHEN YOU ARRIVE AT YOUR NEW DESTINATION (REGARDLESS OF WHETHER YOU FLEW EAST OR WEST)

- Eat like a local. Adopting the local meal schedule when you arrive—eating breakfast, lunch, and dinner more or less at the same times as the locals, even if you're not hungry—will help you shift your circadian clock to the new time zone. Personally, I hate to eat when I'm not hungry (and love to eat when I am!), so I opt out of this protocol. If it works for you, go for it. The sooner you can get on the local meal schedule, the better. Many find it effective to fast for a period of fourteen to sixteen hours before the local breakfast time so they are very hungry when that time arrives.
- Time your light exposure correctly by knowing your temperature minimum. This is the real power tool to shift your circadian clock fast, but it involves learning your temperature minimum first. See Sleep Protocol 3 and Sleep Protocol 5 for more on that topic. For now, know that it occurs approximately two hours before your usual wake-up time. If you typically wake up at 7:00 a.m., your

temperature minimum is very likely to be between 4:30 a.m. and 5:30 a.m. Here's the key: If you get bright light exposure in the one to two hours after your temperature minimum, you will adjust your circadian clock such that it will make you want to go to sleep earlier on the subsequent nights and wake up earlier on the subsequent mornings. Perfect for traveling east. Conversely, if you get bright light exposure in the one to two hours before your temperature minimum, you will shift your circadian clock such that it will make you want to go to sleep later on the subsequent nights and wake up later on the subsequent mornings. Perfect for traveling west. But remember we are referring to the time of your temperature minimum before you leave! This is vital to understand. Get it right and you'll shift fast across time zones. Get it wrong and it can be a mess. Why? Well, think about this. Imagine you fly from San Francisco to London (+8 hours). You land in the morning London time and the sun is out. I said get morning sunlight in your eyes, right? Yes, but for the first two to three days in London before you adjust your clock, viewing that London morning sunlight is the same to your brain's clock as getting a giant blast of light in your eyes at midnight San Francisco time. So you just made your body want to go to bed much later and wake up later. You got it backward. Many people sleep well their first night

after flying east, wake up, see sunlight, and the second night is a mess. They can't sleep! It would be better to wear dark sunglasses during the two to three hours before your temperature minimum for the first two days in London. After those first two or three days, just view sunlight per the recommendations earlier in this chapter. Obviously, I am just using London as an example. You must figure it out based on your own travel destination. I know it sounds complicated, but if you know the time of your temperature minimum back home, and follow the instructions above, you'll nail this protocol easily.

- Use artificial light and blue and green (short wavelength) blocking glasses as tools. Of course, it's often very cloudy, or you may land at night, and we can't control the time of sunrise and sunset—though if you can, please contact me. What you can do, however, is travel with a portable bright light device (you'll want one with 5,000- to 10,000-lux brightness) to shift your circadian clock. Several are the size of a book and thin enough to fit in a carry-on bag. I have no affiliation with any companies that make these, but I use the Verilux Happy Light when I travel. They cost about $50. At home I use a larger countertop unit, the Carex Daylight (also no affiliation) and Ra Optics or Bon Charge full-spectrum LED bulbs in the daytime, as discussed earlier. Remember that even when it is cloudy, sunlight through cloud cover is still

almost always far brighter than artificial lighting. Get outside and view that sunlight at the correct times and you'll shift your circadian clock fast. Then, you can return to the standard protocols to stay on the local schedule. One more thing we can all control is the brightness of our evening and nighttime environment. As you get ready for bed, reduce exposure to bright lights—especially blue and green, or short-wavelength, light. Alas, as you now know, all LED lights are rich in blue and green wavelengths. For that reason, I wear blue and green (short wavelength) blocking glasses after sundown to help adjust to the local schedule. This protocol is known to protect against the melatonin-reducing effects of bright light and to prevent the cortisol-increasing effects of bright light.

- Maintain a regular exercise schedule. While traveling, try to maintain a consistent early-in-the-day exercise routine (adjusted to the local schedule). That, too, helps shift your circadian clock.
- A final tip. For trips that last fewer than two days, staying on (or near) your home sleep-wake, eating, and exercise schedule can be beneficial to minimize disruptions to your circadian rhythm. These tips will help you quickly overcome any jet lag in your future travels—they've certainly helped me over the years, as well as people in

the military, medical professions working night shifts, and more.